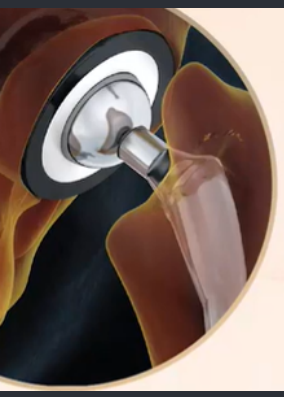
COURSE

Hip Arthroplasty

31 lectures



Kasr Al Ainy
Hip Arthroplasty Course

From Basics to Practice

Watch course

15 October 2026

TRANS PRO

Developed by

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IMPORTANT

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This work was created using a computer program developed by us through Vibe Coding, together with several other AI-assisted tools. Using these tools, we were able to create transcripts of lectures, including lectures delivered in Egyptian Arabic, and organize them into a more academic, book-like format.

The aim is to help the listener focus on listening without having to write notes, while having an organized transcript they can easily return to for quick revision.

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- Click any line in the Contents to jump to that lecture.
- The time at the bottom right of each page (» 04:57) opens the original lecture on YouTube at that exact moment.
- "↑ Chapter" returns to the lecture's contents; "↑ Book" returns to the book's contents.
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- "(see Fig.)" jumps to the figure; "↑ back to text" under the figure brings you back to where it was cited.
- "Ask AI" beside each heading opens a page that turns that section into a ready question for an AI: simplify it, clarify what the recording left unclear, or check for newer guidelines.
- Orange notes such as [C1] and [Q1] are AI corrections and doubts - mostly where the recording was hard to transcribe. Each lecture lists them after its index of figures.

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Biomechanics of THA

(Forces, Offset, Version, Range of Motion, Stability and Templating)

Prof Dr Mohamed Gobba

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Why Biomechanics of THA Matters Ask AI

- Question: importance of **biomechanics of the total hip**
- Last 60–70 years → formal **analysis** of every defect and of performance
- Science behind any craft → very large difference
- Residents implant joints before understanding them
- Implanting without knowing what fits into what, and why
- Every unexplained clinical finding → a **biomechanical explanation**
- Pre-op: discuss head options with patient
- Intra-op: choose which set of heads to use
- Typical problems
  - **Head size:** rep has no 28 mm head, only 32 or 36 → implant or postpone?
  - Post-op leg turns inward
  - New limp absent before surgery
  - **Lateral pain** → cannot sleep on that side
  - Urgent 2 a.m. WhatsApp call from patient
- Question: could anything have been done?
- **Six biomechanical points** → introduction only; detail in textbooks
  - Forces around the hip
  - Offset
  - Version
  - Range of motion
  - Stability
  - Templating

Forces Around the Hip Ask AI

- Two principles
  - Principle 1 — door and hinge
    - Push at point A vs point B
    - Farther from hinge → less force needed
  - Principle 2 — hip forces behave like a **seesaw**
    - Hip = fulcrum in the middle; two sides move
    - Light loads on both sides → hinge survives
    - Heavy loads → problem
- All loads summed at hip → opposed from below by **joint reaction force**
- **Newton's third law:** every force → equal and opposite force
- Heavier loads → joint reaction force increases considerably
- Weight on both sides ↑ → **failure**

Single-Leg Stance and Abductor Deficiency Ask AI

Normal Single-Leg Stance Ask AI

- Double stance → body weight distributed over both legs
- **Single stance** → different loading
- Single stance = half of the gait cycle
- Body weight → acts downward
- **Abductors** → counteract body weight → level the pelvis
- Joint reaction force from below → opposes both, like the seesaw
- Normal single stance: body weight = one force
- Abductors act from **insertion to origin**
  - Level opposite side of pelvis → prevent pelvic drop
- Lever arms balance forces: **A × B = C × D**

Abductor Damage Ask AI

Abductors

- Scenario: THA via **lateral approach**, inexperienced hand → abductors completely damaged
- Equation disturbed → A × B ≠ C × D
- Must reduce C and D → re-balance → level the pelvis
- Weight loss by tomorrow → not possible
- Patient reduces **D** instead
- Mechanism: leans instead of standing upright
- Lean → **center of mass** moves inward toward joint → D ↓ → equation balanced
- Result: equilibrium, but forced to lean
- Complaint: "I limp whenever I walk" → **Trendelenburg gait**

Correction with a Cane Ask AI

Cane

- **Cane** in the **contralateral hand**
- Cane → redistributes the load
- Cane balances equation → no need to lean
- Cane → opposite hand, not same side as operated hip

High Yield

- Abductor loss → A × B ≠ C × D → lean → Trendelenburg gait
- Lean → center of mass toward joint → D ↓
- Cane → contralateral hand

Femoral Offset Ask AI

- **Offset:** distance between **center of the head** and femoral **axis**
- Adjustment → requires **modular stems**
  - Most → no comparative data → almost nobody uses them

Offset	Mechanism	Result
Too small	No offset → **greater trochanter** strikes	**Bony impingement;** hip instability
Too large	Friction ↑ between **IT band** and **trochanteric bursa** → constant pressure	**Trochanteric bursitis;** continuous inflammation

- Complaint of constant lateral pain → offset chosen incorrectly
- Prevention → correct choice through **templating**

Femoral Version Ask AI

- Lateral approach → view with **patella facing the ceiling**
- Main fear → dislocation

Approach	Dislocation direction	Stem version	Aim
Lateral	Anterior	More **retroverted**	Head travels longer distance before dislocating
Posterior	Posterior (opposite)	More **anteverted**	Head travels longer distance before dislocating

Excess Stem Anteversion Ask AI

Version

- Stem excessively anteverted → head recentered in cup only with leg internally rotated
- Pediatric analogue: **increased femoral anteversion** (**in-toeing**)
- Anteversion ↑ intra-op → leg comes out **internally rotated**
- In-toeing not corrected after two months → stem excessively anteverted

Clinical Pearl

- Persistent post-op in-toeing → excessively anteverted stem

Range of Motion Ask AI

- ROM in THA → not for sport/activity
- ROM in THA → for **total hip survival**
- Needed for daily movements → range not small
- **Extremes of range → impingement**
  - → **polyethylene wear** on impingement side
  - → or on opposite side
  - → or **complete failure**
- ROM determinants
  - **Head–neck ratio** factors
  - **Cup factors**

Head–Neck Ratio Ask AI

- **Head–neck ratio** = head diameter : neck diameter
- Larger head → greater ROM
- Same neck, small vs large head → large head → higher ROM
- Step `1` → very large head
- Same head, thick vs thin neck → **thin neck** → higher ROM
- Apparent solution → large head + thin neck
  - Manufactured widely → after a year → prosthesis **fracturing**
  - Thin neck cannot carry large head

Trapezoidal Neck Ask AI

Neck

- **Trapezoidal neck**: wide in one plane → no impingement in that area
- Impingement area removed → neck becomes trapezoidal
- Modern prosthetic neck → not completely round → slightly flattened

Skirted vs Unskirted Heads Ask AI

Heads

- Options: **skirted** and **unskirted** heads
- Scenario: stem grips well → trial shows need for **long head** → needs **skirt**
- Skirt → thickens **neck diameter**
- Thicker neck → head–neck ratio ↓ → ROM ↓

> **Pitfall**
> - Long skirted head → neck diameter ↑ → head–neck ratio ↓ → ROM ↓

Cup Factors: Range vs Stability Ask AI

Cup

- **Spectrum**: range at one end, **stability** at the other
  - Toward range → ROM ↑, stability ↓
  - Toward stability → stability ↑, ROM ↓ considerably

Cup option	Design	ROM	Stability	Consequence
Liner augmentation	Added piece on liner	↓	↑	—
Captive cup	Augmentation above and below liner	Very small (e.g. `60°`)	Most stable component known	High **failure rate**; young patient → **loosening** after a year

- Moving beyond head and neck → **liner offset**
- **Liner offset**
  - Normal: head **center of rotation** moves with **center of rotation of the cup**
  - Liner offset → head center pushed slightly outward
  - → head travels greater distance
  - Reasonable idea, with its own issues

Stability Ask AI

Head Size and Jumping Distance Ask AI

Stability

- Stability → depends on **head size**
- Head size → **jumping distance**
  - Ball exits completely → half its diameter must pass over rim
  - **Jumping distance** = head **radius**
- Larger head → greater distance to travel → more stable
- Video: normal range → continued motion → dislocation = jumping distance
- **3–4 mm** of jumping distance → major role in stability

Soft Tissue Repair Ask AI

- **Posterior capsule** repair after **posterior approach** → matters for jumping distance
  - Head reaches jumping distance + capsule closed → stays
  - Capsule not closed → dislocates

Component Alignment Ask AI

- Cup cannot be placed anyhow
- Patient's native cup position → range patient is accustomed to
- Fear of dislocation → cup over-anteverted, further with **broach** → ROM altered
- Departure from native ROM → disturbed
- Goal → reproduce patient's **native anatomy**
- **Primary arc of range** of total hip
  - Cup within patient's native range → 100%
  - Excessive anteversion or retroversion → **cup impingement** → implant lifespan ↓

High Yield

- Larger head → larger jumping distance → more stable
- Posterior capsule repair → retains head at jumping distance
- Cup outside native range → cup impingement → lifespan ↓

Templating Ask AI

Principle Ask AI

- **Templating**: pre-op planning together in the office
- Example: retained plates → **hardware removal** → template, see design, plan
- Whole operation performed in the office
- Best operation = one performed in the office
  - Choose and request everything needed, including templates
- Templating **X-ray** → specific requirements
  - Covered in the course by **Dr. Mahmoud Abdel-Karim**

Leg Length Ask AI

Templating

- Cup center not coinciding with head center → leg lengthened; offset ↑ or ↓
- **Stem 9 mm higher** → leg lengthened 9 mm on reduction into cup
- **Neck cut** calculated relative to **lesser trochanter**
- Neck cut 15 mm above lesser trochanter → stem seated
  - → stem **center of rotation** higher than cup center of rotation
- Result: normal length restored → lengthening known exactly in millimeters
  - → via the template

Offset and Implant Design Ask AI

Templating

- Case: patient with **arthritic hip**
- Arthroplasty performed neatly → check **AP** view
- Finding → **offset increased**
- Avoidable? Not if no templating and simply using prosthesis supplied to hospital
- Template: offset slightly ↑ → lengthening; measured offset segment slightly smaller
- Matching **centers of rotation** after implantation → offset ↑
- Second case: templating pre-op would have shown the problem
- Template offset vs **actual** prosthesis → actual jumps further → offset ↑ considerably
- → that **stem** unsuitable → different design needed
- Templating → not only for **size**
- Templating → choose **design**: this company's set suits the patient or not

Must-Know

- Templating → size, leg length, offset and implant design
- Neck cut level relative to lesser trochanter → predicts lengthening in mm

Closing Principles Ask AI

- Most THA problems → **scientific basis**
- Complaining patient → do not blame patient or nature → investigate cause
- Every complaining patient → find out why

Tribology

(Friction, Lubrication and Wear in Hip Arthroplasty Bearings)

Prof Dr Mohamed Abo El Soud

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Tribology: Definition and Scope Ask AI

- Lecture first given in a **Basic Science** course, about 5 years ago
- Slide picture: time needed just to recall what **tribology** covers
- **Tribology** = science of friction
  - **Tribos** = friction (rubbing)
  - Not "three"; not the science of camera tripods
  - English began theorising → **David Tabor** first to speak about it
- Three components
  - **Friction**
  - **Lubrication**
  - **Wear**

The Tribosystem Ask AI

- **Tribosystem** = system discussed; all theories applied to it
  - Body + another body
  - **Lubricant** between them = interfacial material → opposes friction
  - Environment: temperature + humidity
  - Pressure on system = **load**
- Outputs of the system
  - **Wear particles**
  - Surface shows the effect of wear

Element	Tribosystem role
Two bodies	Rubbing surfaces
Lubricant	Interfacial material, opposes friction
Temperature, humidity	Environment
Load	Pressure applied to system
Wear particles, worn surface	Outputs

Friction Ask AI

- **Friction** = rubbing of two bodies
  - Resistance to motion between two bodies moving on each other
- Friction equation: $F = \mu \times W$
  - F = **friction force**
  - μ = **coefficient of friction**
  - W = **load** applied
- Equation has no term for **surface area** or contact area
- **Apparent contact area** → the two surfaces seen touching
- **Real contact area** → surfaces never touch completely, however polished

  - Cause → surface is rough
- Polished surface stays rough → high spot + low spot
- **Surface roughness** of hip materials

Material	Surface roughness
Polyethylene	1 micrometre
Polished metals	0.05 micrometre (most cases)

- Surface roughness → leads into lubrication

High Yield
- F = μ × W → no contact-area term
- Apparent contact area ≠ real contact area
- Polyethylene roughness 1 micrometre; polished metal 0.05 micrometre

Lubrication Regimes and the Lambda Ratio Ask AI

Regimes Ask AI

Lubrication

- **Asperities** = surface roughness
- Asperities completely touching → **boundary lubrication**
  - Lubricant = substance added to smooth motion
  - Present only as a very thin layer
  - Layer still allows all asperities to touch
- More lubricant → asperities moved apart
- Asperities separated → **fluid film** forms between the two surfaces
- Halfway between → **mixed lubrication**
- Surfaces completely separated → fluid film; surfaces touching → boundary
  - Expressed by an equation → lambda ratio

Lambda Ratio Ask AI

Lambda

- **Lambda ratio (λ)** = minimum film thickness (h-minimum) / surface roughness of the two surfaces
  - h-minimum = minimum distance between the two surfaces
- Bigger fluid film → h-minimum exceeds surface roughness → λ rises → better lubrication
- λ versus **coefficient of friction**
  - λ rising → boundary → mixed → fluid film lubrication
  - Coefficient of friction falls → lubrication improving
  - Shortly after λ passes 3 → coefficient of friction high again
- Reason for rise at λ greater than 3
  - Film that large needs a very **viscous** (sticky) lubricant
  - Lubricant itself resists motion
- λ greater than 3 → lubrication decreases again → coefficient of friction rises again
- Review of the two extremes
  - **Boundary lubrication** → surfaces touching, total asperity contact
    - Lubrication at the level of molecules

    - Single lubricant molecule **adsorbed** on the two surfaces → smooths motion
  - **Fluid film** → no asperity contact at all, complete separation of surfaces

Regime	Asperity contact	Lubricant	Coefficient of friction
Boundary	Total	Very thin layer; single molecule adsorbed on surfaces	Highest
Mixed	Partial (halfway)	Intermediate	Falling
Fluid film	None; complete separation	Film thicker than roughness	Lowest
λ greater than 3	None	Very viscous, resists motion	High again

Exam Pearl

- λ = h-minimum / surface roughness
- Rising λ → boundary → mixed → fluid film
- λ just past 3 → friction rises again (viscous lubricant resists motion)

Fluid Film Lubrication: Hydrostatic versus Hydrodynamic Ask AI

- **Fluid film lubrication** → two types
- **Hydrostatic lubrication** → factories
  - Fluid film pumped between two surfaces by a piston → keeps them apart
  - Not present in the body
- **Hydrodynamic lubrication** → present in the body
  - Most important part of the lecture

Type	Mechanism	Setting
Hydrostatic	Film pumped by piston, surfaces kept apart	Factories; not in body
Hydrodynamic	Film generated by motion	The body

Water-Skier Model Ask AI

Hydrodyn.

- Man gliding on water on skis, boat in front (unseen)
- Factors making him glide better
  - **Speed** of boat → higher speed, better gliding
  - **Load** → tall, fat man sinks into water
- Lighter man → better gliding
- **Angle** between ski and water
  - Larger angle → acts as resistance
  - Smaller angle → faster glide
  - Zero or negative angle → sinks into water
  - Needed: angle present, as small as possible

- Rough, uneven ski undersurface → acts as a brake → **surface roughness** matters
- Four factors: speed, load (applied by patient), surface roughness, angle

Factor	Better gliding	Worse gliding
Speed	High	Low
Load	Light	Heavy → sinks
Angle	Present, as small as possible	Large → resists; zero or negative → sinks
Surface roughness	Smooth undersurface	Rough, uneven → brakes

Hydrodynamic Lubrication in the Hip Ask AI

Wettability and Surface Finish Ask AI

Hip

- **Ceramic head** "loves water" → water spreads over its surface
  - High **wettability**
  - Surface finish completely smooth
  - Both help fluid film formation
- Next factor → angle between the two surfaces

Radial Clearance Ask AI

Hip

- Large angle between surfaces → polar contact, no fluid film
  - Fluid escapes out entirely
  - Two surfaces touch each other
- **Radial clearance** = radius of liner − radius of head
  - Must be present
  - Must be small
  - Result → polar contact with high conformity

Entrainment Ask AI

Hip

- Polar contact + high conformity → fluid film via **entrainment**
- At rest → polar contact
- At rest → fluid film only slight, at the sides
- Motion begins with boundary lubrication
- Head moves → head itself spreads fluid film over the other surface = **entrainment**
- Achieved by small radial clearance

Must-Know

- Radial clearance: present but small
- Large clearance angle → polar contact, fluid escapes, no film
- Entrainment = moving head spreads fluid film over opposing surface
- Sequence: rest (polar contact) → boundary lubrication → fluid film

Head Size and Velocity Ask AI

Hip

- Does size matter? Bigger head →
- → more lubrication
- **Velocity** = Δd / Δt
  - Δd = distance moved
  - Thigh flexed 90 degrees: 28 head versus 36 head
  - Δt constant
- Distance moved differs → 36 head moves more
- 36 head → greater distance → higher velocity → better lubrication
- Remaining factors
  - Patient's weight
  - **Synovial fluid** present

Skier factor	Hip equivalent
Speed	Head velocity → larger head (36 versus 28) faster
Load	Patient's weight
Surface roughness	Surface finish, wettability (ceramic head)
Angle	Radial clearance → small

Wear Mechanisms Ask AI

- **Wear** = third component of tribology
- Many types; three best known
  - Abrasive wear
  - Adhesive wear
  - Fatigue wear

Abrasive Wear Ask AI

Wear

- **Abrasive wear** = ploughing
  - Analogy: kid with a nail scratching cars as he walks
- Occurs in metal-on-poly
  - Steel, however polished, has asperities
  - Asperities scratch the **polyethylene** with walking
- Also the type produced by **third body wear**
  - Cement piece embedded in polyethylene → scratches femoral head
  - Or scratches femoral component of the knee

Adhesive Wear Ask AI

Wear

- **Adhesive wear** → bonds form between the two surfaces
  - Intermolecular bonds resist motion → produce friction
  - Responsible for **spot welding**, e.g. head–trunnion
- Spot welding at head–trunnion → as in **trunnionosis**

Fatigue Wear Ask AI

Wear

- **Fatigue wear** → repeated stress on polyethylene surface
  - Microscopic cracks form on the surface
  - Cracks coalesce → large piece
  - Piece eventually breaks off

Type	Mechanism	Example
Abrasive	Ploughing; asperities scratch softer surface	Metal-on-poly; third body wear (cement in polyethylene)
Adhesive	Intermolecular bonds between surfaces → spot welding	Head–trunnion, trunnionosis
Fatigue	Microscopic cracks coalesce → large piece breaks off	Polyethylene surface

Linear versus Volumetric Wear and Head Size Ask AI

- **Linear wear** versus **volumetric wear;** small head versus large head
- **Pressure** = load (patient's weight) / surface area
  - Small head → smaller surface area → higher pressure
  - Linear wear → small head sinks into polyethylene faster
- **Volume** involves πr^2 → radius to the power of 2
  - Large head → higher volume of particles generated
  - Large head → more wear particles
- Trade-off of the large head
- Stability + lubrication on one side
- Linear or volumetric wear on the other

Feature	Small head	Large head
Surface area	Smaller	Larger
Pressure	Higher	Lower
Linear wear	Faster penetration into polyethylene	Slower
Volumetric wear	Lower	Higher → more wear particles
Stability, lubrication	Less	Better

High Yield

- Pressure = load / surface area
- Small head → high pressure → more linear wear
- Large head → πr^2 → more volumetric wear, more particles
- Large head trade-off: stability and lubrication versus wear

Wear Rate Ask AI

- **Wear rate** → last point
- Joint implanted → wearing in occurs
- Asperities begin to bed onto each other
  - Analogy: running in a new car → parts still bedding in
  - Continues until asperities worn down and flattened
- Beginning → very high wear rate = **wearing in** (running in)
- Then → steady
- Finally → approaches end of life of the prosthesis
- Wear rate rises a second time → **catastrophic failure** of the implant

Phase	Wear rate	Event
Wearing in (running in)	Very high	Asperities bed in, worn down, flattened
Steady state	Steady	—
End of prosthesis life	Rises again	Catastrophic failure of implant

Biomaterials in hip arthroplasty

(Material Properties, Metals, Ceramics and Polyethylene)

Prof Dr Mahmoud Fahmy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Biomaterials: Definition and Classes Ask AI

- **Biomaterial** → any substance or combination, synthetic or natural
  - Used for any period of time
  - As a whole or as part of a system
  - Treats, augments or replaces any tissue, organ or body function
- Materials in **orthopaedic implants**
  - **Metallic alloys** → stainless steel, titanium, cobalt-chromium
  - **Non-metallic materials** → ceramics, polymers (polyethylene)

Stress–Strain Curve Ask AI

- Basis of material behaviour → **stress–strain curve**
  - Relationship between stress and strain
- Applied stress → **strain** → change in shape and dimensions
- Two key zones
  - Elastic zone
  - Plastic zone of deformation

Elastic and Plastic Zones Ask AI

- **Elastic zone** → load removed → reversible recovery → original shape
- **Plastic deformation** → irreversible recovery → no return to original shape

Young's Modulus Ask AI

- Elastic zone slope
  - Steeper slope → **stiffer** material
  - Flatter slope → less stiff, more flexible material
  - Slope = **Young's modulus of elasticity**
    - Specific to each material
- **Young's modulus** ranking, highest → lowest
  - Ceramic → metal alloys → cortical bone → polyethylene, bone cement

High Yield

- Young's modulus = slope of elastic zone
- Steeper slope → stiffer material
- Ceramic → metal alloys → cortical bone → polyethylene, bone cement

Material Behaviour Terms Ask AI

Brittle and Ductile Ask AI

- **Plastic zone** → internal breakdown without gross fracture
  - **Gross fracture** → at failure point
- Very little plastic deformation → **brittle**
  - Example → glass; in orthopaedics → ceramic

- Large plastic deformation → **ductile**
  - Example → stainless steel

Notch Sensitivity Ask AI

- **Notch sensitivity** → susceptibility to fracture with crack or scratch
  - Glass or ceramic cutter → notch first → tap → easy break

Hardness Ask AI

- **Hardness** → resistance to surface scratching and indentation
  - Unrelated to stress–strain curve
  - Hardest material overall → diamond
  - Hardest in orthopaedics → **ceramic**

Toughness Ask AI

- **Toughness** → area under the stress–strain curve
  - = energy absorbed before failure
- More precise than **ultimate tensile strength**
  - Ultimate tensile strength → highest stress point on curve
  - Same ultimate tensile strength, larger area → ductile vs brittle material

Metal Alloys Ask AI

- **Pure metals** → soft, ductile → alloys developed
- **Alloy** → metal + one or more other elements
  - Stronger, harder
  - Corrosion resistant
  - Better surface finish
- Key alloys → stainless steel, cobalt-chromium, titanium

Stainless Steel Ask AI

- Composition → on slide
- Advantages
  - **Very stiff**, fracture resistant
    - Use → **cemented stems** → less bending stress on cement → cement protected from failure
- Problems
  - **Stress shielding**
  - Susceptible to **corrosion**

Cobalt-Chromium Ask AI

- Composition → on slide
- Better corrosion resistance, lower **wear rate**
  - → bearing surface, e.g. **femoral head**
- **Highest modulus of elasticity** among metals, least ductility
  - → also used in **cemented stems**

Titanium Ask AI

- Composition → on slide
- **Very biocompatible**

  - No tissue irritation
- Corrosion resistant
- **Low modulus of elasticity** → very close to bone
  - Directly above cortical bone in Young's modulus ranking
  - No stress shielding, bone loss or resorption → **uncemented systems**
    - Uncemented stems, uncemented cups
- Problem → **poor wear** → metal debris
  - → not used in **bearing surfaces**

Tantalum Ask AI

- Newest metal
- **Low modulus of elasticity** → closer to bone
- **Extremely porous** → bone ingrowth
- Promising → **acetabular reconstruction,** revision of acetabular cups

Clinical Pearl

- High modulus → stainless steel, cobalt-chromium → cemented stems
- Low modulus → titanium, tantalum → uncemented stems and cups
- Titanium → poor wear → not a bearing surface

Modes of Material Failure Ask AI

Wear and Fatigue Ask AI

- All materials fail by **wear** and fatigue failure
  - **Fatigue failure** → failure below ultimate tensile stress repetitive loading
- Metals → additional mode: **corrosion** (chemical reaction)

Corrosion Types Ask AI

Type	Mechanism	Key point
Galvanic	Two different metals coupled → cathode + anode → electrochemical reaction	Best-known pair → stainless steel + cobalt-chromium
Crevice	Crack or scratch on surface	Stainless steel most susceptible; avoid surface cracks at insertion
Fretting	Micromotion between two materials	Always at head–neck junction

Metal Debris and Its Effects Ask AI

- Wear or corrosion → metal ions, particulates, **metal debris**
- Debris deposited in tissues → **dose-related allergic response**
  - Local or systemic
- Local → **ALVAL** syndrome, pseudotumours, bone and tissue destruction

- Systemic → metal ion elevation, organ toxicity, **carcinogenesis**
- Consequence → **metal-on-metal** bearings now much less used

Ceramics Ask AI

- **Ceramics** → solid inorganic compounds, metallic + non-metallic elements
  - Bound by ionic and covalent bonds

Advanced Ceramics Ask AI

- Current **advanced ceramics** → alumina, zirconia, alumina matrix composite

Ceramic	Wear	Fracture risk
Alumina	Very low wear rate	Brittle, susceptible to fracture
Zirconia	Higher wear rate	Less risk of fracture
Alumina matrix composite	No (minimal) wear	Very low fracture risk

Generations Ask AI

- Generations → 1974 to 2003 → current **BIOLOX delta** = alumina matrix composite
  - Aim → lower fracture risk via smaller **grain size**

Material Characteristics Ask AI

Ceramic

- **Stiff** → first in Young's modulus ranking
  - No deformation under high load
- **Brittle** → very little plastic deformation → higher fracture risk
- **Hard** → scratch resistant → resists **third-body wear**
- Strong in compression, poor in bending
  - → used as **bearing surface**
  - → not used in stems → would fracture
- **Biologically inert**
  - Particles → no allergic, immunologic or inflammatory response
  - No tissue response → **bioinert**
- **Hydrophilic** → wettability
  - Uniform synovial fluid distribution → lower **coefficient of friction**

Advantages Ask AI

Ceramic

- **Very low wear rate**
  - → less osteolysis, especially less **aseptic loosening**
- Very small, bioinert particles → no tissue response
- Longer **implant longevity**
  - → still recommended bearing surface in **young adults**

Disadvantages Ask AI

Ceramic

- **Squeaking** → noisy sounds, up to 30% of patients
  - Exact cause unknown; suggested causes
    1. Insufficient lubrication
    2. Neck impingement on metallic rim

        - Localized stripe wear
- **Fracture** → very rare in new generations
    - Increased by poor surgical technique
        - Cup **retroversion** → markedly accelerated fracture
- **Stripe wear** → see figure
    - Head–liner separation on loading
    - Taper or neck impingement on acetabulum
- **Not forgiving**
    - Requires learning curve
    - Requires proper surgical technique and component insertion
- **Very expensive**

Revision Drawbacks Ask AI

Ceramic

- Ceramic liner removal difficult → specific **suction instrument**
- Head removal → **taper damage**
- **"Once a ceramic, always a ceramic"**
    - Ceramic particles remain in tissues, not easily removed
    - → revision only to ceramic, not to other bearing surfaces

Pitfall

- Cup retroversion → markedly accelerated ceramic fracture
- Ceramic not forgiving → proper technique and component insertion
- Once a ceramic, always a ceramic

Polyethylene: Structure and Manufacture Ask AI

- **Polyethylene** → polymer of ethylene molecule
- Types depend on → molecular weight, cross-linking

Molecular Weight and Structure Ask AI

- **Molecular weight** → number of monomers (ethylene molecules) per polymer molecule
    - Low, medium, high, ultra-high molecular weight
    - **UHMWPE** → more than `200,000` monomers per molecule
- Structure
    - Parallel lines, no connections between monomers → **linear homopolymer**
- **Cross-linking** → connections between monomers
    - Resists **intermolecular mobility**
    - → more resistant to wear and deformation

Manufacture of UHMWPE Ask AI

- **Ethylene** (C_2H_4) → polymerization → polyethylene
    - = chain of carbon atoms + bonds between them
- Processing → one of two methods

Method	Process	Wear
Ram extrusion	Rods → cut and machined into cup shape	—
Direct compression moulding	Mould in exact cup shape → cup produced directly	Better wear characteristics

Sterilization and Cross-Linking of Polyethylene Ask AI

Sterilization Methods Ask AI

- **Sterilization** → key step after processing
- Ethylene oxide or gas plasma → no cross-linking → **inferior wear** properties
- **Irradiation** → dose approx. 2–4 Mrad

Irradiation and Oxygen Ask AI

- Irradiation outcome → depends on **oxygen availability**

Environment	Result	Effect
Vacuum or inert gas (no oxygen)	**Cross-linking**	Chains connected to each other
Oxygen present	Chain scission + oxidation → **oxidative degradation**	All properties spoiled; ↑ wear, ↑ fracture risk

Free-Radical Mechanism Ask AI

- Irradiation → breaks **covalent bond** between carbon atoms
  - → unpaired electron → **free radical**
- Free radical → strong **affinity to oxygen**
  - Oxygen present → binds oxygen → polyethylene degradation
  - Oxygen absent → **cross-linking** between carbon atoms → resists intermolecular mobility

Higher-Dose Cross-Linking Ask AI

- More cross-linking
  - → higher radiation dose, up to 8–10 Mrad
- Benefits → resists intermolecular mobility
  - More resistant to deformation, wear, **osteolysis**
  - Improved implant longevity
- Cost → worse mechanical properties
  - Lower **fatigue** and fracture resistance

Free Radical Removal Ask AI

- Oxygen re-entry after sterilization still possible
  - **Shelf storage**
  - **In vivo** environment
- Free radical removal → two options

	Remelting	Annealing
Process	Plastic melted again	—
Free radicals	All removed	Not all removed
Crystalline part	Affected	Not affected
Fatigue cracking	Prone	Not prone

- **Crystalline part** → responsible for mechanical strength of polyethylene
- Alternative strategies
  - Irradiation + **antioxidant**, e.g. vitamin E
  - Sequential irradiation and annealing

Must-Know

- Irradiation without oxygen → cross-linking
- Irradiation with oxygen → oxidative degradation
- Oxygen re-entry → shelf storage, in vivo
- Remelting → all free radicals removed, prone to fatigue cracking
- Annealing → crystalline part spared, free radicals remain

HXLPE vs UHMWPE Ask AI

	UHMWPE	HXLPE
Wear	Reference	↓ approx. 55–90%, metal or ceramic head
Larger **head diameter**	↑ volumetric wear	No increase in volumetric wear

Polyethylene: Advantages and Disadvantages Ask AI

Advantages Ask AI

PE

- **Biocompatible** → not toxic or carcinogenic like metal ions
- No **squeaking**, unlike ceramic
- **High wear resistance** → especially HXLPE
- Less **fatigue failure**
- **Easy revision**
- **Familiar** → no specific learning curve
  - Same learning curve for HXLPE and UHMWPE
- **Adaptability** → liner options
  - Extended liner, enhanced posterior wall, offset liner, constrained liner
- **Forgiving** → two key situations
  - **Impingement** → stem neck vs polyethylene edge → wear
    - Catastrophic in metal and ceramic; forgiving in polyethylene
  - Increased **cup abduction** → edge loading, head vs polyethylene → wear
    - Catastrophic in metal and ceramic

    - Forgiving in polyethylene
- **Lower cost**

Disadvantages Ask AI

PE

- **Wear** → osteolysis → aseptic loosening
    - Vicious circle, nothing to stop it

Bearing Surface Comparison Ask AI

Bearing	Wear
Cobalt-chromium on polyethylene	Approx. 200 µm per year
Ceramic head on polyethylene	Half
On cross-linked polyethylene	Approx. 10%
Metal-on-metal, ceramic-on-ceramic	Lower still

- Metal-on-metal, ceramic-on-ceramic → better wear than polyethylene (agreed)
- Polyethylene still favoured for
    - Intra-operative liner options → **adaptability**
    - Forgiveness
    - Easy revision
    - Cost
    - Familiarity
    - Avoids other bearings' specific disadvantages
        - Metal-on-metal → **ALVAL** syndrome → now rarely used
        - Ceramic → squeaking, fracture, other problems
- **Metal-on-polyethylene** → gold standard, preferred by most surgeons worldwide
    - National joint registry, America, 2003–2013 → most used
    - Swedish Hip Arthroplasty Register, 2005–2013 → lion's share

Exam Pearl

- Metal-on-polyethylene → gold standard bearing
- Cobalt-chromium on polyethylene → approx. 200 µm per year
- Cross-linked polyethylene → wear approx. 10%

Summary Ask AI

Metals Ask AI

- **Alloys** → stronger, harder than pure metals

Alloy	Modulus of elasticity	Key point	Use
Stainless steel	Very high	Less bending stress on cement → cement protected	Cemented stems
Cobalt-chromium	Very high	Very good wear properties	Cemented stems; femoral head
Titanium	Low, closer to bone	No stress shielding or bone resorption	Uncemented stems and cups
Tantalum	Low, closer to bone	No stress shielding or bone resorption	Uncemented stems and cups; acetabular reconstruction (very promising)

- Metal ion effects, local and systemic → **metal-on-metal** use much decreased

Ceramics Ask AI

- Superior in **wear** and osteolysis
- **Fracture** → much decreased in new generations, still present
  - → needs precise surgical technique

Polyethylene Ask AI

- **UHMWPE** → more than 200,000 monomers per molecule
- Irradiation in oxygen-free environment → **cross-linking**
  - Higher dose → **highly cross-linked polyethylene** (HXLPE)
- Up to 90% decrease in **wear**
- Allows **large heads** → greater stability
- Still biocompatible, familiar, adaptable, forgiving, low cost

How to choose your Bearing

(Wear, Lubrication and Bearing Surface Selection)

Prof Dr Khaled Fawzy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 40) “activation of the osteoblasts” → “osteoclast activation”

Queried - kept as spoken, but looks wrong

Q1 (p. 40) as spoken; force × coefficient of friction gives frictional force, not wear (wear depends on load and sliding distance)

Case: The Painful Hip After Arthroplasty Ask AI

Initial Presentation Ask AI

Case

- Patient with **arthritis** → previous technique, options, balance between them
- Relatively young, newly presenting arthritis
- Decision → **osteotomy** vs arthroplasty
- Next question → which bearing?
- At this age → **metal-on-polyethylene**
- Fixation → cemented vs uncemented; cemented proposed, rationale questioned
- Key question → bone **osteoporotic**? → yes
- Treated with this implant

Symptomatic Years Later Ask AI

Case

- Symptomatic 3 years later, at age 75 → **prosthesis type**?
- Second scenario → symptomatic 10 years after replacement → prosthesis type?
- Question → which **bearing** was used?
- Cause of pain → suggested: **loosening**
- Beyond loosening → other obvious findings?
- Suggested cause of loosening → **cement**
- Component **not cemented** → still osteolysis
  - Osteolysis → several zones; then inspect other component
- Question → osteolysis from cement, or another cause?
- Never-cemented component + **osteolysis** → cause?
- **Instability** proposed → questioned as cause of this change

Eccentric Head Ask AI

Case

- **Head not centred** in cup
- Position unchanged → good position at implantation 10 years earlier
- Normal → head sits **concentrically** within cup
- Eccentric head → not shortening → process inside joint
- Cause → **polyethylene wear**

Clinical Pearl

- Eccentric head in cup on X-ray → polyethylene wear
- Osteolysis around uncemented component → not cement-related

Polyethylene Wear and Osteolysis Ask AI

Particle Disease Ask AI

PE wear

- Polyethylene → **wear particles** → osteolysis
- Mechanical ball-and-socket → rusts
- Prosthetic joint → no rust → **polyethylene wear** instead
- **Polyethylene wear particles** → very small

  - → engulfed by **macrophages**
  - → **cytokines** released
  - → **osteoclast** activation → osteolysis *[C1 · transcript: "activation of the osteoblasts"]*

Wear Generation Ask AI

PE wear

- Head smooth to eye; electron microscope → **asperities**
  - Asperities scratch polyethylene → shed **polyethylene particles**
- After wear → cup cavity **larger**
- Case X-ray → worn polyethylene → particles → **osteolysis**
- Weakest link → **polyethylene** → reason for alternative bearings

High Yield

- Wear particles → macrophages → cytokines → osteoclasts → osteolysis
- Head asperities → scratch polyethylene → particles
- Weakest link of the bearing → polyethylene

Friction and Wear Mechanisms Ask AI

Coefficient of Friction Ask AI

- Wear → force × **coefficient of friction** *[Q1 · as spoken; force × coefficient of friction gives frictional force, not wear (wear depends on load and sliding distance)]*
- **Metal-on-polyethylene** → very low coefficient of friction
  - **Charnley** → low friction arthroplasty
  - Precisely → **low frictional torque arthroplasty**

Mechanisms of Wear Ask AI

- Four mechanisms:
  - **Abrasion**
  - **Adhesion**
  - **Third-body wear**
  - Polyethylene **fatigue**
- Microscopic asperities rub → mutual abrasion → **abrasive wear**
- **Ceramic** → **surface pits** → rub on bearing surfaces → wear

Joint Load Ask AI

- Hip stresses → up to 5× **body weight**
  - 80 kg patient, walking up/down hill → ~400 kg across hip

Lubrication Ask AI

Types of Lubrication Ask AI

- Requirements → low coefficient of friction, low force, good **lubrication**
- Three types:
  - **Fluid film lubrication** → occurs with movement
  - **Boundary lubrication** → **lubricin** molecules in synovial fluid

  - **Mixed lubrication**

Head Size and Lubrication Ask AI

Head size	Dominant lubrication
Large head	Fluid film lubrication
Small head (e.g. 28 mm)	Boundary lubrication

- Boundary → **lubricin** lines surface asperities; contact persists
- Fluid film → **fluid layer** separates the two surfaces
  - Analogy → car oil between two metal surfaces
- Car engine → additive applied to metal → **boundary lubrication** also needed
- Oil build-up between surfaces → **fluid film** → minimal friction
- Basis → mechanics and biomechanics

Improving Ultra-High Molecular Weight Polyethylene Ask AI

Oxidation Ask AI

UHMWPE

- **UHMWPE** particles → osteolysis → need for improvement
- Cut section, cup left long on shelf → **oxidation**
- Oxidation → **accelerated wear**
- Unrecognised when low friction arthroplasty evolved → discovered later
- Oxygen + **irradiation** → oxidation
- Irreversible → oxygen not removable afterwards → act before oxidation

Processing Ask AI

UHMWPE

- **Sterilisation** → required
- **Annealing** → part of processing sequence
- Sterilise in **inert gas** or vacuum
- **Cross-linking** → high-dose radiation
  - After radiation → **anneal** or melt
- Most up-to-date → **vitamin E**
- Results:
  - Improved **wear rates**
  - Long-term **longevity**
  - Allows **large head**

Conventional vs Cross-Linked Ask AI

UHMWPE

Feature	Conventional PE	Cross-linked PE
Large head → volumetric wear	Increased (possibly ↑ linear wear)	Not increased
Fatigue strength, fracture toughness	—	Decreased
Risk–benefit	—	Benefit outweighs risk

Exam Pearl

- Oxygen + irradiation → oxidation → accelerated wear
- Sterilise in inert gas or vacuum
- High-dose radiation → anneal or melt → cross-linked PE
- Cross-linked PE → ↓ fatigue strength and fracture toughness
- Latest → vitamin E

Resurfacing Ask AI

- Operation → **resurfacing**
- Bearing → **metal-on-metal**
- Questions → osteolysis with MoM? common? what **fractured**?
- Fracture predisposition → small **neck** size, not stem size
- **Females** → contraindicated; no longer used
  - Reason → **smaller necks** and heads → fracture; not only childbearing
- **Notching the neck** during insertion → fracture
- **Technique-sensitive** → training required

Metal-on-Metal Bearings Ask AI

Advantages and Toxicity Ask AI

MoM

- Advantage → femoral **bone stock** likely preserved
- Problem identified later:
  - **Toxicity** → metal ions pass into blood

Run-In Phase Ask AI

MoM

- Engine/car analogy → joint **runs in**
- First year → irregularities wear each other off → **smoother surfaces**
- Run-in → ↑ blood **cobalt** and chromium; then decrease
- Graph → very high **cobalt and chromium levels** at implantation → decline after first year
- Levels **rising** on analysis → something wrong

Contraindications Ask AI

MoM

- **Renal impairment** → cannot be used; harmful
- **Women of childbearing age** → teratogenic

The Painful Metal-on-Metal Hip Ask AI

MoM

- Painful, no **fracture** → design fault?
- Option → **aspiration**
- Painful, nothing wrong, **no gap**
- Imaging → **MARS MRI** or other imaging
  - MARS → Metal Artifact Reduction Sequence
- Finding → large collection → **ALVAL**
  - ALVAL → Aseptic Lymphocyte-dominated Vasculitis-Associated Lesion
- **Metal debris** clearly visible
- Metal ions + metal hypersensitivity → tissue destruction → **pseudotumour**
- Concerns:
  - Metal ions in **blood**
  - **Renal impairment**
  - Possible **carcinogenic** effects
  - **Pseudotumour**
- Pseudotumour → majority of MoM implants **withdrawn from market**

Must-Know

- Resurfacing fracture → small neck, female sex, neck notching
- MoM contraindications → renal impairment, women of childbearing age
- Rising cobalt/chromium after first year → something wrong
- Painful MoM, normal X-ray → MARS MRI → ALVAL / pseudotumour
- Pseudotumour → most MoM withdrawn from market

Ceramic-on-Ceramic: Squeaking and Fracture Ask AI

Squeaking Ask AI

CoC

- **Squeaking** → fracture of something?
- Prosthesis → **ceramic-on-ceramic**
- CoC → well known for **squeaking**
  - Very noisy, distressing → especially going **upstairs**
- Another CoC → **fracture**
- Evolving problems:
  - **Squeaking**
  - **Fractures**

Identifying the Bearing on Radiographs Ask AI

Radiograph	Bearing surface
Left side	Metal-on-polyethylene
Other side	Ceramic-on-polyethylene
This side	Ceramic-on-ceramic

- Advanced countries → **implant records** document the prosthesis

- Cup appearance → probably **polyethylene liner**, not ceramic; head visible
- **Half ball, broken** → not metal (very hard, does not break in half) → ceramic

Ceramic Fracture Ask AI

Ceramic

- Question → why does ceramic break? (glass analogy)
- **Micro-cracks** → first, second, third generations → propagate
  - → **fracture**
- **Crack propagation** + trauma → fracture
- If using ceramic → fourth generation **BIOLOX delta**
  - **Pinkish** colour
  - Very low **fracture risk**
- Diagnosis by **X-ray**?
- Often not visible on **X-ray**
- Diagnosis:
  - **Pain** complaint
  - **Squeaking**
  - **CT**
  - Possibly **synovial fluid** analysis
- Revision timing → **as soon as possible**

Exam Pearl

- Fourth generation BIOLOX delta → pinkish → very low fracture risk
- Ceramic fracture → often invisible on X-ray → pain, squeaking, CT
- Fractured ceramic → revise as soon as possible

Revision of a Fractured Ceramic Ask AI

Choice of Revision Bearing Ask AI

Revision

- Revise to → **ceramic**
- Revision to **polyethylene-on-metal:**
  - Broken ceramic debris → **third body** → wear
  - Debris scratches **metal head** → rougher → accelerated wear
- Real option → **ceramic-on-ceramic**

Taper and Stem Ask AI

Revision

Finding	Action
Taper intact, stem stable	Metal sleeve
Taper damaged	Remove and revise stem → damaged taper damages ceramic → fracture

Pitfall

- Fractured ceramic → metal head on polyethylene → third-body wear
- Ceramic debris scratches metal head → accelerated wear
- Damaged taper retained → ceramic damage → fracture

Ceramic-on-Ceramic: Advantages and Concerns Ask AI

Advantages Ask AI

CoC

- **Hard-on-hard** bearing
- Low **surface roughness**
- Low **coefficient of friction**
- Small, inert **wear particles**
- **Least volumetric wear**
- **Wettable, hydrophilic** → encourages fluid film lubrication
- Low **scratch profile**

Scratch Behaviour and Wettability Ask AI

Property	Metal	Ceramic
Accidental scratch	Asperities → more polyethylene wear	V-shaped pit → less wear
Water contact angle	86°	45°

- Smaller **contact angle** → more wettable surface

Concerns Ask AI

CoC

- **Fracture**
- **Squeaking**
- No **lipped inserts**
- **Limited modularity** → head sizes, neck lengths
- Needs **accurate positioning**
- **Expensive**
- **Stripe wear**

Choosing the Bearing Ask AI

Options and Wear Ask AI

Bearings

- Options:
  - **Metal-on-polyethylene**
  - **Ceramic-on-polyethylene**
  - Metal or ceramic on **highly cross-linked polyethylene**
  - **Metal-on-metal**
  - **Ceramic-on-ceramic**
- Metal-on-polyethylene → **largest volumetric wear**
- Ceramic-on-ceramic, metal-on-metal → **least volumetric wear**

Decision Factors Ask AI

Bearings

- **Sensible choice**, not absolute → based on:
  - Research **evidence**
  - **Benefit–risk ratio**
  - **Price**
  - Patient age, activity, comorbidities, **life expectancy**

Choice by Age Ask AI

Bearings

Patient group	Preferred bearing	Alternative
Young, active, below 60	Ceramic-on-ceramic, large head (32 or 36 mm)	Ceramic-on-cross-linked PE
60–65, active	Ceramic-on-ceramic	Ceramic-on-cross-linked PE
65–75	Ceramic-on-cross-linked PE	—
Above 75	Cobalt-chrome head on cross-linked PE	—

Resurfacing Indications Ask AI

Resurfacing

- **Limited indications:**
  - First → **males**
  - **Large head size** → 56–64 mm; small sizes not used
  - **McMinn hip** → Birmingham hip; other hips withdrawn from market
  - Needs **expertise** → otherwise conventional hip replacement

Caution with Ceramic-on-Ceramic Ask AI

CoC

- Choice → depends on many **factors**
- **Ligamentous laxity** → CoC with care
- ↑ Preoperative **range of motion** → CoC with care
- **Anatomical deformities** → CoC with care

Summary: Classification of Bearing Surfaces Ask AI

Category	Bearing	Key point
Hard-on-soft	Metal-on-polyethylene	—
Hard-on-soft	Metal-on-cross-linked polyethylene	—
Hard-on-soft	Ceramic-on-polyethylene	—
Hard-on-hard	Metal-on-metal	Major concerns; withdrawn from market; limited indications
Hard-on-hard	Ceramic-on-ceramic	Better wear rates and longevity vs other bearings

High Yield

- Below 60, active → ceramic-on-ceramic, large head (32 or 36 mm)
- 65–75 → ceramic-on-cross-linked PE
- Above 75 → cobalt-chrome head on cross-linked PE
- Resurfacing → males, 56–64 mm head, Birmingham hip, expert surgeon
- Laxity, ↑ preop ROM, deformity → ceramic-on-ceramic with care

Implant fixation

(Cemented and Cementless Fixation)

Prof Dr Mahmoud Fahmy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 56) “grade C radiolucency 50% to 90%” → “grade C radiolucency 50–99%”

C2 (p. 56) “grade D: absence of the stem” → “absent cement below stem tip”

C3 (p. 58) “extensive coating loads proximal diaphysis” → “extensive coating loads diaphysis”

Implant Fixation: Definition and Types Ask AI

- Topic → **types of implant fixation**
- **Implant fixation** → coupling of prosthetic component to musculoskeletal system
  - Prosthetic + natural elements → act together harmoniously
- Goal → eliminate motion at implant–bone interface
- Two types
  - **Cemented fixation**
  - **Cementless fixation** → biologic fixation
    - Bone growth into or onto porous surfaces

Indications: Cemented vs Cementless Ask AI

Schools of Thought Ask AI

- Many schools on indications
  - Some surgeons → cement in every patient
  - Others → cementless in every patient
- Cup → **porous-coated cementless cup** preferred by most surgeons
  - Reason → reliable long-term results
- Cemented → still acceptable in elderly, **poor bone stock**
  - Both cemented and cementless → acceptable options

Dorr Classification Ask AI

Dorr

- **Dorr classification** → major guide to fixation choice
- Dorr type → largely decides cemented vs cementless stem

Factor	Cementless stem	Cemented stem
Dorr type	A and B	C (stovepipe femur)
Age	Young	Elderly
Activity	High-activity	—
Other	—	Irradiated bone (summary)

- Any patient → either fixation type remains possible

High Yield

- Dorr A and B, young, high-activity → **cementless stem**
- Dorr C (stovepipe femur), elderly → **cemented stem**
- Cup → porous-coated cementless preferred

Bone Cement: Composition and Polymerization Ask AI

Composition Ask AI

PMMA

- Cement → polymer: **polymethyl methacrylate (PMMA)**
- Supplied to theatre → powder + liquid
- **Powder**
  - **Polymer** → two carbon atoms
    - First carbon → two hydrogen atoms
    - Second carbon → methyl group + methacrylate group
  - **Initiator** of polymerization → benzoyl peroxide
  - **Radio-opacifier**
    - Gives cement radio-opacity on X-ray
  - Antibiotics
- **Liquid**
  - **Monomer** → methyl methacrylate
  - **Accelerator** of polymerization → dimethyl-p-toluidine
  - **Stabilizer** (inhibitor) → prevents monomer polymerization inside liquid
  - Additives
- Why not liquid alone
  - Very long setting time intra-operatively
  - Large energy (heat) release
  - **Shrinkage**
- Compensation → **pre-polymerized powder**

Polymerization Ask AI

- Initiator + activator → free radicals → break carbon double bond
- Large energy release → **exothermic reaction**
- In vitro → 80–85 °C
- In vivo → much lower
  - Thin cement layer → heat dissipated via stem + bone blood circulation
  - Cement **imbibes water** → further cooling

Exam Pearl

- Powder initiator → **benzoyl peroxide**
- Liquid accelerator → **dimethyl-p-toluidine**
- Liquid alone → slow setting, high heat, shrinkage
- Polymerization exothermic → in vitro 80–85 °C

Cementing Generations Ask AI

- Successive generations → better canal preparation, pressurization, mixing

Generation	Features
First	Hand mixing; no bone preparation, no plug, no pressurization → blood/void inclusion, ↓ penetration
Second	Plug (cement restrictor); retrograde filling with **cement gun**; pressurization → ↑ mechanical interlock
Third	Improved pressurization with **rubber seal**; **vacuum mixing**
Recent advances	Aim → ↓ mixing time

Mechanical Properties of Cement Ask AI

- Two groups → short-term, long-term properties

Short-Term Properties Ask AI

- **Brittle**
  - No significant plastic deformation before fracture
  - Brittle → **notch-sensitive**
    - Notch → **stress riser** → easy breakage
  - Revision → crack cement first → then remove

Load	Cement strength
Compression	Strong
Tension	Poor
Shear	Moderate

Long-Term Properties Ask AI

- Susceptible to **creep**
- **Stress relaxation**
- **Fatigue failure**

Creep Ask AI

Cement

- **Creep** → ↑ plastic deformation under constant load over time
- In vitro experiment
  - Cement inside **serrated metal cone**
  - One specimen → loaded long period; other → unloaded
  - After set period
    - Unloaded → no serration imprint
    - Loaded → serrations imprinted
  - → cement undergoes creep with time

Fatigue Failure Ask AI

Cement

- Last long-term property
- **Fatigue failure** → failure below ultimate tensile strength under cyclic loading
- Site → points of high stress concentration
- Protection → **stress relaxation**
- Creep + stress relaxation → key clinical role (below)

Cement Function and Stem Design Philosophies Ask AI

Cement as Grout Ask AI

- Function → macro- and micro-interlock + **interdigitation** with trabecular bone
- Cement → **grout, not glue** (mechanical grout)
  - Fills space → forms **mantle** → load transfer to bone
  - More load to bone; likely more even stress distribution
- Function requires preparation of both surfaces
  - Bone surface → cleaned of blood + debris → ↑ penetration → interdigitation
  - Stem surface → one of two design philosophies → different stem behavior

Stem Design Philosophies Ask AI

- Two cemented stem philosophies

Feature	Composite beam (shape-closed)	Loaded taper
Surface	Rough	Polished
Geometry	—	Tapered in two planes
Cement interface	Strong bond with cement	—
Subsidence	Not allowed; subsidence → failure	Designed to subside

- Loaded taper subsidence → controlled by **air-filled (hollow) centralizer**
- Axial loading → subsidence
  - → **radial compressive forces** in cement
  - → transferred to bone as **tensile hoop stresses**

Clinical Significance of Long-Term Properties Ask AI

Taper

- Long-term properties → clinical relevance
  - Axial load → stem subsides → radial compressive forces
  - **Creep** → ↑ interdigitation of cement within medullary canal
  - Radial compression → transmitted as tension (hoop stress) in bone → harmful
- Relief of bone tension
  - Sleep/rest → no load change; stress persists
  - → **stress relaxation**
    - Likely → bone stress dissipates; compressive force holding stem remains

Must-Know

- Cement → **grout, not glue**
- Composite beam → rough, no subsidence; subsidence → failure
- Loaded taper → polished, double-tapered, designed to subside
- Subsidence → radial compression in cement → hoop stress in bone
- Stress relaxation → protects against fatigue failure

Cement Mantle Ask AI

Mantle Stresses Ask AI

- Load transfer → through **cement mantle**
- Mantle → weakened by voids (porosity); located at high-stress site
- Forces
  - Compression
  - Shear + tension at **cement–metal interface**

Avoiding Mantle Failure Ask AI

- **Mantle thickness** > 2 mm → ↓ fracture risk
- **Uniform** mantle
- ↓ **Porosity** → fewer stress points
- Good **pressurization** → ↑ interdigitation
- **Clean, dry bone** bed → ↑ interdigitation
- **Stiff stem** (biomaterials lecture) → ↓ bending stress on mantle
- No **sharp edges** on stem
  - → avoids stress concentration

High Yield

- Mantle > 2 mm, uniform
- ↓ Porosity, good pressurization
- Clean, dry bone bed
- Stiff stem, no sharp edges

Radiographic Grading of Cementing Ask AI

Barrack & Harris Grading Ask AI

Barrack

- **Barrack & Harris grading** → grades A, B, C, D

Grade	Finding
A	Complete filling → "white-out"
B	Radiolucency up to 50%
C	Radiolucency 50–99% (stated as 50–90%) *[C1 · transcript: "grade C radiolucency 50% to 90%"]*
D	100% radiolucency, or absent cement below stem tip *[C2 · transcript: "grade D: absence of the stem"]*

Zones Ask AI

- **Cup** → **DeLee and Charnley zones**
  - Numbered lateral → medial
- **Stem** → **Gruen zones**
  - Lateral → medial: zones 1–7
  - Anterior → posterior: zones 8–14

Cementless Fixation and Bone Ingrowth Ask AI

Ingrowth vs Ongrowth Ask AI

Cementless

- Cementless → **biologic fixation**
  - Bone growth into porous surfaces → secures implant
- Two ways
  - **Bone ingrowth** → into porous surface
  - **Bone ongrowth** → onto rough surface
- Image → ingrowth into porous surface vs ongrowth onto rough surface

Bone Ingrowth Ask AI

- **Porous-coated surface** → cellular/physiological response
  - Resembles healing cascade of cancellous defect
  - → similar to **primary fracture healing**
- Porous coating types
  - **Sintered beads** (most popular)
    - Titanium or cobalt-chrome
    - Sintered to substrate at high temperature
  - **Titanium fiber mesh**
    - Cut titanium wires, bonded (probably diffusion bonding)
  - Likely **plasma-sprayed titanium alloy**
  - Probably **porous tantalum** (trabecular metal; biomaterials lecture) → highly porous

- Factors facilitating ingrowth
  - Implant → likely optimal pore size + porosity
  - Surgeon
    - Viable bone (probable)
    - Good contact; likely minimal gaps
    - ↓ **Micromotion** → via initial rigid fixation
- **Initial rigid fixation** → critical for ingrowth

Initial Rigid Fixation Ask AI

- Two methods → **press-fit, line-to-line**

Feature	Press-fit	Line-to-line
Principle	Minimal motion; rigid fixation via compression	Bone contour = implant size
Acetabular cup	No screws needed	Screws required
Femoral stem	—	Extensively porous coated → initial frictional fit

- Press-fit → achieved by **under-reaming** of bone

> **Clinical Pearl**
> - Press-fit cup → compression alone, no screws
> - Line-to-line cup → screw fixation required
> - Line-to-line stem → extensive porous coating for frictional fit

Bone Ongrowth and Hydroxyapatite Coating Ask AI

Bone Ongrowth Ask AI

- **Bone ongrowth** → onto roughened, non-porous surface
- Microscopic picture → shows rough-surface production
- Production methods
  - **Plasma spray** → probably molten metal particles sprayed onto substrate
  - **Grit blasting**
    - Particles propelled onto surface → roughened
    - **Aluminum oxide** particles → rough surface + **microdivots**
- Ongrowth requirements
  - Rough surface with microdivots
  - Stem design → **double-wedge taper**
  - Initial rigid fixation → must be **press-fit**

Hydroxyapatite Coating Ask AI

HA

- **Hydroxyapatite (HA) coating** → osteoconductive
- Adjuvant → porous-coated + grit-blasted surfaces
- Function → ↓ component **migration**
  - Mechanism → bidirectional closure of bone–prosthesis gap

- Osteoblasts adhere to HA → grow toward bone → bone ingrowth

Femoral Stem Loading and Fixation Signs Ask AI

Stem Loading by Coating Ask AI

- Load transfer → depends on coating: proximal vs distal (extensive)
- **Proximal coating** → load to metaphysis + proximal diaphysis
- **Extensive coating**
  - Load → diaphysis *[C3 · transcript: "extensive coating loads proximal diaphysis"]*
  - **Stress shielding** → proximal femur
  - **Spot welds** (endosteal consolidation) → near stem end

Signs of Well-Fixed Cementless Components Ask AI

- Radiographic signs of well-fixed cementless components

Femoral stem	Acetabular cup
Spot welds	Intact acetabular screws
Proximal stress shielding	No progressive radiolucent lines
No radiolucent lines around stem	No migration on serial radiographs
No subsidence on serial radiographs	—

High Yield

- Extensive coating → proximal stress shielding + distal spot welds
- Spot welds → sign of well-fixed cementless stem
- Serial radiographs → no subsidence (stem), no migration (cup)

Summary Ask AI

- Two types → cemented, cementless
- **Cemented**
  - Cement → polymer; grout, not glue
  - Brittle, **notch-sensitive**
  - Stress relaxation + creep
  - New cementing generations → better function
  - Stem mechanisms → loaded taper, composite beam
  - Indications → elderly, Dorr C, **irradiated bone**
  - Factors improving stem fixation in cement
- **Cementless**
  - Ingrowth (porous surface) + ongrowth (rough surface)
  - Indications
    1. Acetabular cups
    2. Dorr A and B

    - Young, high-activity
- Stem loading → depends on stem coating

Stem designs

(Composite Beam and Taper-Slip Concepts in Femoral Stems)

Prof Dr Ahmed Hazem

Watch Original Lecture

Lecture transcript processed with AI assistance.

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C1 (p. 68) “collum type cuts the whole neck” → “collum type retains whole neck”

Queried - kept as spoken, but looks wrong

Q1 (p. 66) As spoken; contradicts earlier statement that Types 1 and 2 are designed for Type A femur; double wedge suits Dorr A/B.

Lecture Scope Ask AI

- Continuation of Dr. Mahmoud's lecture → focus on stem specifications
- Key topic: choosing a stem design within the available **portfolio**
  - Whole portfolio reviewed → how to choose a design
  - Stem designs reviewed → recognise the design at patient follow-up
  - Order: **cemented** designs → cementless designs

Cemented Stems: Engineering Basis Ask AI

Two Philosophies Ask AI

- **Two philosophies** for cemented stems → from engineering, not chosen by surgeons
  - Rod inside another material (cement) → two modes of interaction
    1. **Composite beam**
    2. **Loaded taper**

Material Properties and Composite Action Ask AI

- **Cement** → strong in compression; compressive forces make it stronger
  - Shear, tension → voids → cement failure
- **Steel** → better in tension; no elastic stretch when pulled
- Same applies to shear
- **Composite action** → principle used in building construction
  - Concrete (cement) → resists compressive forces
  - Steel (iron) rods within → antagonise tension / shear forces
- Result → structure combining both forces
  - Compression antagonised alongside tension / shear
- Same philosophy → applied to cemented stems

Composite Beam Ask AI

- Cement + stem components → bonded together in this region
- Stem and cement bonded firmly → **primary (initial) stability**
  - Requirement: no micromotion at all
- **Micromotion** → cement cracks
  - Same mechanism used deliberately in revision → cement cracked until failure → extracted
  - Any micromotion → harmful
  - Stem must stay fixed, stable in this zone

Loaded Taper Ask AI

Rationale: Secondary Stability Ask AI

- Problem: cement settles during the **first year**
  - Some micromotion → construct settles into final position

  - Engineers → use this micromotion → beyond initial stability → **secondary stability**
- Origin of the **loaded taper concept**
- Stem loading → forces converted into **compression**
  - → construct progressively tighter, stronger
  - → different kind of stability

Barrel Analogy Ask AI

- Source: **barrels** (e.g., wine storage)
  - Good wooden blocks
  - Surrounded by **steel hoops**
- Barrel filled → more pressure on wooden blocks
  - Stainless steel wires (hoops) → compressive force converted into **hoop stress** (radial force)
  - → components tighter, more stable
- Stem equivalent: subsidence → compressive force converted into **radial force** within cement
  - → construct progressively tighter

Design Requirements Ask AI

LT

- **Tapered** → able to subside
- **Polished (smooth)** → otherwise no subsidence
- **Void at the tip** → space beneath tip in cement mantle for subsidence
- Inserted with a **centraliser** → must be able to subside

Controlled Subsidence Ask AI

- Aim: **secondary stability** after one year
- Expected stem subsidence: ~1 mm
  - Stem at insertion vs at one year → subsided inside cement mantle
- ~1 mm subsidence → termed **controlled subsidence**
  - Rate: ~1 mm per year (first year)

Exam Pearl

- Loaded taper requirements → tapered, polished, void at tip, centraliser
- Controlled subsidence → ~1 mm in first year
- Endpoint → secondary stability after one year

Composite Beam vs Loaded Taper Ask AI

Stability and Load Transfer Ask AI

- Composite beam → stable from start; locks and stays
- Loaded taper → locks, slides, subsides → **secondary stability** after one year
- Composite beam → load shared by stem, bone and cement together
  - Stresses delivered distally
- Loaded taper → **load transfer** to surrounding structures
  - Stem converts shear forces → compressive (radial) forces
    - Transmitted top → bottom, approximately along the stem

Design Comparison Ask AI

Feature	Composite beam	Loaded taper
Surface	Rough → cement interdigitates with stem	Polished
Shape	Anatomical	Straight, tapered
Aim	Stick to cement	Controlled subsidence within one year
Collar	Favoured → transmits stresses, reduces proximal stress shielding	Harmful → prevents subsidence
Void	Harmful → allows subsidence → cracks	Must (distal to tip) → centraliser required
Stability	Initial	Secondary (after one year)
Forgiveness	Not forgiving	More forgiving

- **Centraliser** → allows stem to subside inside cement mantle
- Both concepts → work effectively

Pitfall
- No mixing between the two concepts
- Use one concept fully to its end, or the other fully to its end

Cementless Stems: Fixation Principles Ask AI

Biological Fixation Ask AI

- Goal: **biological fixation** (per Dr. Mahmoud)
  - Requirements:
    - **Immediate mechanical stability**
    - **Intimate bone contact** → as in primary bone healing
  - → stem designed to fit bone shape properly
    - Femur classified → stem fitted correctly to its shape

Dorr Femoral Types Ask AI

- Three femoral types → **Dorr classification**
  - Type A, B, C

Dorr type	Metaphysis	Zone of fixation
A	Very good zone of fixation	Metaphysis
B	Partly lost	Metaphysis (metaphyseal–diaphyseal)

Dorr type	Metaphysis	Zone of fixation
C	Entirely lost; good cortical support only	Diaphysis

- Three fixation groups → metaphyseal / metaphyseal–diaphyseal / diaphyseal
- Analogy: primary stability in **revision surgery** when the classification was devised

Khanuja Classification Ask AI

- **Khanuja classification** (2011) → still used today
  - Basis: zone of fixation + stem geometry
  - Method: all market stem geometries collected → where each manufacturer designed its stem to fix
  - Result: 4 main types + 2 additional types

Khanuja type	Zone of fixation
1, 2	Metaphyseal
3 (3A, 3B, 3C)	Metaphyseal + diaphyseal
4	Diaphysis
5 (additional)	Modular
6 (additional)	Anatomical

Types 1 and 2: Metaphyseal Fixation Ask AI

- Types 1 and 2 → **metaphyseal fixation** stems
  - Designed for the **Type A femur**

Type 1: Single Wedge Ask AI

- Broad mediolaterally above; thin below (other plane)
  - → enters within the narrow cortex, fits the metaphysis
- Broad in one plane, thin in the other
- Depends on **correct positioning**

Type 2: Double Wedge Ask AI

- Opposite of Type 1 → broad in both AP and lateral views
- Termed **metaphyseal filling** → fills metaphysis in AP and lateral
- Major problem: whole metaphysis filled → large amount of good bone removed
  - Likely not feasible through a **minimally invasive** approach
- Impossible in the **Type A femur** *[Q1 · As spoken; contradicts earlier statement that Types 1 and 2 are designed for Type A femur; double wedge suits Dorr A/B.]*
- Works well in the **Type C femur** → broad distally

Type 3: Metaphyseal–Diaphyseal Fixation Ask AI

- Category **most surgeons** use; different devices
- All grip at the **metaphyseal–diaphyseal junction**
  - Initially designed for Type B femur → in practice also work well in Type C femur
- Difference from Type 1 → **straight**, continuous taper
  - No wide-above / narrow-below sudden change
  - Continuous taper top → bottom in AP and lateral views
- Three categories → 3A, 3B, 3C

Type 3A Ask AI

- Simplest in operation
- **Conical**, not cylindrical → straight taper top → bottom
- **Three-point fixation** in AP and lateral views
- Most of today's market

Type 3B: Wagner Design Ask AI

- **Splined**, long stems
- Grip mainly in **proximal diaphysis**, not metaphysis
- Long splines (flutes) → **three-point** resistance
- Splines cut into bone like a knife → **rotational stability**
- **Conical** → any position (version) possible
  - → use reportedly favoured in difficult / deformed femora

Type 3C Ask AI

- Used by **most surgeons**
- Grips **metaphyseal–diaphyseal junction** + proximal diaphysis
  - → also works well in Type C femur
- **Rectangular cross-section**; fully coated
- Fixation in AP view:
  - Rectangular shape → **four-point** rotational stability
  - Flat surface → fits lateral position with **three-point** fixation
  - Rectangular stem → grips metaphysis; fixes approximately in metaphysis in **lateral view**

Clinical Pearl

- Type 3 → category most surgeons use
- 3A → most of today's market
- 3B (Wagner) → splines give rotational stability; any version
- 3C → also works well in Type C femur

Type 4: Diaphyseal Fixation Ask AI

- Grip in the **diaphysis** → metaphysis lost
- Very tough prostheses
  - **Cylindrical columns** → diaphyseal fit
  - Broad enough to fill diaphysis → femur exposed to cracks / fractures

  - Enter and grip diaphysis → **diaphyseal scratch fit**
- **Wedge (lock)** inside diaphysis → diaphyseal problems, cracks
  - → most have declined in market use

Types 5 and 6: Modular and Anatomical Stems Ask AI

Type 5: Modular Stems Ask AI

- Used for **diaphyseal fixation**
- Fixed distally + proximally → two different parts
  - **Distal part** → fixed distally with screws or coating
  - **Proximal part** → grips metaphysis, possibly weak in that patient

Type 6: Anatomical Stems Ask AI

- Back on the market now
- Broad laterally + broad posteriorly → fills **metaphysis**
- **Anatomical contour** → fits proximal anteversion
  - → right and left stems

Short Stems Ask AI

- Newer design type
- Length: standard stem ~14 cm → **short stem** ~half
- **Metaphyseal engagement** only
- Very many types on the market
- Classified by amount of **calcar / neck** removed:
  - **Collum** type → retains whole neck *[C1 · transcript: "collum type cuts the whole neck"]*
  - **Partial collum** type
  - **Trochanteric-harming** type
- All fit in the metaphysis

Summary Ask AI

Cemented Stems Ask AI

Summary

High Yield

- Cemented stems → composite beam or loaded taper
- Composite beam → initial stability; rough, collar, anatomical; not forgiving
- Loaded taper → polished, collar incompatible, straight, tapered
- Loaded taper → more forgiving; secondary stability after initial insertion

Cementless Stems Ask AI

Summary

- Grip metaphysis, metaphyseal–diaphyseal junction, or diaphysis → **four types**
  - Most surgeons use Type 3
- Two additional types → **modular, anatomical**

Cup designs

(Cemented and Cementless Acetabular Components)

Prof Dr Fady Kamal

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 77) As spoken; safe zone is posterosuperior quadrant only; anterosuperior screws endanger external iliac vessels.

Overview of Acetabular Cup Designs Ask AI

- Topic → **acetabular cup design**
- Two groups:
  - **Cemented cups**
  - **Cementless cups**

Cemented Cup Ask AI

Cemented

Material Ask AI

- Made of **polyethylene** → two types (materials lecture):
  - **Ultra-high molecular weight polyethylene (UHMWPE)**
  - **Highly cross-linked polyethylene (HXLPE)**
- Very large difference in **longevity** between the two
- Features discussed:
  - **Cement spacers**
  - Cup **profile** → thick vs thin polyethylene
  - **Internal diameter**

Cement Spacers Ask AI

- **PMMA spacers (cement spacers)** → adequate cement mantle all around cup
  - Cement mantle surrounds cup circumferentially
- **Cement mantle:** 2 to 4 mm
- Approximate value quoted: 2.5 mm

Marker Wire Ask AI

- **Marker wire** → placed in desired direction:
  - Posterior
  - Posterosuperior
  - Superior
- Runs around cup → two functions:
  - Shows **position of the cup**
  - Shows probable **wear** on radiographs
- Possibly also estimates cup **thickness**
  - Ellipse of cup measured along cup diameter
- Ensures **thickness of the cement** around cup

Outer Surface Grooves Ask AI

- **Grooves on the outer surface** → macro-interlock between cement and cup

Profile Ask AI

Cemented

- High vs low profile → adjusts **lateralization** of cup

Goal	Choice
Lateralize cup slightly	**High profile**
Medialize cup slightly	**Low profile**

Internal Diameter Ask AI

- Head sizes: 22 up to 40 mm
- Larger heads not usable with **UHMWPE** → wear
- UHMWPE concern → later **osteolysis**
- **HXLPE** → wear reduced by 90%
- 32, 36 and 40 mm heads → **HXLPE** only, not conventional polyethylene

	UHMWPE (conventional)	HXLPE
Wear	Limits larger heads	Reduced by 90%
Late concern	**Osteolysis**	—
32, 36, 40 mm heads	Not usable	Usable

High Yield

- Cement mantle → 2 to 4 mm
- PMMA spacers → circumferential cement mantle
- High profile → lateralize; low profile → medialize
- **HXLPE** → wear reduced by 90%
- 32, 36, 40 mm heads → HXLPE only

Cementless Cup Material Ask AI

- Cementless features discussed:
  - **Cup material**
  - **Cup shape**
  - **Surface coating** of the material
  - **Liner** and modularity (liner inside cup)
  - **Fixation**
- Options → cobalt-chrome / titanium / tantalum
- Most widely used now → **titanium**
- Titanium advantages:
  - Porous → **bone ingrowth**
  - **Modulus of elasticity** → less stress shielding on surrounding bone
  - Easier design and manufacture

  - Easier insertion → **press-fit** technique
- **Tantalum** → probably higher porosity than other materials used
- Tantalum drawback → somewhat **expensive**

Cementless Cup Shape Ask AI

- Older classification (dated 1983 in lecture) → five types:
  - **Cylindrical**
  - **Square**
  - **Conical**
  - **Ellipsoid**
  - **Hemispherical**
- Now → only **hemispherical** cup used

Generations of Cementless Cups Ask AI

- Cup in general use → **third generation**

First Generation Ask AI

- Before **World War II** (before ~1945–1950)
- **Stainless steel**
- Aim → **mechanical fixation** only
  - Biological fixation concept not yet emerged
- Fixation → **screws**

Second Generation Ask AI

- Appeared → **early 1980s**
- Problems:
  - **Difficult insertion**
  - **Vertical cups** → edge loading on polyethylene → ↑ polyethylene wear
  - **Locking mechanism** problems

Third Generation Ask AI

- Used from early **2000s**
- Improvements:
  - Shape → hemispherical / **hemi-ellipsoid**
  - Polyethylene → **gamma-irradiated not in air** → ↑ longevity
  - **Locking mechanism** problems addressed
  - **Periphery** trimmed (possibly to about one third of cup)
    - → no impingement of periphery with neck of the femur
  - Inner surface highly polished → ↓ **backside wear**
    - Second generation: unpolished inner surface → backside wear between cup (shell) and polyethylene
  - Better engagement between **liner** and cup
- Net result:
  - ↓ **Dissociation**
  - ↓ **Wear**

  - ↓ **Failure of the locking mechanism**

Exam Pearl

- Third generation → cup in general use
- First generation → stainless steel, screws, mechanical fixation only
- Second generation → **vertical cups** → edge loading → polyethylene wear
- Third generation → highly polished inner surface → ↓ backside wear
- Third generation → trimmed periphery → no impingement with femoral neck

Hemispherical Cup Geometry Ask AI

- Hemispherical cup → the one all use now → three geometries

Geometry	Description	Note
Single radius	One diameter, periphery → dome	Obsolete
Dual radius	Smaller internal diameter at dome; much larger at periphery	Obsolete; can fracture the acetabulum
Hemi-ellipsoid	Periphery 2 mm wider than dome	Third-generation shape

- Hemi-ellipsoid → **maximizes bone contact** between cup and acetabulum

Pitfall

- **Dual radius** cup → can fracture the acetabulum

Surface Coating Ask AI

- Covered in detail by **Dr. Mahmoud Fahim**
- Outer layer of cup → **highly porous** material → ↑ bone ingrowth
- Two options:
  - **Porous coated**
  - **Surface roughness** → ↑ bone ingrowth; not used now
- ± **Hydroxyapatite coating** → ↑ osteoconduction
- Porous coating method, most common and most used now → **plasma-sprayed titanium particles**
  - Explained in first lecture

Liner and Modularity Ask AI

Monoblock vs Modular Ask AI

Liner

- Cup comes as:
  - **Monoblock** → now obsolete (many problems)
  - **Metal shell** + liner → polyethylene or ceramic
- Monoblock = one single piece → **no backside wear**
  - Or ceramic cast inside cup → no micromotion between cup and liner

- Modular advantage → many **liner types:**
  - Captive
  - Offset
  - Lipped
- Later liner problem → **liner exchange** possible:
  - Infected
  - Loose
  - Polyethylene wear
- Screws through cup → relevant to **methods of fixation**

	Monoblock	Modular
Structure	One single piece	Metal shell + polyethylene or ceramic liner
Backside wear	None	—
Liner types	—	Captive, offset, lipped
Liner exchange	—	Possible (infected, loose, polyethylene wear)
Screws through cup	Cannot be placed	Can be placed
Status	Obsolete	—

Liner Shapes Ask AI

- **Neutral**
- **Lipped** → lip of about 10 or 20 degrees → stability
- **Offset** → possibly combined with lip
- **Captive**
- Captive → **ring** constrains femoral head inside it

Fixation Methods Ask AI

- Two methods → **press-fit** or line-to-line
- Press-fit → **under-ream** by about 1 mm → insert larger cup → cup held in place
- **Supplementary screw fixation** → if cup suspected not well press-fitted
- **Line-to-line** → screws can be used if:
  - No good initial bone contact
  - Cup not press-fitted after hammering
- Line-to-line or press-fit → **press-fit is better**
- Under-ream rim by 1–2 mm
- Under-reaming → stress on this, no more

Clinical Pearl

- **Press-fit is better** than line-to-line
- Press-fit → under-ream rim by 1–2 mm, then larger cup
- Not well press-fitted → supplementary screw fixation

Acetabular Screws and Safe Zones Ask AI

- Screws placed after cup seated
- **Problems of screws** (three):
  - Source of wear particles
  - Fretting corrosion
  - Neurovascular injury
- Therefore → **four quadrants** around acetabulum:
  - Line from **anterior superior iliac spine** → ischium
  - Line **perpendicular** to it
  - → four zones; screws in **superior zone** *[Q1 · As spoken; safe zone is posterosuperior quadrant only; anterosuperior screws endanger external iliac vessels.]*
- Structures at risk:
  - **Sciatic** nerve → below
  - **Obturator** → in front
  - **External iliac bundle**

Must-Know

- Screw problems → wear particles, fretting corrosion, neurovascular injury
- Quadrants → ASIS–ischium line + perpendicular line → four zones
- Safe screw placement → **superior zone**
- At risk → sciatic below, obturator in front, external iliac bundle

Survivorship: Cementless vs Cemented Ask AI

- Which survives longer → depends on surgeon's **school**
  - Some → cemented better; others → cementless
- Recent study, **cementless cup**:
  - 10-year survivorship → 96%
  - 20-year survivorship → about 90%
- **2013** study → about 27,000 hips
- Design → half cemented, half cementless; 10-year follow-up
- Result → **no difference in survivorship** between cemented and cementless
- Limitation → **older designs** of cemented and cementless cups
  - Further study needed → third-generation cups + HXLPE as used now
- Take-home → cementless cups possibly advantageous in many respects
  - **Technique used** → the key point
- Lecturer's conclusion → **no difference** between cemented and cementless

How to choose your implant fixation

(Choosing Between Cemented and Cementless Fixation in Cups and Stems)

Prof Dr Sherif Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 83) “as in the United States” → “as in the UK”

C2 (p. 88) “literature from the United States” → “UK literature”

Queried - kept as spoken, but looks wrong

Q1 (p. 82) as spoken; fracture at 15 plus AVN 9 years later contradicts arthroplasty at age 15

Opening Case: Fixation by Majority Vote Ask AI

- Per-case question: **cemented vs cementless** cup → audience vote
  - Same question for the stem → second vote
- Majority opinion followed → **cemented cup**; no personal decision
  - Stem probably cementless
  - Options listed: cemented cup, cementless cup, cementless stem
- Reasoning + three points raised by **Dr. Ahmed Hazem** → taken up later

Young Adult with a Previously Nailed Femur Ask AI

Intra-operative Problem Ask AI

Prior nail

- 38-year-old patient; **majority opinion** followed again
- Added **cerclage wire**
- No femoral fracture → wire placed **prophylactically** for fear of fracture
  - Reaming → **thin cortex** at this site
  - Canal previously instrumented: guide wire + **nail reamer** (prior nailing)
    - → previously fractured, **disrupted canal**
- Disrupted canal → no stem hold in **diaphysis** or metaphysis
  - Option: **cementless stem** → feasibility questioned
- Cementless feasible → **extend distally**
  - Component from the **revision** range
  - Distally extending **modular stems**
  - Audience asked for a solution
- Objection: distal (long) stem → ↑ **stress shielding**
  - Cementless advocates asked to solve this
- Answer left open

Trade-offs of Each Option Ask AI

Prior nail

- **Cemented:** only drawback → likely loosening
  - This cup likely to loosen later; done 7 years ago; works in some cases
- Patient **lost to follow-up**
  - Cemented advantages:
    - **Easier**, simpler to perform
    - **Standard stem**, not the suggested long one
    - **Cheaper**
    - Antibiotic-loaded cement (probable)
- **Modular / cementless** advantage: age 38 → some prefer cementless
  - Argument not entirely correct: **England** → cemented fixation
    - Cemented in all cases from age 14 or younger
  - Cementless may still benefit this patient
- **Modular stem** option: Reef or Revitan
  - → acceptable **cementless implant** for the patient

Audience Debate Ask AI

Prior nail

- Why not a **cementless cup**? → valid; lecturer cannot call it wrong
- Better choice → **cementless cup**
- Canal very small, narrow, thin → valid option, not considered at the time
- Case incomplete as shown → older, more **problematic** history
  - Pre-operative limb shortening (probable)
    - Cause: **sciatic nerve** possibly displaced downwards
- → **sciatic nerve** had to be explored (probable)
  - Harder case than shown; focus kept on **fixation technique**

Registry Trend and Head Size Ask AI

Prior nail

- One opinion: **cement both components**
  - Current trend → more cemented, per **English National Joint Registry**
  - Question: why cement the cup?
- Probable **survivorship** at this level with hybrid, cementless, all-cementless
  - Question: this head size or a **larger head**?
- Surgeons judged this the **correct head size**
  - Unusual stem design; lecturer did not recognise it
  - Identified as **CPT**
- CPT confirmed — shoulder (probable) differs from CPT; probably not an **Exeter**
  - Cemented fixation point: large head probably had greater effect than cemented fixation itself

Take-Home Message Ask AI

Prior nail

- Most opinions, incl. operating surgeon, if treated today → **cementless cup + cementless stem**
  - **Dr. Ahmed Galal:** cementless cup placed more medial + cementless stem
  - **Dr. Tarek:** short stem + cementless cup
- Others: **cementless** cup + cementless stem

Post-Traumatic AVN in a 15-Year-Old Female Ask AI

- One opinion: would not operate → **refer** to a professor
- Age 15 → probable **acetabular fracture**; operated by lecturer, 2006 *[Q1 · as spoken; fracture at 15 plus AVN 9 years later contradicts arthroplasty at age 15]*
  - **Femoral neck fixation** also performed
- 9 years later → **AVN**
  - → **osteoarthritis** → needs total hip
- One vote for cemented; operated 9 years after AVN
- Interval ~8–9 years → i.e. 6–7 years after **fixation**; age cited as 15
  - **Dr. Ahmed Galal** → would do nothing for her

Course Ask AI

AVN 15F

- Step 1 → **screw removal**; screw cultures → negative
  - Waited: screws suspected as pain source → pain persisted → **arthroplasty**
- Result not good: joint probably stiff; probable 3 cm **shortening**
  - Soft-tissue **releases** not needed → simple procedure
  - Majority followed → **cemented fixation**

Cemented at 15? Ask AI

AVN 15F

- Cemented at 15 → acceptable per guidelines and **registries**
- Counter: cemented **easier to revise** → no excuse at age 15
  - Longer survival → **cementless**
  - Age 15 → aim for longer **survivorship**
  - Exception: surgeon trained in very good cemented technique, as in the **UK** *[C1 · transcript: "as in the United States"]*
- Only then → well-performed **cemented prosthesis** for her

Post-Acetabular Fracture AVN with Superolateral Defect Ask AI

- 46-year-old male; **post-acetabular fracture AVN**
  - **Superolateral defect** on CT (CT not shown)
  - Vote: cemented vs cementless cup
- Reconstruction: **graft** vs ring
  - Defect too small for a ring → **femoral head graft**
- Graft fixation → **two screws**, not one

Cup and Stem Choice Ask AI

46M defect

- **Cementless cup** in such a defect → yes
  - Previously and now → many such cases receive a cementless cup
  - Some cups (covered in this course) → work **without graft**
- These cups → much **easier**, reliable if trained; more expensive
  - Stem: **cementless**
- Reason → **Dorr type A**
  - Age 46 → either fixation acceptable; Dorr A bone → cementless

Systemic Lupus with Osteoporosis Ask AI

- 36 years; **systemic lupus**
  - X-ray → **osteoporosis**
  - Vote on fixation type
- Audience votes → **cemented** cup
- Done: **cement for cup and stem**
  - Very osteoporotic → fear of losing a **cementless cup** (probable press-fit)
  - → **cemented cup** preferred in such cases

Offset and Neck–Shaft Angle Ask AI

SLE

- Audience remark: **offset** very high
- Near-horizontal → not a **radiographic artefact**; placed deliberately
  - Always **imitates patient anatomy** (approx. 105, uncertain)
  - Egypt: anatomy often not reproducible → check available **implant sets** when planning
- Stem **neck–shaft angle** ≈ offset options
  - Each company → **standard** and offset versions
    - Offset version = **varised** or lateralized (name varies)
  - Each company → **spectrum** of options

  - → **pre-operative planning**: know what the case needs (next case)

Inflammatory Arthritis in Young Females Ask AI

Two Rheumatoid Cases Ask AI

RA

- **Juvenile rheumatoid arthritis**, 16-year-old female
  - Votes split: cemented / cementless
- Abstaining → not an acceptable **answer**
- **Wheelchair**-bound; bilateral involvement
  - JRA → accepted **indication** for cemented → cemented arthroplasty
- **Rheumatoid arthritis**, 30-year-old female → cementless
  - Reasons:
    1. **Age**
    2. **Good bone quality**
    3. **Dorr** ratio / type
- Same disease, opposite fixation:

	JRA	RA
Patient	16-year-old female	30-year-old female
Status	Wheelchair, bilateral	—
Fixation	Cemented	Cementless
Reason	JRA = accepted cemented indication	Age, good bone quality, Dorr type

No Modular Stem or Dual Mobility in a Primary Ask AI

RA

- Not only very young or 46-year-olds → more cases follow
  - Primary case → no role for **modular** stem (Reef, Revitan)
- **Dual mobility** → only for problems needing dual mobility
  - Long stem better? → no
  - Most cases shown → **young females**
- Ages 15 and 16 → **females**, not males

Why Not Arthrodesis Ask AI

RA

- **Rheumatoid** disease → arthrodesis not possible
  - **Arthrodesis** prerequisites:
    1. Normal spine
    2. Normal ipsilateral knee
    3. Normal contralateral hip
    4. **Non-progressive** disease
  - **Rheumatological** disease → no chance
- Counter: consider **arthrodesis** → if patient agrees and it fuses → good
  - Arthroplasty course → arthrodesis case to be shown → fixation choice asked
  - Aim → apply the **fixation lecture** to slide cases

- Arthrodesis case → **cemented total hip** planned
  - Arthrodesis avoided → would end up **converted** to total hip
- Everything must be considered
  - Contralateral **knee flexion deformity**
    - This side → **hip first,** then knee
    - Other side → **knee first,** then hip
  - Difficult case; outside the presentation's focus

Exam Pearl

- Arthrodesis needs normal spine, ipsilateral knee, contralateral hip
- Disease must be non-progressive → rheumatological disease excludes it
- JRA → accepted indication for cemented fixation
- Dual mobility → only for patients with problems needing it

Coxa Valga: Planning and Implant Availability Ask AI

- Seven minutes left; many cases remaining
  - Vote: **cementless cup**? cementless stem?
- Many votes → **both possible**
- Routine **planning** → check available implants
  - Measure **distance** on the other (not fully normal) side
  - Assess: distance, cup, angle, **neck–shaft angle**
  - Other side → **coxa valga**
  - → cup and stem must match the high **shaft angle**
- **Varised stem** → not usable here
- Most stems in **Egypt** → 135° or 137.5°
  - Using one → wrong **offset,** altered greater–lesser trochanter distance
    - → patient **limping forever**
  - No planning → painful patient unable to walk → **unable to move**
- Implant unavailability → never lose the operation's **concept**
  - 145° stem needed but unavailable → tell patient → **wait** for suitable implant
- Final goal → **imitate real anatomy** with real implants
  - Here: cup + **cementless** stem
  - **Big head** → small patient
- Difficult case; earlier → would have **cemented**
  - Case from 2018 → experience increased since

Pitfall

- Most Egyptian stems: 135° or 137.5°
- Wrong neck–shaft angle → wrong offset → permanent limp
- Needed stem unavailable → wait, don't compromise
- Always plan with the implant sets actually available

Cup in Protrusio Without Graft Ask AI

Coxa valga

- If worried → **graft** behind the cup
  - **Dr. Tarek Abdel-Shafy:** cup (probably resurfacing) in protrusio without graft
  - Large series, 2-year follow-up → cup–protrusio gap entirely **filled with bone**

Offset and Limb Length Ask AI

Coxa valga

- **Offset** visible here
- **Offset** → must be imitated
  - **Limb length** easy here → other side diseased
    - Other side → needs surgery within 2–3 years
    - → limb length restoration not mandatory; offset is

Indications: Cemented vs Cementless Ask AI

- Cemented indications drawn from the last few (**rheumatological**) cases
- Cemented **survivorship**: 97% at 10 years, 89% at 15 years

Cemented	Cementless
Rheumatological diseases	Very young patients
Older patients	Good bone stock
Osteoporotic bone	Dorr type A
Dorr type C	All acetabular components (most evidence; Dr. Mahmoud Fahmy)
Low activity	—

High Yield

- Cemented survivorship: 97% at 10 years, 89% at 15 years
- Dorr type A → cementless; Dorr type C → cemented
- Acetabular components → cementless per most evidence
- Osteoporosis, rheumatological disease, low activity → cemented

Older Patients: Quick Cases Ask AI

- 62-year-old male, **osteoarthritis** → cemented; no disagreement
- **Dorr type C** (probably osteoporotic) → both components cemented
  - All his cemented stems → **loaded taper**
  - Prefers loaded taper over **composite beam**
- Has composite beam cases; still prefers loaded taper
  - 60-year-old female, **osteoarthritis** → very tempting for cementless
- Done → **cementless cup + cemented stem**
  - Next: 75-year-old male
- **Femoral neck fracture** at 75 → cemented bipolar

Patient	Diagnosis	Fixation
62-year-old male	Osteoarthritis	Cemented
Dorr type C	Probably osteoporotic	Cemented cup + stem, loaded taper
60-year-old female	Osteoarthritis	Cementless cup + cemented stem
75-year-old male	Femoral neck fracture	Cemented bipolar

Bipolar Hemiarthroplasty: Age Limit Ask AI

- **Bipolar** at 75 → safe
  - Younger patients → no bipolar
  - Age 60–70 → bipolar may not last beyond 10–15 years
    - → **acetabular erosion** after that
    - Some → eroded probably before 15 years (incl. their own studies)
- **Tripolar** cases → coming next
- Bipolar instead of total → financial?
  - At 75 or short life expectancy → depends on **physiological** status
  - Short life expectancy + **acetabular pain** → probably total
  - At 75 now → bipolar or total, probably to avoid dislocation
  - Implant must **hold** in this 75-year-old

Clinical Pearl

- Bipolar safe at 75
- Age 60–70 → bipolar may erode acetabulum within 10–15 years
- Avoid bipolar in younger patients

Tripolar / Dual Mobility in the Elderly Ask AI

- Joint possibly done 16 years ago → must **hold**
  - Votes: all cemented / cementless / long stem / short stem
- Done: **long stem**, cemented, tripolar / dual mobility cup; age 75
  - **Revision** may be needed → correct limb length discrepancy
  - Corrects **limb length discrepancy** as much as possible → possibly over-restored
- Cause → raised **hip center**
  - Correct level → tip of **greater trochanter**
  - Slightly raised; proper **hip center** used
- 76 years old: cemented vs **cementless** → audience votes cementless
- **Tripolar** outcomes → very good results
  - Egypt: only **dual mobility** = cemented, single company (SERF); not routine
- Probable **tumour** case → cement

Intra-operative Periprosthetic Fracture Ask AI

- **Intra-operative fracture** of the hip
  - Fixed → **cerclage wires**
  - Classified → **intraoperative type B**
  - Stem → cementless modular **Wagner**
  - Second option → **cemented**
- Third option → **distal fixation**
- Similar case (courtesy **Hatem Galal**) → same approach
  - **Periprosthetic fracture** → plate + cerclage wires

Conclusions: Femoral and Acetabular Fixation Ask AI

Femoral Component Ask AI

Conclusion

- Cemented **femoral component:**
  - **Elderly osteoporotic** patients
  - **Irradiated bone**
  - **Dorr type C** / stovepipe femur
  - **Deformed femur**

Acetabular Component Ask AI

Conclusion

- **Acetabular components** → cementless
- Cementless cups favoured → components fail more under **shear force** (Fadi's lecture)
  - **Cement** → poor shear resistance
  - Indications: **younger patients**; older patients with good bone
  - Cementless acetabulum in all cases except:
- Poor bone → risk of damage on **impaction** → cemented component

Must-Know

- Components fail more under shear; cement resists shear poorly
- → cementless acetabulum in all cases except poor bone
- Cemented femur: elderly osteoporotic, irradiated, Dorr C / stovepipe, deformed

Literature Bias Ask AI

Literature

- UK literature → always the **cemented camp** *[C2 · transcript: "literature from the United States"]*
  - UK surgeons → very well trained in **cemented** fixation; trend changing lately
- → cemented at any age → very long **survivorship**
  - **American literature:** nobody trained in cemented; younger generation never saw it
- → **biased literature** in both directions → read carefully

Bearing Surfaces: Lecturer's Practice Ask AI

- Preferred: **ceramic on poly** or metal on poly
  - No metal on metal (probable)
  - No **ceramic on ceramic** → problem cases at ages 15, 12, 14 were all CoC

- Over ~20 years (uncertain) → **ceramic on poly**, comfortable
  - Older patients → **metal on poly** fine
  - **Registries:** metal on poly → ~30-year survival
  - Ceramic on poly → also 30-year, longer **survivorship**
- CoC → probably 16 years so far; none reached 30
  - Metal on poly → still **valid**
  - Ceramic on poly → no reason to fear
  - General preference → **hard-on-soft**
- **Wear** → theoretically lowest with ceramic-on-ceramic
  - Its **problems** → greater
  - Egypt → no good formal, specialised 8-year training
- CoC suits a **consultant** using the same setup for 40 years:
  - Same stem, cup, surgical setting, instruments, company
  - Same nurse, same assistance
  - Such a surgeon → CoC as he wishes; in **Egypt** → no

Bearing	Lecturer's use	Registry survival
Metal on poly	Older patients	~30 years
Ceramic on poly	Over ~20 years	~30 years, longer
Ceramic on ceramic	Not used	Probably 16 years so far; none at 30
Metal on metal	Not used (probable)	—

Perioperative Optimization

(Pre-operative Risk Assessment, Infection Prevention and Patient Optimization)

Prof Dr Mohamed Saeed

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 98) as spoken; 30 mg vitamin D is implausible for mice, and the dose has no stated referent

Rationale and Framework of Optimization Ask AI

- Aim → better performance → better arthroplasty outcomes
- Three guiding questions:
  - **Why** perform patient optimization
  - **How** to perform it
  - **What** to do to optimize the patient

Why Optimize Ask AI

- Improves patient outcomes
- Reduces complications and readmissions
- Better outcomes → direct patient benefit

How — Lasting Practice Change Ask AI

- **Protocols** and methods → improved process → good evidence
- Evidence → shows the process is beneficial
- Target: lasting changes in practice
- Improve three domains:
  - Ourselves
  - The process
  - The technology used
- Applied across the **perioperative periods:** pre- → intra- → post-operative

Multidisciplinary Team Ask AI

- **Multidisciplinary team** delivers care:
  - Arthroplasty surgeons
  - Anesthesiologists
  - Infectious disease management
  - Nurses
  - Therapists
  - Nutritionists
  - Administrative co-workers
- Whole hospital cooperates → good operative outcome
- Goal: optimize patient outcome after arthroplasty

Pre-operative Framework Ask AI

- Pre-operative steps:
  - **Checklist** of patient risk factors
  - Assign **level of infection risk**
  - Formulate **medical optimization plan**
- **Bundles** = optimization tool; pre-operative example named in lecture as "workforce" bundle

Pre-operative Risk Factors Ask AI

- Pre-operative period → discussed in detail

Non-modifiable (Patient-related) Ask AI

- Non-modifiable risk factors:
  - Inflammatory arthritis
  - Immunosuppression
  - **Diabetes mellitus**
  - Prior surgery on the joint
  - Previous joint infection
- Other comorbidities listed alongside:
  - Metabolic
  - Allergic
  - Cardiac
  - Renal disease
  - Neurological disease
- Effect → increased risk → worse joint replacement outcome
- Patients with these factors → do worse than others

Modifiable Ask AI

- Definition: changeable → improved outcomes
- Modifiable risk factors:
  - **Poor dentition**
  - **Body mass index** > 40
  - **Malnutrition**
  - **Diabetes mellitus**
  - **Tobacco use**
  - **MRSA screening**
  - **History of open wounds or infections**
  - **Anemia**
  - **Obstructive sleep apnea**
- All → worse operative outcomes

High Yield

- Modifiable factors = targets of pre-operative optimization
- BMI threshold: > 40
- Diabetes mellitus listed among both factor groups

Management of Modifiable Risk Factors Ask AI

Factor	Consequence	Management
Poor dentition	**Septic focus** → ↑ infection risk	Dentist referral → dental clearance
BMI > 40	DVT, readmissions, infections	Do not operate; weight loss or bariatric surgery
Uncontrolled DM (high **HbA1c**)	Wound complications, infections; longer stay; ↑ mortality	—
Smoking > 10 cigarettes/day (half a pack)	More significant post-op complications; more wound complications	—
Obstructive sleep apnea	Longer hospital stay; post-op complications	—

- Diabetes control marker → **Hemoglobin A1C**
- Smoking → direct relationship with post-operative complications

Clinical Pearl

- BMI > 40 → do not operate
- Poor dentition → dentist before arthroplasty
- Obesity → weight loss or **bariatric surgery** first

Malnutrition Ask AI

- Major risk factor for infection

Mechanism Ask AI

Malnutr

- Impaired wound healing
- Impaired **collagen formation**
- Impaired inflammatory response
- ↓ **Lymphocyte** production
- ↓ Macrophage function
- ↓ Ability to fight infection

Assessment Ask AI

Malnutr

- Assess before the procedure
- **Albumin** < 3.5 g/dL → malnourished
- Other markers → more expensive, harder to obtain
- Follow-up with **anthropometric measurements**
- Any method → **nutritional assessment** before surgery

Evidence Ask AI

Malnutr

- Bohl et al., 2018
  - ~49,000 arthroplasty patients

  - **Hypoalbuminemia** → complications, readmissions, longer hospital stay
- Gu et al., 2017
  - 11,000 arthroplasty patients
  - 83% of infected patients → irrigation and **drainage** for infection
  - Only 5% improved — attributed to malnutrition

Exam Pearl
- Malnutrition marker: **albumin** < 3.5 g/dL
- Hypoalbuminemia → complications, readmissions, longer stay (Bohl 2018)

Intra-operative Optimization Ask AI

- Sequence before theatre: workup → risk assessment → medical optimization → physician clearance → admission → operating room

Skin Preparation Ask AI

- At home: **no shaving** — not the night before
- **Chlorhexidine** wash
- **Alcohol-based** skin preparation
- Antiseptic draping

Theatre Attire Ask AI

- **Body exhaust suit** — new arthroplasty attire, used worldwide
  - **Ventilation system** outside the room
  - Complete **mask** → isolated from patient and environment → no contamination

Instruments and Closure Ask AI

- **Glove exchange** before implantation
- **No flash-sterilized instruments**
  - Forgotten items not brought in by flash sterilization
  - All instruments regularly sterilized
- After wound closure → **dilute povidone-iodine** solution wash
- **Silver-impregnated dressing**

Pitfall
- Shaving the night before at home → avoid
- Flash-sterilized instruments → not used
- Gloves not changed before implantation → contamination risk

Post-operative Optimization Ask AI

- **Dedicated joint unit** → ambulation and nutrition
- Early rehabilitation + ambulation → shorter stay

Early Mobilization Evidence Ask AI

- Peterson et al., 2019
  - Mobilization from post-op day 2 → three hours
  - Then three more hours every day
  - → better outcomes, eating and activity; reduced length of stay
  - → greater post-operative value for the patient

Post-operative Bundle Elements Ask AI

- **24 hours** of antibiotics
- **Chlorhexidine wash**
- **Aspirin** → lower-risk patients
- **Physiotherapy** available in the unit
- Wound follow-up via **phone photographs**

Evidence for Optimization Bundles Ask AI

- Van Horn et al., 2019 → significant improvement in patient outcomes
- Bundle effective in practice → reduced infection risk

Study	Design	Result
2018 study	~3,114 patients; 3 years before vs 3 years after bundle	**SSI** ↓ 62%; **PJI** ↓ 92%
Gottschalk et al., 2017	Single center	Infection 12.9% → 1.9% (↓ 87%); length of stay ↓ 20%
Marker-only bundle	Before: 520 patients; after: 4,075 patients	Complications: 35.14 → 14%; 33 → 15%; 42 → 14%; 4.4 → 1.3

- Centers built own bundles from parts → shared with every site
- Marker-only bundle:
  - Markers: **hemoglobin A1C**, BMI, **hemoglobin**, **albumin**
  - Assessment only → no intervention, surgery not stopped
- All → significant reductions in complications and rates

High Yield

- Bundle → SSI ↓ 62%, PJI ↓ 92% (2018)
- Gottschalk 2017: infection 12.9% → 1.9%
- Assessment alone → significant complication reduction

Vitamin D Deficiency and Infection Ask AI

Emerging Modifiable Targets Ask AI

- Under study for pre-/post-operative protocols:
  - **Vitamin D** — not yet in protocols, but used
  - **Patient education**

Design Ask AI

Vit D

- **Hegde et al., 2018 — mouse model**
- Vitamin D–sufficient vs vitamin D–deficient mice
  - 40 mice → deficient diet
  - 20 mice → sufficient diet
- Dose: 30 mg *[Q1 · as spoken; 30 mg vitamin D is implausible for mice, and the dose has no stated referent]*
- Deficient group → infection inoculated at stainless-steel implant → followed three weeks
- Sacrificed → studied **neutrophil infiltration** and **bacterial burden** in cultures

Findings — Deficient Group Ask AI

Vit D

- ↑ **Bacterial bioluminescence** and neutrophil fluorescence
- ↑ **Colony-forming units**
- ↓ **Activated macrophage recruitment**
- Vitamin D replacement → significant effect on bacterial burden and neutrophil infiltration

Patient Education Ask AI

- 2018 study: pre-operative education + phone follow-up for arthroplasty

Pre-operative Phone Education Ask AI

Education

- Phone education starts **one week** pre-op
  - Calls: one week → four days → four days → two days before operation
- Content:
  - Description of the operation
  - Preparation for the operation
  - **NPO** and relevant instructions

Post-operative Calls Ask AI

Education

- Start post-op day 1 → continue two weeks
- Topics:
  - Wound care
  - Medications
  - Pain control
  - Symptoms of concern
  - Mobilization goals
- End point: mobility level 4 in orthopedic clinic
- After two weeks → surgeon visit → outcome assessment

Results Ask AI

Education

- ↑ **Patient satisfaction** in intervention group
- 87% → calls helped preparation for surgery
- 100% → kept more informed
- 97% → would participate again
- ↓ Unplanned calls and visits to orthopedic surgeon and clinic
- Patient satisfaction = end goal

End Points of Optimization Ask AI

- Reassurance of the patient
- Improved patient satisfaction
- Improved patient outcomes
- **Reimbursement** from insurance companies
- After the program → ↓ infection rates, ↓ overall complications, ↓ costs

Preoperative evaluation & Patient Counselling

(History, Examination and Counselling Before Hip Arthroplasty)

Prof Dr Ihab Desouky

Watch Original Lecture

Lecture transcript processed with AI assistance.

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The Road from Clinic to Operating Room Ask AI

- Pathway: **outpatient clinic** → operating room
- Three questions answered along the whole road
  - Is surgery **indicated** now?
  - Is this patient ready for surgery now?
  - Expected outcomes → do we have the capacity to achieve them?
- Sequence of evaluation
  - **Patient interview** (history taking)
  - **Data analysis** → technical considerations
  - **Counselling** → consenting, immediately before surgery

High Yield
- Is surgery indicated now?
- Is the patient ready now?
- Expected outcomes, and capacity to achieve them?

Patient Interview Ask AI

Basic Data Ask AI

History

- Age, sex, **activity level**, occupational habits
- **Chief complaint**
  - Usually pain
  - Others → difficulty walking / limping

Past History Ask AI

History

- **Past medical history**
  - Previous hospital admissions
  - Stability of chronic conditions
  - Comorbidities → hypertension, **diabetes mellitus**, anticoagulant use
  - All → affect subsequent management decisions
- **Past surgical history**
- Cardiac and pulmonary diseases
- **Anticoagulant therapy** → affects:
  - Type of anaesthesia
  - Placement of neuraxial block
  - Prophylaxis given beforehand → patient at risk of **DVT**
- Psychiatric (mental) conditions

Social and Functional History Ask AI

History

- **Social history**
  - Lives alone vs with family
  - Expected support
- **Daily activities**

  - Working or not
  - Sports participation
- Patient **expectations**
- All taken before examination

Patient Factors Affecting the Decision Ask AI

- **Age**
  - Younger than 50–60 years → need longer-lasting implant
  - Risk multiplied: young + **inflammatory** or psychiatric conditions
- **Sex**
  - Women → small **acetabulum** → smaller cup
  - No chance of larger cup / head
  - **Dislocation** rate about 10–12% higher in women than men
    - Cause → smaller acetabular component
- **Activity level** → must be known
  - May be limited → reduced bone stock, **osteoporosis** (previous lecture)
  - Affects mobility problems

Pain Assessment and Contraindications Ask AI

Characterising the Pain Ask AI

Pain

- **Pain** limiting patient's activity → must be understood
- Day pain vs **night pain** → important tools
- First, most important question: pain source
  - **Hip** itself
  - Referred → **lumbar spine**
  - Referred → sacroiliac joint
- Response to previous treatment
- Further questions
  - Pain severity → changes daily activities
  - Rest pain during the day
  - **Night pain** → very significant
  - **Analgesic** use and its duration
  - Hip analgesics, anti-inflammatory drugs, assistive devices
- Previous surgery around the hip; previous procedures

Contraindications Ask AI

Pain

- Skin infections / lesions relevant to surgery
  - More than three foci
  - Large **open wounds / ulcers**
- **Neuropathic** condition
- Permanent **paralysis**
- All → **contraindications** to hip surgery

Physical Examination Ask AI

Inspection and Gait Ask AI

Exam

- Order: **inspection** → gait examination
- **Limping,** if present
- **Attitude** of limb → standing and sitting
- **Previous incisions**
- **Sinuses**
- Skin infection in same limb
- **Leg ulcer**
  - All → dealt with before surgery

Palpation, Motion and Power Ask AI

Exam

- **Palpation** → bony prominences
- **Range of motion**
  - Normal range of motion (figure)
- **Abductor** muscles → must have good power
  - Muscle power assessment
- **Neurovascular status**
- Finally → clear limb length analysis

Limb Length Assessment Ask AI

LLD

- **Apparent shortening**
  - From **pelvic obliquity** and scoliosis
  - No true shortening in the limb
  - Cause may be the **spine**
- **True limb lengths** → detect true **limb length discrepancy**
- **Fixed flexion deformity**
  - Measuring true length → sound limb in same degree of flexion
  - Allows correct true-length measurement
- **Block test** → apparent length
  - Wooden blocks; patient stands on blocks
  - Pelvis levelled on blocks
  - Large difference → may be true discrepancy from hip pathology

Clinical Pearl

- Apparent shortening → pelvic obliquity, scoliosis; cause may be spine
- True lengths → confirm true discrepancy
- Fixed flexion deformity → sound limb placed in same flexion

Special Tests Ask AI

Exam

- Hip special tests → **Thomas test,** Trendelenburg (abductor) test
- **Trendelenburg test** → tests the abductors
  - Watch the side of the **anterior superior iliac spine (ASIS)**
  - Patient stands slowly on one leg → pelvis observed

  - Pelvis does not rise → **abductor insufficiency**
  - Insufficiency → addressed during surgery

Scoring Systems Ask AI

- **Scoring system** → patient's general function
- Two types

	Patient-reported outcome	Observer / physician-reported outcome
Filled by	Patient, on paper	Doctor
Use	Follow-up of improvement	Physician-assessed items
Example	Arabic, English, German, other translations	**Harris Hip Score**
Key point	Must be in patient's own language	Deformity, range of motion → filled by physician

- **Patient-reported outcome**
  - Patient fills paper per his answers → followed afterward
    - Shows his improvement
    - Study (2019) → score must be translated into patient's language
    - Translations: Arabic, English, German, several others
    - Must be in patient's **own language**
- **Observer / physician-reported outcome**
  - Filled by the doctor
  - **Harris Hip Score** → includes deformity, range of motion
  - Deformity, range of motion → filled by physician

Radiographic Evaluation Ask AI

- **Radiographic evaluation** → nature of hip pathology
- **Bone quality** → comparison with contralateral side
- **Radiographic templating**
- **Bone defects** assessment → two parts
  - **Loss of substance**
  - Defect in **bone alignment**
- Beyond bone loss → **spinopelvic motion**

Spinopelvic Mobility Ask AI

Normal Mechanics Ask AI

Spinopelvic

- Ask about recent **spine surgery** and recent spine problems
- Standing → **lumbar lordosis** + pelvis tilts anteriorly
- Sitting → **posterior pelvic tilt** → allows flexion

	Standing	Sitting
Lumbar spine	Lordosis maintained	**Flattening** of lordosis
Pelvis	**Anterior pelvic tilt**	Posterior pelvic tilt
Acetabulum	—	Opens anteriorly

- Anterior acetabular opening on sitting
  - Prevents **anterior impingement**
  - → prevents **posterior dislocation**
- After **total hip arthroplasty** → same sequence, standing and sitting (figure)
- Lateral **X-ray** → standing and sitting positions

Disturbed Mobility Ask AI

Spinopelvic

- Lumbar spine problem, e.g. **fusion**
  - Pelvic tilt locked anteriorly during sitting
  - → **anterior impingement** → **posterior dislocation**
  - Occurs even with correct cup orientation
  - Cause → disturbed **spinopelvic mobility**
- Opposite direction
  - **Lost lumbar lordosis** → no anterior tilt on standing
  - Anterior part of head uncovered → **anterior dislocation**
- More stability needed → **dual-mobility** arthroplasty
  - Better than conventional

	Lumbar fusion	Lost lumbar lordosis
Position at fault	Sitting	Standing
Pelvic tilt	Locked anteriorly	No anterior tilt
Mechanism	Anterior impingement	Anterior head uncovered
Dislocation	**Posterior**	**Anterior**

Pitfall

- Correct cup orientation → still dislocates if spinopelvic mobility disturbed
- Fusion → posterior dislocation; lost lordosis → anterior dislocation
- Solution → dual-mobility arthroplasty

Medical Fitness and Implant Planning Ask AI

Medical Fitness Ask AI

Fitness

- Preoperative medical reports, consultations → covered by **Dr. Mohamed Saeed**
- **ASA classification**

ASA class	Decision
I, II, III	Operable after **optimisation**
4 or 5	Not fit → unsuitable for **elective surgery**

Data Analysis and Technical Considerations Ask AI

Planning

- All data → **radiographic** + **clinical** → assessed
- Determines **approach** and **implant**, according to:
  - Age
  - **Bone quality**
  - Previous surgery
  - **Availability of implants**

Patient Counselling Ask AI

Definition and Goals Ask AI

Counsel

- Second part of the lecture
- **Definition:** providing information about performing surgery
  - **Orally** or in **written form**
  - To patient or his **representatives**
- Clear and understandable
- Patient knows what the procedure involves → clarity essential
- **Complications** that may occur
  - Measures to prevent them, as far as possible
- Simple, plain words
- Goal → patient realises role of surgery in improving his life

Preoperative Patient Education Ask AI

Counsel

- **Patient education**, preoperative stage
  - **Disease** → his condition, problems of the natural course
  - **Benefits** of surgery → improving his condition
  - **Risks** → related to operation itself and to prosthesis
    1. Dislocation
    2. Fracture
    3. Limb length discrepancy
    4. Nerve injury
    5. Infection
- Patient signs **consent** before surgery
- **Anaesthesia**
  - Regional (spinal)
  - **General anaesthesia**
- **Postoperative pain control**
- **Postoperative rehabilitation**
  - Ongoing **physiotherapy** programme → restores function

  - Condition → patient committed to it
- **Patient expectations** → discussed before talking about the operation

Home Preparation and Preoperative Medications Ask AI

Family Support and Home Modification Ask AI

Home

- Functional challenges can be overcome → patient needs support
- **Family support** group around the operation → family caring for patient
- **Home modifications** → easy walking without falls
  - Remove obstacles (loose rugs)
  - Install **handrails** → no falls
  - **Walking aid**
- **Assistive devices** → activities without falls; many other tools

Preoperative Medications Ask AI

Drugs

Drug	Action before surgery
Antiplatelet drugs	Stop 7–10 days before
NSAIDs	Stop 7–10 days before
Methotrexate	Continue as is
Hydroxychloroquine	Continue as is
Other drugs	Discuss with **rheumatology** specialist

Patient Instructions Before Surgery Ask AI

- Continue regimen before surgery
- Food rich in **iron** and **calcium**
- Stop **smoking** → 6 months before and 6 months after surgery
- Day before surgery
  - Remove **nail polish**
  - Shower → antibacterial soap over the body
  - Sleep on clean sheets
  - Good night's sleep
  - Nothing to eat or drink after designated time
  - No lotions
  - No **shaving** before surgery

Must-Know

- Antiplatelets, NSAIDs → stop 7–10 days before
- Methotrexate, hydroxychloroquine → continue
- Smoking → stop 6 months before and 6 months after
- No shaving, no lotions before surgery

Take-Home Message Ask AI

- **Preoperative evaluation** for arthroplasty affects:
  - Decision → operate or not
  - **Timing**, implants, incision / approach
  - Risk of future complications
- Answer the **three questions** in preoperative assessment
- Honest, clear information to patients
- Accept these responsibilities → best outcomes and preparation

Rehabilitation in THA

(Hip Precautions, Early Mobilization and Functional Recovery)

Prof Dr Molham Mahmoud

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 114) “cementless: immediate weight bearing” → “cemented: immediate weight bearing”

C2 (p. 117) “wait 4 to 6 months before driving” → “wait 4–6 weeks before driving”

Rehabilitation in THA: Overview Ask AI

- **Rehabilitation** → key, often forgotten part of **total hip arthroplasty (THA)**
  - Common attitude: operate → send to physiotherapy → leave the rest
  - Understandable, but must change
- Definition → **restoring function** after surgery
- Performed by a **team**
  - Orthopaedic surgeon = **head of the team**
  - Surgeon responsible for patient until the very end
- Starts from first patient contact → pre-operative + post-operative parts
- **Hip arthroplasty** → among the best procedures for pain relief and function
- Aim → pain relief + function within expected time course
  - Ideally via a **rehabilitation protocol**
- Components
  - **Preoperative education** and exercises
  - Post-operative **functional activity**
  - In-patient **post-operative exercises**
  - In-patient rehab → **home discharge** → **outpatient rehab**
  - **Avocational activities**

High Yield

- Rehab = team; surgeon = **team head**
- Starts at first contact, not after surgery

Preoperative Education and Exercises Ask AI

Education Ask AI

- Benefits
  - **Prevents complications** → patient understands early movement and early rehab
  - **Reduces anxiety**
    - Major operation → prepare patient → less anxiety before and after surgery

Exercises Ask AI

- Hip muscles usually impaired → **preoperative conditioning** required
- Significant **pre-op vs post-op function correlation**
  - Poorer pre-op function → lower post-op score than good pre-op function
  - Poor pre-op function → poorer post-op function

Exam Pearl

- Pre-op function predicts post-op function

Functional Activity: Weight Bearing Ask AI

- Two main goals
  - **Weight bearing**
  - Adherence to **hip precautions** (dislocation precautions)
- Key determinant → **cemented vs cementless** implant
  - **Cemented** → immediate weight bearing possible *[C1 · transcript: "cementless: immediate weight bearing"]*
  - **Cementless** → in the past, no immediate full weight bearing → gradual progression

Hip Precautions Ask AI

- Many factors → covered in a dedicated lecture
- Practical message → leg positions to avoid → prevents dislocation
- Positions → **depend on surgical approach**

Posterior / Posterolateral Approach Ask AI

Posterior

- Includes posterior variations
- Avoid
  - **Internal rotation**
  - **Adduction**
  - **Flexion** beyond 70–90°
- Combination = post-op dislocating position
- Easy triggers → low seat, bending to pick something up
  - No excessive hip flexion in any of these
- Maintain **hip angle**
- Low seat / picking up → tendency to over-flex hip
  - Solution → move the whole body, or use a **reacher** (same idea as Dr. Ehab)
- Avoid **crossing the legs** → adduction + internal rotation

Anterior / Anterolateral Approach Ask AI

Anterior

- Opposite of posterior → dislocating movements
  - **External rotation**
  - **Hyperextension**
  - **Adduction**

Approach	Movements to avoid
Posterior / posterolateral	Internal rotation, adduction, flexion > 70–90°
Anterior / anterolateral	External rotation, hyperextension, adduction

Commonest Dislocating Positions Ask AI

- **Crossing the legs** (one leg over the other)
- Lying in **external rotation**

Must-Know

- Precautions depend on **surgical approach**
- Posterior → avoid internal rotation, adduction, flexion > 70–90°
- Anterior → avoid external rotation, hyperextension, adduction
- Commonest → **crossing legs**, lying in external rotation

Functional Restriction and Assistive Devices Ask AI

Functional Restriction Devices Ask AI

- **Abduction pillow**
- **Knee immobilizer**
- **Abduction brace**
  - Simple items (**pillow**) → sufficient
  - Cumbersome **abduction brace** → poor compliance
  - **Randomised controlled trial** → abduction brace: no proven benefit, only complicates matters

Pitfall

- **Abduction brace** → no benefit on RCT + poor compliance; pillow suffices

Assistive Devices Ask AI

- **Walker, cane, crutches**
- Benefits
  - **Stability**
  - Less load on the operated hip
  - Prevent **falls**
  - Wider base of support
  - **Psychological and physical confidence** → moves without falling
- Choice depends on
  - **Stage of rehabilitation**
  - Patient **ability** to use device
  - Patient **age**

Post-operative Exercises Ask AI

- Start → **immediately**
  - Operated in the morning → **mobilisation** + exercises in the **afternoon**
- Exercise **chart** → lists all exercises
  - Given to **physiotherapist** or patient
  - Simple; covers lower-limb muscles **from ankle to hip**
  - Each exercise easy to perform
- Dose → 3–4 times/day, 10 repetitions each
- Progression → **distal to proximal**
  - **Ankle pumps** → dorsiflexion / plantar flexion + ankle rotation (internal / external)

  - **Knee bending** + **quadriceps isometrics** → **hip abduction** — all supine
  - Patient gains control →
  - **Weight bearing** → standing abduction, extension, flexion
    1. Flexion → leg not above the **waist**
    2. Initial maximum ≈ 70°, even if patient can flex further

High Yield

- Exercises start **same day** of surgery
- 3–4 times/day × 10 repetitions
- Progression **distal** → **proximal**, supine → standing

Home Discharge Ask AI

- **Discharge criteria**
  - Walks 15–30 m with an appropriate **assistive device** → acceptable; without one → excellent
  - **Transfers** independently
  - Manages some **activities of daily living** at home
  - **Understands precautions**
  - **Medically stable** → above all
- After discharge → **home therapy** continues
  - **Post-operative rehabilitation** not to be forgotten after leaving
  - **Home rehabilitation** → not widespread in Egypt; available abroad
- Exercise stage complete → next stage: avocational activities

Clinical Pearl

- Discharge → walks 15–30 m, transfers, ADLs, knows precautions, **medically stable**

Avocational Activities Ask AI

- Definition → hobbies, sports, activities beyond ordinary daily tasks
- **No randomised controlled trial**
- Evidence → **survey** of **American Association of Hip and Knee Surgeons (AAHKS)**
  - Each surgeon gave his preference → compiled recommendations
  - ≈ 550 surgeons responded
  - Wait at least 3–6 months before **sports activities**
- Sports chart → three categories

Category	Examples	Rule
High-impact	Jogging, squash	Not allowed; no reason to do them
Low-impact	Golf, swimming, walking	Allowed; no problem with load
Intermediate	—	Surgeon-dependent; prior experience; time since surgery

High Yield

- AAHKS survey (≈ 550 surgeons), not RCT
- Sports → wait ≥ 3–6 months
- **High-impact** (jogging, squash) → not allowed

Driving Ask AI

- Key avocational activity
- Return to driving → **independence** + **psychological relief**
- Questions to answer
  - Can he operate the **pedals**?
  - Car → **automatic or manual**?
  - **Reaction time** → most important
    - Prolonged **braking time** / slow leg use
- Recommendation → wait 4–6 weeks before driving *[C2 · transcript: "wait 4 to 6 months before driving"]*
- **Left THA** → faster return
  - Left leg not needed for driving
  - Right leg → operates the pedals

Exam Pearl

- **Reaction time** → most important driving question
- Left THA → faster return to driving

Sexual Activity Ask AI

Importance Ask AI

- Very important for the patient who needs it
- 75% of arthroplasty candidates → difficulties with sexual activity
- Same 75% → significant improvement in **sexual function**
  - After the arthroplasty
- Very **sensitive** topic
  - Only 10% ask their doctor (Western countries)
  - 90% want advice without asking
  - Don't wait for patient to ask
  - Don't assume irrelevant because of age or other factors

Recommendations Ask AI

- Resume sexual activity 1–3 months after arthroplasty
- Patient → **passive** partner ("P-passive"), not the active one
- Avoid positions with excessive hip flexion → beyond
  - 70°
- Positions by sex

Patient	Position
Female	Supine; passive
Male	Leg in extension, or standing; passive

Clinical Lesson: Indirect Questions Ask AI

- Lecturer's early THA case
- Follow-up for **suture removal** at week 2–3
  - Man from Upper Egypt, ≈ 75 years, building contractor
  - Asked: "Can I take a bath?" → wound perfect → told yes
  - Next morning → hip **dislocated**
  - "Bathing" = euphemism for intercourse
- Lesson → patient asks indirectly, in simple terms
  - Patient won't ask directly → **you must explain it**

Sexual Counselling Ask AI

- Pictures not necessarily needed
- **Sexual counselling** → very important
- Start with **open discussion** → patient + partner

Clinical Pearl

- Only 10% ask; 90% want advice without asking
- Raise the topic proactively → **sexual counselling**
- Patient = **passive** partner; avoid flexion > 70°
- Resume 1–3 months post-op

Summary Ask AI

- Rehab starts pre-operatively
- Continues after **discharge home** with follow-up
- **Avocational and sexual activities** → very important

Guidelines for VTE prophylaxis

(Risk Assessment, Agents and Duration of Thromboprophylaxis)

Prof Dr Mahmoud Abdel Karim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 125) as spoken; half dying overstates hip fracture mortality (about 10% at 30 days, 20-30% at 1 year)

Opening Case: Clopidogrel Before Surgery Ask AI

- Patient on **Plavix (clopidogrel)** alone → dose taken this morning
- Injury → needs **replacement,** not fixation
- Questions posed:
  - VTE prophylaxis plan?
  - Stop the Plavix?
  - What is known about Plavix?
  - Core question → **when** to operate
- Answer revisited at lecture end

Scope and Sources Ask AI

- ~Half of orthopaedic surgeons → same answer to opening case
- Scope:
  - **Elective THA and TKA**
  - **Trauma** — hip fracture
  - **Controversies** — e.g. opening case
- Basis → **NICE guidelines**
  - **NICE** = National Institute for Health and Care Excellence
- Training place → shapes tools carried into practice
  - Know what you know, and from where
- Other sources:
  - **ACCP** — American College of Chest Physicians
  - **Cochrane group** — very important source
- NICE versions:
  - Earlier → 2010
  - Current → March 2018

Why VTE Prophylaxis Matters Ask AI

- Theoretical topic, but very important → not to be neglected
- **VTE events** in orthopaedic surgery — DVT, PE:
  - No prophylaxis → `40–60%`
  - With prophylaxis → `~1–2%`
  - Europe / developed countries → reported `~0.1–0.2%`
- Some cases still lost
- VTE → **preventable cause of death**
- First step → **risk assessment:**
  - **VTE risk**
  - **Bleeding risk**
- Both often needed in same patient
- High **bleeding risk** → may limit prophylaxis

High Yield

- No prophylaxis → VTE 40–60%
- With prophylaxis → ~1–2%
- VTE → preventable cause of death
- Risk assessment → VTE risk + bleeding risk

VTE Risk Factors Ask AI

- Full table in guideline resources → no need to memorise
- Five to remember:
  - **Previous clot** — patient or relative, especially first-degree
    1. Yes → "unfortunately, sir…"
    2. → high risk of major clot, above average
    3. → **most important factor**
  - **Active cancer** or cancer treatment
  - **Age** > 60 (sometimes quoted > 40)
  - **Obesity** — subject of panel debate
  - Known **thrombophilias**

Must-Know

- Previous clot (self or first-degree relative) → most important
- Active cancer or cancer treatment
- Age > 60
- Obesity
- Known thrombophilias

Bleeding Risk Factors Ask AI

- Second assessment → **risk of bleeding**
- Long list → prefer a **thinking algorithm**
  - Asked in American College training programme since 2006
- Factors:
  - **Active bleeding**
  - **Hepatic disease** → be careful
  - Acquired **bleeding disorders**
  - Inherited bleeding disorders:
    1. **Haemophilia**
    2. **Von Willebrand disease**
    3. Outside orthopaedic expertise

General Measures: Hydration and Mobilisation Ask AI

- Elderly relatives get clots → partly inadequate **fluid intake**
- Intake → 2–3 L/day

  - Unless restricted by **cardiological** status
  - Never allow **dehydration**
- Postoperative **mobilisation** — even in bed:
  - Move upper and lower limbs → ankle pump exercises
  - Warn patient in advance

Elective Total Hip Arthroplasty Ask AI

Risk Balance Ask AI

THA

- Two risks weighed → **thrombosis risk** vs bleeding risk
  - THA, TKA, all procedures here → thrombosis risk higher
  - → VTE risk usually exceeds **bleeding risk**

Pharmacological Prophylaxis Ask AI

THA

- Most cases → **LMWH**
  - Generic names used
- Guidelines:
  - 2010 → duration 28 days
  - Recent years → **aspirin** option, incl. ACCP
- **Aspirin** dose:
  - Initially → 150 mg
  - Now → 75–150 mg
  - Duration → ~1 month (28 days)
- Routine → **LMWH** from start × 28 days
- Speaker's preference → **LMWH**
  - **Aspirin** → reserved for specific cases
  - Selection → after **risk stratification**
- Worldwide → aspirin **approved** for THA and TKA
- Alternatives:
  - **Rivaroxaban**
  - **Apixaban**
  - **Dabigatran etexilate**

Mechanical Prophylaxis Ask AI

THA

- Combine pharmacological + **mechanical** prophylaxis
  - e.g. **anti-embolism stockings**
- Contraindication to drugs → **mechanical alone** from start
  - Typical in trauma → active bleed elsewhere → decision needed
- Same approach if VTE risk high

High VTE Risk with High Bleeding Risk Ask AI

- Both risks high → **multidisciplinary group**
  - Hepatologist
  - Neurologist

  - Haematologist
  - Vascular surgeon
- **Combined meeting** → best decision
- **Involve family:**
  - e.g. LMWH given → heavy bleed → must stop
  - → clots may occur
- No avoidance → disclose openly

Elective Total Knee Arthroplasty Ask AI

- Same approach as THA; shorter duration:
  - **Aspirin** 75 or 150 mg × 14 days — NICE wording
  - **LMWH** × 14 days
  - Plus **anti-embolism stockings** / other mechanical methods
- Same alternatives → rivaroxaban, apixaban, dabigatran etexilate
  - Generic names only
- Contraindication → start **intermittent pneumatic compression (IPC)**
  - Unavailable in most local hospitals
  - Continue → **until patient mobile**

	THA	TKA
LMWH	28 days	14 days
Aspirin	75–150 mg × 28 days (~1 month)	75 or 150 mg × 14 days
Alternatives	Rivaroxaban, apixaban, dabigatran etexilate	Same
Mechanical	Anti-embolism stockings	Stockings / other mechanical
Drugs contraindicated	Mechanical alone	IPC until mobile

Exam Pearl

- THA → 28 days prophylaxis
- TKA → 14 days prophylaxis
- Aspirin → 75–150 mg, now an approved option
- Always combine with mechanical prophylaxis

Hip Fractures Ask AI

VTE Risk and Mortality Ask AI

Fracture

- **Fractures** → very high stakes; ~half of patients die (as quoted) *[Q1 · as spoken; half dying overstates hip fracture mortality (about 10% at 30 days, 20-30% at 1 year)]*
- Saying → "come from the hip, go from the hip"
  - Anecdote → speaker's patient died before surgery
  - Advice → prophylaxis × one month → persistent **VTE risk**

- Bleeding risk also present → calculate, discuss with patient

Postoperative Agents Ask AI

Fracture

- **LMWH** → 6–12 h post-op; ~12 h
  - Patient left 24 h → clot → common; share responsibility
- **Fondaparinux** → 6 h post-op
  - Condition → low **bleeding risk**

Timing and Preoperative Prophylaxis Ask AI

Fracture

- Trauma patient → may take time to prepare
- Elective → no **preoperative** prophylaxis needed
- Best practice → surgery within 36–48 h; now **36 h**
- Surgery delayed → start prophylaxis
  - Stop **Clexane** 12 h pre-op
  - Stop **fondaparinux** 24 h pre-op
- Surgery delayed → **mechanical** prophylaxis until surgery (ACCP)

Agent	Post-op start	Stop pre-op
LMWH (Clexane)	6–12 h (~12 h)	12 h
Fondaparinux	6 h, if low bleeding risk	24 h

IVC Filter Ask AI

Fracture

- Often written → "don't forget **IVC filter**"
- Since 2012 → apparently limited to specific indications, not primary prophylaxis

Evidence and Caution Ask AI

Fracture

- DVT evidence → no certainty for or against
  - Study; invest in training
  - Follow guidelines **dogmatically**
  - Track new evidence
- Some agents → efficacy possibly offset by **bleeding complications**

High Yield

- Hip fracture → surgery within 36 h
- LMWH → 6–12 h post-op; stop 12 h pre-op
- Fondaparinux → 6 h post-op if low bleeding risk; stop 24 h pre-op
- Delay → mechanical prophylaxis until surgery
- Duration → one month

Perioperative Antiplatelets Ask AI

Aspirin Ask AI

Aspirin

- Patient on aspirin → stop before surgery? **Never**
  - Primary prevention (no specific disease) → continue
  - Cardiovascular indication → continue

- Involve **cardiologist** from start → "no need to stop aspirin"
- Don't stop aspirin or Plavix on **own judgement**
  - Advice against guidelines → **their responsibility,** not yours
- **Spinal / epidural** → allowed on aspirin
- Start **routine preoperative regimen**
- **Platelets** prepared → in case of bleeding
- Final decision → surgeon's own

Clopidogrel (Plavix) Ask AI

Plavix

- Question → **continue or discontinue** Plavix on admission
  - Revision surgery
  - Stent + trauma
- Stopping unilaterally → serious complications
  - **Bare-metal stent**
  - **Drug-eluting stent**
- **Neuraxial block** on clopidogrel
  - → **contraindicated**
- Plavix stopped or not → operate **early** (next day), not after a week
  - Safe technique + confident **haemostasis** → same or next day
- **General anaesthesia** → proceed
  - Anaesthetist view → GA fine, no problem
  - Requires quick, confident technique
- Stent > 6 months old (corrected from 6 weeks):
  - Stop **Plavix**
  - **Continue aspirin**
- Inform patient of risk; **platelets** prepared

	Aspirin	Clopidogrel (Plavix)
Stop before surgery	Never — primary prevention or CV	Not unilaterally; stent > 6 months → stop, continue aspirin
Spinal / epidural	Allowed	Contraindicated
General anaesthesia	—	Fine
Timing of surgery	—	Early — same / next day
Prepare	Platelets	Platelets

Pitfall

- Stopping aspirin or Plavix on own judgement
- Stopping Plavix with a stent → serious complications
- Neuraxial block on clopidogrel → contraindicated
- Waiting a week on Plavix → operate early instead

Orthogeriatric App Ask AI

- **Orthogeriatric App** — by AO Trauma
  - Android and iPhone → recommended download
- Covers:
  - **Osteoporosis**
  - **Delirium**
  - **Perioperative pain management**
  - **Anticoagulation**
- Anticoagulation section:
  - **Perioperative thromboprophylaxis**
  - **Management of pre-existing** anticoagulation
  - Next page → yes / no questions
  - Agent list → select → when to stop, when to give, what to add
  - Plus risk assessment

Opening Case Resolved Ask AI

- Professor of anaesthesia's mother:
  - Fracture at 9 a.m.; **Plavix** at 9 a.m.
  - Operated at 3 p.m. → did well
  - **General anaesthesia**
- Incorporate into practice
- Prophylaxis → one **month**

Lateral Approach

(Direct Lateral (Hardinge) Approach: Technique and Surgical Anatomy)

Prof Dr Hazem Abdel Azeem

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Introduction and Historical Perspective Ask AI

Hip Surgery Then and Now Ask AI

- Past hip practice → very heavy patients, many difficulties
  - Classic scenario → **fracture neck of femur**, patient left bedridden
- Speaker's early career → 1974 or 1975 start
  - Probable standard then → **hemiarthroplasty** for fracture neck of femur
- Arthroplasty since → developed; teaching expanded
  - Solutions now for hip problems once thought unsolvable
- Childhood hip conditions threatening later life
  - Previously → diagnosis possible, **treatment options** limited
- Speaker's pride in progress of hip surgery at this university

Egyptian Hip Society Ask AI

- Announcement → April, 8th or 9th
  - Annual meeting of the **Egyptian Hip Society**
  - Society founded ~8 years ago → forum for communication and development
  - Next conference → Cairo
    - Attendees invited to participate, contribute ideas and presentations

Lateral Approaches to the Hip Ask AI

- **Lateral approaches** → a group of quite different approaches
- Common feature → **centered on the greater trochanter**
  - Usually directed anteriorly
- Members of the group
  - **McFarland approach** (McFarland and Osborne)
  - **Modified McFarland approach**
  - **Hardinge approach**

Hardinge Approach: Position and Incision Ask AI

Position Ask AI

Hardinge

- **Hardinge approach** → most commonly used lateral approach → covered first
- Session focus → surgical technique, not theory
  - Indications and contraindications → not discussed
- Position options
  - **Lateral position**
  - Supine position
    - Supine → sandbag under the buttock; limb slightly tilted
- Speaker's preference → **lateral position**
  - Easier for trainees to get used to

Incision Ask AI

Hardinge

- Length → usually >10 cm
- Runs over the **greater trochanter**
  - ~Half of incision over trochanter (centered on it)
  - Other half likely extending proximally
- Mark **anterior** and **posterior borders** of the greater trochanter
  - Trochanter → middle of the field
- Incise skin → subcutaneous tissue
- Visible in wound → **tensor fasciae latae**

Hardinge Approach: Superficial Dissection Ask AI

Fascial Layer Ask AI

Hardinge

- **Gluteus maximus** fibres → from posterior side, inserting posteriorly
- Interval between TFL and gluteus maximus
  - Contains **fibro-fatty tissue** or appears as a yellowish line
  - Follow this line → deeper
- Self-retaining retractor → then incise the **fascia**
- Fascial incision → extended distally ~10 cm below tip of greater trochanter
- Orientation → imagine the destination
  - Area exactly opposite → **base of the neck**
  - **Head of femur** → at the pointer's position
  - Gives impression of where the approach leads

Trochanteric Bursa Ask AI

Hardinge

- **Trochanteric bursa** → usually easy to deal with
  - Exception → thick **trochanteric bursitis** containing fluid
    - Undesirable → possible bleeding or contamination
    - Management → quick excision
- Then clear the tendinous part in front

Gluteus Maximus Insertion Ask AI

Hardinge

- Part of **gluteus maximus** → inserts onto posterior part of greater trochanter
- Options
  - Retractor beneath it → preserve it
  - Or extend the fascial incision distally
- Either way → facilitates retraction
- Anatomy here → **varies between individuals**
- Retract TFL and gluteus maximus → **trochanter** seen beneath

Abductor Exposure Ask AI

Hardinge

- Identify **gluteus medius** tendon and **vastus lateralis**
  - Essential → next step is the **gluteus medius split**

Gluteus Medius Split and Superior Gluteal Nerve Ask AI

The Split Ask AI

Hardinge

- **Superior gluteal nerve** lies in this region → direct the split accordingly
- No single fixed split pattern
  - Gluteus medius fibre groups → **anterior**, middle, part of posterior
  - Anterior group distinct from middle group

Trendelenburg Concern Ask AI

- Goal → preserve **abductor function**
  - Assessed by → **Trendelenburg test**
  - Weakness → **Trendelenburg gait**
- Claim: lateral approach causes postoperative Trendelenburg gait → **not correct**
  - Not difficult to avoid
- Therefore → split carried downward (distally)
- Pass between fibres → some gluteus medius fibres encountered → carry split distally
- Image description
  - Part that was **reflected**
  - **Abductor part**
  - Tenotomize → push anteriorly
- **Gluteus minimus** → in depth of the wound
- Major fear → subsequent **Trendelenburg gait**
  - Even if transient, for some hours
  - Usual cause → **nerve concussion** (neuropraxia)

Safe Zone Ask AI

SGN

- Anatomical **safe zone** → from greater trochanter upward
  - Extent → roughly one-third of the distance
  - Within it → superior gluteal nerve avoided
- Course of **superior gluteal nerve**
  - Probably above the piriformis → **between gluteus medius and gluteus minimus**
  - Main trunk → runs a distance on gluteus minimus muscle
  - Enters **posterior part** of muscle → branches proceed to anterior part
- Preserve **medius–minimus plane** → nerve preserved
  - Avoid excessive retraction, traction, stretching
- Always stay in the plane between gluteus medius and minimus

High Yield

- Trendelenburg gait after lateral approach → avoidable, not inevitable
- Usual cause → neuropraxia (nerve concussion) of superior gluteal nerve
- Nerve → between gluteus medius and minimus; enters posterior part
- Protection → stay in medius–minimus plane, no excessive traction
- Gluteus medius split → carried distally

Hardinge Approach: Tenotomy and Capsule Ask AI

Inverted-T Tenotomy Ask AI

Hardinge

- After **gluteus medius** split
  - **Gluteus minimus** found → probably not split; runs from above downward
- **Tenotomy of gluteus medius tendon** → anterior part
- Distal tenotomy also needed → **inverted T** shape
  - Opens space between gluteus medius tendon and underlying **capsule**
- **Capsule** now seen beneath
  - Gluteus medius previously covering it from above → now retracted out of the way

Capsule Management Ask AI

Hardinge

- Capsular opening → surgeon's preference; any type acceptable
  - **T capsulotomy**
  - **Capsulectomy**
  - **Square capsulotomy**
  - Inverted pattern → transverse at the neck, vertical above

	Past	Current
Total hip arthroplasty	**Excision of capsule** + synovectomy first	**Without capsulectomy,** except where needed
Total knee arthroplasty	Included synovectomy	—
Debridement	Much wider field	Narrower
Options	—	Capsulotomy, synovectomy, **labral excision** as needed

Dislocation, Acetabular Exposure and Closure Ask AI

Dislocation Ask AI

Hardinge

- Patient → lying on the side
- **External rotation** of the hip + simultaneous control of the leg
  - Leg held in a sterile leg bag or similar
  - **Assistant** → moves the leg from the other side
- Alternative → leg brought over the table edge → **not preferred**

Knee Caution Ask AI

- Caution in hip surgery patients
  - Elderly without reserve, or **knee osteoarthritis**
  - Leg twisted across the other + force by hand → **knee pain**
    - Persists long postoperatively → troubles walking
- Possibly only the assistant applies the manoeuvre, from above

Pitfall

- Forceful twisting of leg during dislocation → persistent knee pain
- At-risk → elderly without reserve, knee osteoarthritis
- Leg over the table edge → not preferred

Acetabular Exposure Ask AI

Hardinge

- After dislocation
  - Neck osteotomy probably completed → work on **acetabulum**
- Two or three pointed retractors
  - Placed between labrum/acetabulum and capsule
  - Many surgeons do not know where to place them
- Layers → **labrum**, socket beneath, **capsule** attached above it
  - Retractor site → **between labrum and capsule**
- Incision → allows **extension** in every needed area

Clinical Pearl

- Acetabular retractors → between labrum and capsule
- Two or three pointed retractors

Femoral Preparation and Closure Ask AI

Hardinge

- Then **femoral side** preparation → facilitates later steps
- After preparation and **reduction** → everything back to normal
  - Unless a **capsulectomy** was performed
- Repair sequence
  - **Capsule** (if capsulotomy performed)
  - **Gluteus minimus**
  - **Gluteus medius group** → reattached
    - Onto the **trochanter**
  - Close in layers
- Closure → layer by layer

McFarland Approach Ask AI

Concept Ask AI

McFarland

- Speaker learned it and continued using it
- **Simple approach** → does not damage muscles
  - **No tenotomies**
- Position → **lateral position**
- Incision
  - Not necessarily centered on trochanter
  - For a **limited hip procedure** → centered on trochanter, as in Hardinge
- Incision carried in same line → **tensor fasciae latae**
  - → **iliotibial band** distally

Original McFarland Ask AI

McFarland

- One unit taken → **gluteus medius** + tensor fasciae femoris + band
- **Gluteus medius–vastus lateralis** conjoint tendon → kept in continuity

Reflecting the Common Tendon Ask AI

McFarland

- Through iliotibial band incision → reflect **common tendon** from greater trochanter from above
- Two options
  - **Subperiosteal** reflection (raising the periosteum)
  - **Osteotomy** → part of greater trochanter taken
- Then clear gluteus and TFL
  - From above
- Exposure → upper part of the **neck**
- Upper part of **head** → covered by layers
  - **Gluteus minimus** → capsule beneath → head beneath capsule
- Image review
  - Line of incision of **gluteus medius**
  - **Trochanter**
  - **Tensor fasciae latae**
- Reflection of common tendon → either **subperiosteal**
  - Or **osteotomy of greater trochanter** with the common tendon
- View → **gluteus medius, common tendon, vastus lateralis**
  - Reflected anteriorly

Modified McFarland Approach Ask AI

Rationale Ask AI

Mod McF

- Original McFarland → **whole gluteus medius** taken anteriorly
  - Endangers **nerve supply** → arrives from posterior part
- Solution → **modified McFarland;** incision made anteriorly
  - Between **anterior third** and **posterior two-thirds** of gluteus medius
  - Posterior two-thirds → left in place (as in Hardinge)
  - Anterior part → reflected forward

Technique Ask AI

Mod McF

- Split → made at the corresponding site
- Flap reflected
  - **Anterior part of gluteus medius**
  - Part of vastus lateralis, likely its posterior part
- Successive steps beneath → **gluteus medius**

	Hardinge	Original McFarland	Modified McFarland
Position	Lateral (speaker) or supine	Lateral	Lateral
Incision	Centered on greater trochanter	Centered on trochanter for limited procedure	Same line, extensile

	Hardinge	Original McFarland	Modified McFarland
Gluteus medius	Split; anterior part tenotomized	**Whole** muscle reflected anteriorly	**Anterior third** reflected; posterior two-thirds left
Tenotomy	Inverted-T	None	None
Continuity	—	Gluteus medius + conjoint tendon + vastus lateralis	Gluteus medius + vastus lateralis sleeve
Nerve risk	Superior gluteal nerve in split	Nerve supply endangered	Nerve supply spared

Extensile Exposure and Extended Trochanteric Osteotomy Ask AI

Extensile Value Ask AI

Mod McF

- **Extensile** from above downward → proximally and distally
  - Usable in **revision** surgery
- Extendable → probably to the femoral shaft
  - Relevant for shaft view → e.g. **long stems**
  - Extension follows the same structures
- All → continuous lateral structures kept linked

Extended Trochanteric Osteotomy Ask AI

ETO

- Detailed in next lecture → **Dr. Mahmoud Abdel Karim**
- Requirements for **extended trochanteric osteotomy**
  - Osteotomy and release of **trochanteric fragment**
  - **Muscle group** remains **attached** to fragment
    - Not taken out in the hand and put back loose → in continuity
- This approach → fragment kept as a **flap** during osteotomy
  - As seen in the diagram
- Osteotomy fragment → **vastus lateralis** part attached
- Enter the **canal** → work → return **fragment flap** to its place
- Now performed widely → **revisions** increased greatly
  - Not very difficult
  - Used in difficult situations, likely with cemented stems
- Prerequisite → **extensile exposure**
  - **Modified McFarland** approach
  - Or **posterior approach** → has some limitations

Speaker's Experience and the Sleeve Principle Ask AI

- This approach → more advantages than posterior approach
- Long experience with **McFarland / modified McFarland**
  - Since assisting late professor (El-Hommos), with modifications
  - Work with Dr. Maher in England
  - Many cases in Egypt, including health insurance cases
- **Extensile exposure** → any level, limited or extended, needs nothing else

  - Extended as needed → **muscle sleeve** remains continuous
  - Distally or proximally → sleeve in continuity
- **Distal extension** → after **osteotomy of the neck**
  - Probably retract trochanter to another position
  - Reflect posterior flap instead of **anterior flap**

Must-Know
- Modified McFarland → anterior third of gluteus medius reflected with vastus lateralis
- Extensile proximally and distally → usable in revision
- ETO → trochanteric fragment kept attached to muscle group as a flap
- Muscle sleeve → kept in continuity

Disadvantages and Complications Ask AI

Exposure Limitations Ask AI

Lateral

- Disadvantage → other side probably not included in the exposure

Gluteus Medius Complications Ask AI

Lateral

- Reopening needed on 2–3 occasions
  - **Gluteus medius tendon tear**
  - **Gluteus medius denervation**
- Not **100% safe**
  - Gluteus medius complications → whether tendon detached or kept in continuity
  - Dependent on **technique**

Trochanteric Osteotomy Nonunion Ask AI

- **Osteotomy of greater trochanter** → may fail to unite (**nonunion**)
  - Do not rely on **suture only** → more secure fixation

Leg Length Ask AI

Lateral

- Another disadvantage
  - Cannot observe leg lengths → no **leg length equalization**
  - Possible in **posterior approach**

Exam Pearl
- Lateral approach complications → gluteus medius tear, denervation
- Trochanteric osteotomy → nonunion risk; more than suture fixation
- Leg length equalization → posterior approach, not lateral

Posterior Approach

(Positioning, Incision and Soft-Tissue Repair in the Posterior Approach)

Prof Dr Mohamed Abo Elsoud

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Introduction Ask AI

- Topic: **posterior approach** to the hip
- Context: follows Dr. Hazem's lateral approach talk
- First seen performed by **Dr. Tarek Abdel Shafy**
  - Speaker then a resident, just back from the army
- Format: surgical video + running commentary
- Style → *Science and Faith* programme, late **Dr. Mostafa Mahmoud**
- Upcoming lecture: **Choice of the Approach**
  - Literature → which approach is better
  - Lateral approach advantages vs posterior approach disadvantages

Patient Positioning Ask AI

- **Lateral position** — no alternative
- **Pelvic support**
  - Anterior → **symphysis pubis** or anterior superior iliac spine (ASIS)
  - Posterior → **sacrum**
    - Not on the lumbosacral junction
- Leg moves freely anteriorly + posteriorly → full **hip flexion**

Pitfall

- Leg striking the anterior post → **false impression** of easy dislocation
- Posterior support on lumbosacral junction → wrong; sacrum only

Skin Incision Ask AI

- **Classical incision:** curved posteriorly
  - Centred on the **greater trochanter**
- Upper (proximal) + lower (distal) limbs — as in most hip approaches
- **Extensile** → proximal or distal extension as required

Superficial Dissection Ask AI

- Skin → subcutaneous tissues → fascia over **gluteus maximus** + fascia lata
- **Fascia lata** opened → diathermy or knife, per preference
- **Gluteus maximus** fibres → split bluntly by hand, in fibre line
- Video = speaker's own case
- Next layer: **fat over the sciatic nerve** + bursal surface (term unclear in recording)
- **Constant bleeder** at this angle
  - Muscular branch → gluteus maximus supply
- Sciatic nerve usually lies within this fat

Bursal Layer Handling Ask AI

Bursa

- Options (bursa — term unclear in recording)
  - Simply opened
  - Scissors passed beneath → opened over them with **diathermy**
- Diathermy applied onto scissors → **scissor tip points toward sciatic nerve**

Short External Rotators and Sciatic Nerve Ask AI

- **Gauze** → sweeps fat + sciatic nerve posteriorly
  - Left in place → protects sciatic nerve
- Routine posterior approach → sciatic nerve **not exposed**
- **Short external rotators** now clear
  - Obturator internus tendon + two gemelli
  - Below → fleshy **quadratus femoris**
  - **Piriformis** hidden → beneath posterior border of gluteus medius
- Piriformis exposure → retract **posterior edge of gluteus medius**
- Identification
  - Fleshy muscle → **quadratus femoris**
  - Tendon with muscle above + below → **obturator internus**
    - Superior + inferior gemellus
- Same patient, gauze removed → **sciatic nerve** + obturator internus clearly seen
- Retractor → holds posterior border of gluteus medius
- → **Piriformis tendon** revealed (next video)
- Muscle deep to piriformis tendon → **gluteus minimus**

Release of Rotators and Capsule Ask AI

- **Gluteus minimus** → protect; never cut
- Piriformis tendon identified
- Rotators released → **diathermy** or knife
  - **Tagging sutures** on rotators → surgeon preference
- **Rotators + capsule** opened together
  - Gluteus medius → not opened
  - Gluteus minimus → not injured
- **Capsule** = white tissue beneath → opened too
- **Upper quadratus femoris** released → lesser trochanter + femur well seen
- Capsule opened → gluteus minimus visible
- **Hohmann retractor** beneath gluteus minimus → retracts + protects it
- Upper capsule → **capsulotomy** completed
- **Synovitis** → gush of fluid/synovium on capsular opening
- Hohmann passed **superiorly** → carries gluteus minimus with it
  - → **superior capsule** seen → capsulotomy completed
- Upper quadratus femoris completed + opened → **lesser trochanter** seen
- **Quadratus femoris** incision completed → superiorly + inferiorly
- Opened capsule visible

- Glistening **iliopsoas tendon** → on lesser trochanter

Dislocation and Neck Exposure Ask AI

- **Dislocation** → in total hip
- Fracture neck of femur → only **stump of the neck** delivered

	Lateral approach	Posterior approach
Dislocation manoeuvre	External rotation + figure-of-four	Maximum internal rotation

- This case: fracture neck of femur → neck stump delivered
- Landmarks: **lesser trochanter** + iliopsoas tendon
- **Blunt Hohmann** → beneath iliopsoas tendon
- Second Hohmann above neck stump → mostly **sharp** → neck well exposed
- **Rotator + capsule stump** → always remains in piriformis fossa
  - However close to greater trochanter the release
- **Piriformis fossa** = stem entry point
  - → stump removed → **box osteotome** → femoral canal
- Arthritis case: femoral head, lesser trochanter, Hohmann
  - Hohmann on **upper neck** → rotator stump removed same way

Exam Pearl

- Lateral approach → external rotation + figure-of-four
- Posterior approach → **maximum internal rotation**
- Rotator/capsule stump in piriformis fossa → blocks stem entry → remove

Neck Cut and Head Extraction Ask AI

- **Neck cut** → two cuts
  - **Horizontal** → along rotator stump line
  - **Vertical** → second cut
- Horizontal cut → **greater trochanter** must not fracture — key point
- Osteotome in horizontal line alone → exits → **greater trochanter fracture**
- **Vertical cut** → protects greater trochanter
- Residual bone piece → **safety**, not timidity
  - → removed with box osteotome
- Fracture neck → head still inside → **corkscrew** → head extracted
- **Ligamentum teres** → usually attenuated in elderly
  - This case: very thick → scissors → easy head delivery
- Head delivery → leg manipulation, head taken forward (wording unclear in recording)
- Head still tethered → **inferior capsule** + ligamentum teres
  - Divided → diathermy or scissors
- Elderly, very tough tissues → **scissors**
- **Anterior capsule** + ligamentum teres still attached → head delivered

High Yield

- Neck cut = horizontal + **vertical** cut
- Vertical cut → prevents greater trochanter fracture
- Leftover bone → removed with box osteotome
- Fracture neck → head retrieved with **corkscrew**

Anterior Capsular Resection Ask AI

- **Anterior capsular resection** — very important step
- Older arthroplasty patients → **anterior capsule** often very hypertrophied
  - → femur held stuck against acetabulum
  - → femur cannot go forward → acetabulum not seen
- **Acetabular reaming** → hand must be free anteriorly
  - Otherwise → reaming in **retroversion**
- Anterior capsule = structure holding femur back

Removal method	Technique
From below	Cut with **scalpel**
Hohmann-assisted (easier)	Hohmann between **gluteus minimus** and anterior capsule

- Capsule in this case → very hypertrophied
- Hohmann → takes **minimus** forward
- **Kocher** → holds anterior capsule for resection
- White capsule → Kocher slips → re-applied
- Whole anterior capsule resected → **anterior mobilisation of the femur**
- Anterior **Hohmann** → takes femur forward
  - Femur then "forgotten" → leg straight, no further attention
- Hohmann placement → **labrum–capsule plane** (per Dr. Hazem)
- Anterior + posterior Hohmanns → both between labrum and capsule
- Posterior retractor seen entering between white **labrum** and capsule

Labrum and Inferior Capsule Release Ask AI

- Capsule + retractor within it identified
- Capsular release completed
  - **Inferior capsule** → incised
  - **Labrum** → cut + removed (per Dr. Tarek Abdel Shafy)
    - Mostly removable at cup preparation

Labrum	Tool	Reason
Anterior	Diathermy or scalpel	—
Posterior	**Scalpel** preferred	Diathermy heat → sciatic nerve

- Posterior labrum removed → residual tissue also removable
- Once removed → view opens; tissue brought downward
- **Inferior capsule** → incised
- Anterior tissue removed → comes out very easily
- **Transverse acetabular ligament** → appears at inferior acetabulum
- Inferior **Hohmann** → also possible

> **Clinical Pearl**
> - Posterior labrum → **scalpel**, not diathermy
> - Diathermy near sciatic nerve → onto scissors, tip toward nerve
> - Gauze left posteriorly → protects **sciatic nerve**

Acetabular Exposure and Preparation Ask AI

Retractor	Placement
Posterior Hohmann	Between labrum and excised capsule
Second Hohmann	Anterior → femur taken anteriorly
Inferior Hohmann	Inferior acetabulum

- **Transverse acetabular ligament** identified

Surgeon Position Ask AI

Surgeon

Hip	Surgeon position	Benefit
Right	Behind the patient	—
Left	Other side	Straight hand; clearer reaming view

- This case: right hip → same side

Acetabular Preparation Ask AI

Cup

- **Acetabular reamers** → cup bed prepared
- Reaming → **acetabular cup** inserted
- ± 1–3 **acetabular screws** as needed (unclear in recording)
- Cup alignment → **transverse acetabular ligament** at lower cup margin

Posterior Repair Ask AI

- Femoral side completed → **posterior repair** mandatory
- Repaired: **piriformis**, rotators, capsule

Fixation	Rationale
Gluteus medius tendon	More durable → withstands external + internal rotation
Drill holes in bone	Possible; rotation, esp. internal → pull-out

- → Some surgeons prefer gluteus medius tendon
- Repair complete → joint reduced + covered, no longer visible
- All **posterior structures** repaired → non-absorbable sutures preferred

Take-Home Message Ask AI

- **Posterior approach** → easy + reproducible
- **Pelvic support** → patient does not roll forward or backward
  - Video: support slightly high, on abdomen
  - Correct → pubis or anterior superior iliac spine
- **Preserve gluteus minimus**
  - → avoids posterior + lateral approach disadvantages combined
  - i.e. no added **abductor** injury
- **Retractor placement** → protects sciatic nerve + anterior structures (vessels)
- **Anterior capsular release** → anterior mobilisation of the femur
  - Esp. elderly with hypertrophied capsule

Must-Know

- Pelvic support → pubis/ASIS anteriorly, sacrum posteriorly
- **Gluteus minimus** preserved throughout
- Anterior capsular resection → avoids reaming in retroversion
- Posterior repair → gluteus medius tendon, non-absorbable sutures

Direct Anterior Approach DAA

(Patient Selection, Technique and Learning Curve of the Anterior Approach)

Prof Dr Sherif Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 156) “abduction first” → “adduction”

C2 (p. 159) “2 crutches for 2 months” → “2 crutches for 2 weeks”

Queried - kept as spoken, but looks wrong

Q1 (p. 152) As spoken; learning curve of 5-10 years conflicts with 'first 100 cases' and 'first two or three years'.

Q2 (p. 158) As spoken; standard posterior stability test is flexion + internal rotation + adduction, not abduction.

Intervals of the Direct Anterior Approach Ask AI

- Second lecture of the DAA series
- Two intervals:

Interval	Between	And
Superficial	Sartorius	Tensor fasciae latae (TFL)
Deep	Rectus femoris	Tensor fasciae latae (TFL)

- Remaining steps → as in other approaches

Patient Selection Ask AI

- DAA → applicable to all patients
  - Some patients better suited → surgeons who use it say so
  - Some surgeons build a whole patient-selection programme around it
- **Ideal patient:**
  - **Primary hip** → not used in revisions
  - BMI < 30
  - Small-built → i.e. female
  - Non-osteoporotic bone
  - Preferably **coxa valga** / valgus neck
- **Learning curve** → operate only on such cases:
  - First five to ten years *[Q1 · As spoken; learning curve of 5-10 years conflicts with 'first 100 cases' and 'first two or three years'.]*
  - Or first 100 cases → in females
- Reason → muscles weaker in females
- Trained group: the lecturer, Ahmed Hazem, Dr. Fouad, Mahmoud Abdel-Karim
- First two or three years → stay with favourable cases

High Yield
- Ideal DAA patient: primary hip, BMI < 30, small-built female
- Non-osteoporotic bone; coxa valga / valgus neck preferred
- Not used in revisions
- Learning curve: favourable cases only, first 100 cases

Concept and Advantages Ask AI

- **Minimal incision** (7 cm, even 2 cm) → not the idea at all
- Core idea → **intermuscular / internervous planes**
  - Nothing cut → surgeon goes between muscles
  - Only **anterior capsule** opened
  - Posterior structures respected:
    1. Posterior capsule

        - Short external rotators
        - Sciatic nerve
        - Glutei
- Therefore:
    - **Dislocation** → much lower, almost zero
    - **Trendelenburg gait** → zero
    - **Heterotopic ossification** → almost zero
    - **Postoperative restrictions** → none

Disadvantages and Requirements Ask AI

- Disadvantages → special instruments + slow learning curve
- Ordinary instruments → DAA impossible; dedicated set mandatory
    - **Offset reamers** / **offset inserters** → not essential; simple one acceptable
    - Without the instrument set → nothing can be done
    - **Six retractors** → important; work still difficult with them
- Ordinary instruments → acetabulum perhaps manageable, femur not
    - **Technically demanding** approach
    - Demanding **learning curve**
- **Special table**?
    - Previously avoided → believed to need special table
    - Team learned DAA in Belgium 4 years ago from operating surgeons

Positioning, Draping and Skin Incision Ask AI

Positioning Ask AI

- **Ordinary table** only
    - Support placed → supports the patient
    - Leg: one third above table break, two thirds below
        - Reason → leg goes into **figure-of-4**
    - Drawing: obese patient; leg on a table; patient near table edge

Draping Ask AI

- Some centres → special drapes for ease
- Here → ordinary lateral draping, drapes side to side
- Assistant on one side → leg on the other

Skin Incision Ask AI

- Landmark → ASIS (as shown)
    - Start → three fingerbreadths distal + three fingerbreadths lateral
- Some Belgian surgeons → **bikini incision**
- Team's **classic incision** → **extensile**, extendable distally
- Second option → **transverse** incision
    - Experienced surgeons only
    - Team: not yet after 4 years; not before 10 years

- Incision oblique, about 8 cm

Superficial and Deep Dissection Ask AI

Superficial Dissection Ask AI

- Skin + subcutaneous tissue → cautery dissection → **TFL**
- TFL appearance → as on slide
- Muscle visible through fascia
  - **Fascia opened in the middle** (over muscle) → away from the lateral femoral cutaneous nerve
  - Fascial flap held
- Medial (front) flap → finger along TFL → falls into **TFL–sartorius interval**
- Interval used → **two retractors,** above and below femoral neck

Neck Anatomy and Retractor Placement Ask AI

- **Femoral neck fracture** → no neck → retractor unusable
  - Therefore → femoral neck fracture **not an indication**
  - Hence valgus neck / long neck preferred → retractors usable
- Short neck / **protrusio:**
  - Protrusio → practically impossible
    - Team's protrusio cases → very, very difficult
    - First or second retractor cannot be placed → do not do it
  - Look for neck valgus

Pitfall

- Femoral neck fracture → no neck → retractors cannot be placed
- Femoral neck fracture is not an indication for DAA
- Protrusio / short neck → practically impossible; avoid

Deep Dissection Ask AI

- Deep fascia opened with scissors → **yellow fat** beneath
- Cautery dissection → **anterior aspect of acetabulum**
- **Iliocapsularis** → attached over anterior acetabulum
  - Its fascia opened → **curved retractor** beneath it
  - Pushed in one direction only (as shown)
  - Effect → lifts iliocapsularis → exposes anterior acetabulum
  - Also → protects **femoral nerve**
    - Femoral nerve lies superficial to iliocapsularis → no risk

Capsule Ask AI

- **Capsulectomy** → as in posterior approach (Dr. Hazem)
- As in lateral approach → capsulotomy or capsulectomy
  - Some → **H-shaped** capsulotomy
  - Some repair capsule; others do not
    - Non-repair rationale → **anterior capsule** has no importance

Neck Osteotomy and Head Removal Ask AI

- **Neck osteotomy** → single osteotomy possible
- Difficult cases → **double osteotomy**
  - Middle fragment removed (corkscrew) → head comes out easily
  - Sometimes too small to extract head as in lateral / posterior approaches
- Head removed → **corkscrew**
- Next retractor → toward the obturator

Acetabular Preparation Ask AI

Exposure Ask AI

Acetabulum

- **Three Hohmann retractors** → needed for acetabular view:
  - Inferior
  - Posterior wall
  - Anterior wall
- Viewing in this direction → very accurate

Cup Orientation Ask AI

Acetabulum

- Cup orientation:

Parameter	Reference	Angle
Anteversion	From the floor	15°
Inclination	Handle opened from midline	40°

Reaming Ask AI

Acetabulum

- **Straight vs curved reamer** → no difference
  - Straight reamer photographed → shows it is not very important
  - Importance of **canal finder** + straight reamer → seen later
- Acetabular component inserted

Fixation Philosophy Ask AI

- All surgeons observed abroad → **cementless,** not cemented
- Team's cemented cases → more difficult
  - Cementing the cup → no problem
  - **Cementing the femur** → difficult
- **Cementless cup and stem** → reduces operative time
  - Reason → **fast-track surgery**
  - Discharge → same day or next day
  - Cementless → easier
  - Cementless → faster

Femoral Exposure and Preparation Ask AI

Exposure Ask AI

Femur

- **Femoral side** → the difficult part
- Trick: adduction (stated as abduction) → Hohmann **lateral to greater trochanter** *[C1 · transcript: "abduction first"]*
- **Bare area:**
  - Gluteus medius attachment → not at tip; 1.5 cm lateral to tip
  - Area between tip and gluteus medius attachment → **bare area**
  - Retractors placed here
  - Very small area → bone cannot be **osteoporotic**
  - Team has fractured this area before
  - Some say area has no value (tendons independent) → still try hard to preserve it
- All retractors placed → small capsular structures released
- **Hook** on femur → pulled to confirm femur comes out
- Reason → **greater trochanter** lies inside, behind acetabular rim
  - Supine patient → femur must be brought out from under acetabulum → very difficult
  - Hook required → another reason osteoporosis excluded from indications
- Femur pulled outward → structures **released**
  - Until hook lifts femur a sufficient distance
  - Before this → do not try; impossible or very difficult
- Femur up → **figure-of-4:**
  - Healthy leg lifted
  - Operated leg in figure-of-4 beneath it → rested on table side

Exam Pearl

- Bare area: between greater trochanter tip and gluteus medius attachment
- Gluteus medius inserts 1.5 cm lateral to the tip
- Osteoporotic bone → bare area fracture risk

Preparation Ask AI

Femur

- Femur rasped → **offset rasp**, a little easier
- Retractors:
  - **Hohmann** → bare area, between trochanter and gluteus medius
  - **Medial retractor** → on the **calcar**
  - Leg → figure-of-4
- **Box osteotome** → **canal finder**
  - Without it → canal location never predictable
  - Canal → very strange place, very strange direction
  - Canal finder inserted → shows the way → canal followed

Calcar and Releases Ask AI

Femur

- Cut high → **calcar planed (reamed);** flat calcar → cut precise
- Release small capsular structures:
  - Superior → release

  - Inferior → release
  - **Posterior** → do not release
    - Releasing them → compromises stabilizers + **posterior capsule**
- Posterior capsule released → approach loses its benefit

Must-Know
- Never release the posterior structures
- Posterior capsule released → DAA loses its benefit

Trial Reduction and Intraoperative Assessment Ask AI

Trial Reduction Ask AI

- **Trial reduction** → traction + internal rotation → reduces very easily
  - Force needed → problem: do not force; go back and find it
  - Reduction should be very easy

Impingement and Telescoping Ask AI

- Impingement checked first; then telescoping:

Test	Manoeuvre	Checks
Impingement	Internal rotation	**Anterior impingement**
Impingement	External rotation	**Posterior impingement**
Telescoping	Distal (as usual)	**Vertical offset**
Telescoping	Lateral	**Lateral offset**

- Femur comes laterally → **offset stem** needed
  - Extended-offset / lateralized (high-offset) stem

Limb Length Discrepancy Ask AI

- **LLD assessment** → very easy
- Reason → patient supine, both legs in front of surgeon
  - **Medial malleoli** → equal length?
  - **Galeazzi test** → hips $45°$ + knees $45°$ flexed → compare in flexion

Combined Version Ask AI

- Check → **cup parallel to head (neck) cut**
- Parallel → **combined version** excellent
  - Cup inclination → very accurate
  - **Combined anteversion** → ideal

Clinical Pearl
- Trial reduction: traction + internal rotation → very easy
- Force needed → stop and look for the problem
- Cup parallel to neck cut → ideal combined anteversion

Stability Testing Ask AI

- Stability tests:
  - External rotation + adduction
  - External rotation + abduction
  - External rotation + abduction + extension
  - Flexion + internal rotation + abduction *[Q2 · As spoken; standard posterior stability test is flexion + internal rotation + adduction, not abduction.]*
- Sequence and reverse:
  - Extension + external rotation → test adduction, abduction, extension
  - Leg flexed → internal rotation + abduction
  - Then the reverse

Final Limb Length and Offset Ask AI

- Finally → limb length checked; in front of surgeon (supine)
  - Advantage of DAA → checking limb length
  - And offset

Approach	Limb length method	Ease
Posterior / lateral	**Schanz pin** in pelvis + **sterile ruler**	Harder
Posterior / lateral	Mark on greater trochanter (stitch) → measured repeatedly	Harder
DAA	Limb length seen directly + **Galeazzi** test	Much easier

- Posterior / lateral example → lengthen 2 cm: distance 9 cm → must end 11 cm

Closure and Postoperative Care Ask AI

Closure Ask AI

- **Tranexamic acid** 1 g → before starting closure
- TFL fascia → closed **continuously**
  - Then skin + subcutaneous tissue
- **No drain**
  - Exceptions → muscular patients, bleeding

No Restrictions Ask AI

- **Postoperatively** → no restrictions
  - Patient can pray, flex, put on shoes, turn in bed
- Clinical example → 58-year-old female
  - **Alzheimer's disease / minimal cognitive impairment**
    - Minimal cognitive impairment → term used in younger patients

Illustrative Case Ask AI

- **Femoral neck fracture**; patient is a doctor
  - Suggestion → give her precautions; concern → poor compliance
  - Team → arthroplasty + precautions against dislocation

- Sleeps **curled up (baby position)** since operation day
- **Never dislocates** → the idea of DAA
  - Suits patients needing it → **no restrictions;** hip will not dislocate
  - Reason → intact stabilizers:
    1. Posterior capsule
    2. Rotators
    3. Glutei
- Only **anterior capsule** opened → almost no value in stability

Rehabilitation Ask AI

- **No physiotherapy** needed
- **2 crutches** for 2 weeks (stated as months) → **1 crutch** for a period → then off *[C2 · transcript: "2 crutches for 2 months"]*

Complication: Femoral Nerve Neurapraxia Ask AI

- One case of **femoral nerve neurapraxia** → one year to recover
  - Very serious complication; happened to the team
  - Patient profile:
    1. Short neck + **protrusio**
    2. About 38 years old
    3. Very obese
  - Nerve conduction study → all wrong; all directions reversed
  - Recovered; no other complications
- More severe (permanent) injury → would not have been easy

Pitfall

- Femoral nerve neurapraxia → one year to recover
- Occurred in short neck + protrusio + very obese patient
- These features → outside DAA indications

Take-Home Message Ask AI

- **Intermuscular and internervous** → natural interval around the hip
- **Ordinary table** → no special table required
- Lower dislocation rates + faster recovery
- **No restrictions** on movement → unlike other approaches
- Patient → very satisfied

Extended Trochanteric Osteotomy ETO

(Indications, Types and Technique of Trochanteric Osteotomies)

Prof Dr Mahmoud Abdel Karim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Introduction and Philosophy Ask AI

- Core question → femur decided to be opened → how to open it
- Aim → know how to open such a case
- **MIS** (minimally invasive surgery) → intention to do a good job
  - Not only the technique or incision used
- Revision goal → adequate exposure → facilitates the operation
- Speaker learning the **anterior approach** with Dr. Sherif
  - Not yet used in a single private patient → reflects current opinion
  - Use for own patients → only after confidence + sufficient cases

Indications of Trochanteric Osteotomy Ask AI

- Improved exposure → **extensile approach**
- Removal of **well-fixed cement**
- Removal of a well-fixed **femoral component**
- Difficult hip, even in primary settings
  - Old ankylosis / fusion
  - Cluster of adhesions
  - **Heterotopic ossification (HO)**
- **Trochanteric malposition**

Types of Trochanteric Osteotomy Ask AI

- Covered previously; beyond the scope of this lecture

Type	Cut	Muscle carried
Standard (A, B in figure)	Trochanteric	Gluteus medius only
Sliding	Slightly lower	Gluteus medius + vastus lateralis

- Topic of this lecture → **extended trochanteric osteotomy (ETO)**
- Dr. Hazem → introduced its characteristics and situations needing it
- Approach (figure) → **posterior** or **lateral**

Extended Trochanteric Osteotomy: Definition and Indications Ask AI

Definition Ask AI

ETO

- Osteotomy of the **lateral one third of the femur** → removes that cortical layer
  - Nickname → **"coffin lid"**
- Opens the femur → straight access to the canal
- >90% of slides → writings of **Wayne Paprosky**
- His material at **ICJR** meetings → makes the procedure much easier

Indications Ask AI

ETO

- Removal of **well-fixed cement**
  - Long cement mantle not otherwise removable, e.g. infection / sepsis
- Reasonably **well-fixed cementless stem** → ETO sometimes required
  - Sometimes combined with cement removal
- Well-fixed cement + **femoral varus deformity** → ETO → **neutral reaming**
  - Reaming in varus / valgus (usually varus) → femoral **perforation**
- **Correction of proximal femoral deformity**
- Removal of **intramedullary hardware**, e.g. a nail
  - Paprosky case → shown later
- Improves **acetabular exposure**
- Allows **abductor tensioning**

Current Indications by Frequency Ask AI

ETO

- Indications above → **complex primary** setting
- Paprosky (2018) ranking:

Rank	Indication
1 — most common	**Infection** (speaker has done such a case)
2	**Periprosthetic fracture**, Vancouver type B2
3	**Femoral loosening**
4 — least common	**Femoral varus deformity**

Exam Pearl

- Most common current ETO indication → **infection**
- Second → periprosthetic fracture, Vancouver B2
- Least common → femoral varus deformity

Contraindications and Advantages Ask AI

Contraindications Ask AI

ETO

- **Cemented femoral revision** → cement may extrude into osteotomy site → fixation difficult
  - Practice declining → used only if no other solution
  - Almost **a lost art apart from Exeter**; uncertain if still done on femoral side
- **Non-union rate** → concern for some authors; arguable either way

Advantages Ask AI

ETO

- **Quite versatile**
- Easy once a few done → learn by attending someone who performs it
- Decreased operative time; decreased complications
- **Cement removal** → dangerous stage, even in the best hands
  - Use **image intensifier**; be careful
  - ETO → makes it much easier

- Improves **acetabular exposure** (view shown)
- Starts as a window → not merely a window → **extensile exposure**
- Decreased operative time, **blood loss**, risk of complications
- Decreased risk of a second problem → perforation / fracture
- Addresses **abductor tensioning**

Preoperative Planning Ask AI

- **Accurate templating** → very, very important
- Common errors (all stages of training):
  - Osteotomy too short or too long
  - No room left for the stem distally
  - Proximal fracture
  - Failure to remove the implant
- **Preoperative templating** → determines **osteotomy location** → improves exposure
- Length must allow:
  - Component extraction
  - Adequate fixation of planned components
- **Femoral deformity** → osteotomy at the **apex** if correcting it

Osteotomy Length Ask AI

ETO

- No fixed number
- Minimum 10 cm from the **greater trochanter tip**
- Minimum 4–6 cm **diaphyseal fixation** of planned prosthesis
- Check available sizes with the company
- In practice ~12 to 15 or 16 cm → varies → derive from **templating** and planning

Must-Know

- ETO length → at least 10 cm from greater trochanter tip
- Distal diaphyseal fixation → at least 4–6 cm
- Typical length → 12 to 15 or 16 cm, from templating
- Deformity correction → osteotomy at the apex

Intraoperative Sequence and Backup Ask AI

- Stem outside planned limits → do not proceed without **backup**
  - Different implants available
- Perform ETO **before dislocation** (Dr. Tarek)
  - In this case mandatory → dislocation → femoral fracture
- Small / osteoporotic bone → shortening or fracture may be more likely
  - May still be the only option; in some cases the only solution
- Stem removed proximally → loose cement or long mantle → remove **distal cement**
- Aim → exposure

Approach Ask AI

- Speaker prefers the **posterior approach** → not difficult
- Must also master posterior approach extensions (possibly **Kocher-Langenbeck**) and related exposures
- Skill acquired with time → all must learn it
- Anterior approach users → must be well trained to **convert** if needed
- One very experienced surgeon's practice ≠ applicable to all
- Posterior approach → versatile, good exposure
- ETO also via **lateral** or anterolateral approach → Daniel, Barry, other groups
- Speaker's preferred technique → posterior approach

Surgical Technique Ask AI

Exposure Ask AI ETO

- Preserve the **musculotendinous sleeve** (Dr. Mohamed Abou El-Saoud)
  - Above → gluteus medius
  - Below → vastus lateralis
- Use a **simple retractor**
- No complete stripping to bare bone (Dr. Hazem) → "it's not an art"
- **Minimal stripping** ~2 cm → **Hohmann retractor** → start osteotomy

Measuring and Marking Ask AI ETO

- Measure **according to the template**
- ~12 to 15 or 16 cm → always measure
- Mark → **methylene blue**, diathermy, or anything else

Posterior Longitudinal Cut Ask AI ETO

- Posterior longitudinal limb → **saw**
  - Saw from the start → if experienced
  - Alternative → **drill holes** first → prevents saw **deflection** → fracture
  - Multiple drill holes; add one at the corner → avoids **fragmentation** / comminution
- Connect the drill holes
- Corners → must be **rounded**
- Sharp / square corner → **fracture**; quite common
- Abroad → **pencil burr** rounds the corner
- Alternative → osteotome or drill holes → round as much as possible

Anterior Limb Ask AI ETO

- Two options (Dr. Tarek with Dr. Hazem):

Option	Method	Caution
Complete anteriorly	Saw, **pencil burr**, etc.	**Anterior vessels**
Controlled fracture	Simple **osteoclasis**	—

- Sequence → complete **longitudinal posterior portion** → anterior part with saw from the front

- Alternative (Dr. Tarek) → move the limb → same step
  - Especially **osteoporotic bone** → cannot take much → may fracture
- Speaker's preference → osteotome or drill hole → **curve** the cut proximally → drill holes
- Connect them, even from the front
- **Vastus lateralis** in front → do not **devascularize** the fragment
  - Minimal handling; no tissue stripping

Levering the Fragment Ask AI

ETO

- Insert multiple **osteotomes** → 2 or 3
- Very gentle; do not rush; not aggressive
- Before levering → release **anterior pseudocapsule** → 100%
  - Not released → fragment fractures → pieces retained
- Final product delivered → existing stem exposed

Pitfall

- Sharp / square corner → fracture → round it
- Saw without drill holes → deflection → fracture
- Anterior saw cut → anterior vessels at risk
- Stripping vastus lateralis → devascularized fragment
- Unreleased anterior pseudocapsule → fragment fracture

Stem Extraction Ask AI

ETO

- Remove the **stem** → different instruments and steps, incl. **Gigli saw**
- Instruments often unavailable locally → learn tricks
- Removal not guaranteed
- Different instruments → flexible / back-cutting **osteotomes** (instrument wish-list slide)
- Train within own institution → develop beyond available instruments

Prophylactic Cerclage and Fixation Ask AI

Prophylactic Distal Cerclage Ask AI

- **Cerclage distally** → before anything else → avoids regret
- Commonly forgotten → reaming proceeds
  - Even in a primary using a **long stem** for any reason
- Missing cerclage → split → **fracture**
- Most important point of the lecture

High Yield

- **Prophylactic cerclage** / cable distally → before reaming
- Applies also to primaries with a long stem
- Omitted → femur splits → fracture

- Then **broaching**, etc.
- **Advancement** → remove a piece distally → described long ago
  - Indications decreased a lot; trick worth mentioning

Fixation of the Osteotomy Ask AI ETO

- **Cables** and wires → at least 2 to 3
- No cables / limited budget (lesson from Dr. Maher):
  - **Double-wire cerclage**
  - Prophylactic cable distally → very important
- Sequence → prophylactic cerclage first → 2 to 3 more proximally, per case

Figure-of-Eight Trochanteric Cerclage Ask AI GT

- Trick published in **JBJS American 2003**
- Often needed → **greater trochanter cerclage** → figure-of-eight technique
  - **Transverse cerclage** → passed **under the muscle**
  - Cerclage over the muscle → muscle necrosis
- Wire under muscle → up over the **trochanter** → vertical limb
- New configuration → **excellent compression** + stability
- Very important; appearance shown

Complications Ask AI

- Often not discussed
- Each complication → how to avoid it
- Tight **anterior pseudocapsule** → must release it
- **Proximal migration** may occur
  - Standard osteotomy → only gluteus medius attached
  - **Sliding osteotomy** → vastus lateralis also attached → **tethering effect** → prevents migration
  - Still might occur in some cases

Case Examples Ask AI

Case 1 — Infected THA (Speaker) Ask AI

- Presented **infected** → stem removal uncertain → needed ETO
- Preoperative templating advised; intraoperative **image** always obtained (instrument / marker) to confirm
  - Especially with a long stem
- **Heavily infected on acetabular side**
- **Femoral component** → quite well fixed (seen after opening)

> **Clinical Pearl**
> - Templating → plus intraoperative image to confirm length
> - Especially with a long stem

- Finished product; **intraoperative** view → no problem anywhere
- **Triple cerclage**
- Double wires if cables unavailable → **cables** very expensive

Case 2 — Intramedullary Implant (Paprosky) Ask AI

- **Wayne Paprosky's** case, not the speaker's
- ETO → very useful, quite valid → **intramedullary implant** extraction

Case 3 — Childhood Deformity Ask AI

- **Deformity** from early childhood
- Procedure → partial ETO + **completed medial osteotomy** → correction
- Paper → nearly six years of **follow-up**

Approaches: Comparative Evidence Ask AI

- Approaches only → evidence piece left to Mohamed
- 2015 paper → compared the approaches discussed
- Scoring → plus / minus or number of pluses → **efficacy** + superiority of results

Domain	Components scored
Acetabular exposure	Anterior, superior, posterior
Femoral exposure	Metaphysis, canal, lateral cortex

- **Trans-trochanteric** approaches → 3 pluses in everything
- Next → **posterior approach**

Further Reading and Conclusion Ask AI

Further Reading Ask AI

ETO

- Details + further reading → shared for whoever wants

Resource	Type	Authors	Source
"Extended Trochanteric Osteotomy in Complex Primary Total Hip Arthroplasty"	Paper — very important	Craig Della Valle, Wayne Paprosky	**JBJS American 2003**
"Surgical Treatment of Hip Arthritis", chapter 37	Book — very good	Paprosky	—
"Extended Trochanteric Osteotomy"	**Campbell's video**	Crockarell, Gregory Dabov	YouTube
"Extended Trochanteric Osteotomy in Complex Total Hip Arthroplasty"	Video interview	**Craig Della Valle** (works with Paprosky)	ICJR website

- Campbell's video → watched by speaker in 2003 as assistant lecturer
- Both videos → everyone advised to watch

Conclusion Ask AI

- ETO → very **versatile** concept
- Very good results → if **adequately planned** + small details followed

- **Preoperative templating** → very, very, very important
- **Instruments** and backup → manage a lost item or major problem

Choice of the approach

(Comparing Posterior, Lateral and Anterior Approaches and Choosing Between Them)

Prof Dr Mohamed Abo Elsoud

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 177) As spoken; lateral approach lowers but does not abolish dislocation or femoral fracture; absolute claim overstated.

Introduction: Determinants of Approach Choice Ask AI

- Original title **Choice of the Approach** → reworked at course organization
  - By Dr. Sherif Khaled and Dr. Mohamed Gabba
  - New title: **Literature and Registry Data Review**
- Difficult topic → careful assimilation of evidence required
- Generally held view, summarizable in ~five slides
  - Approach choice in **total hip arthroplasty (THA)** → highly controversial
  - THA highly successful → **regardless of surgical approach**
  - → Choice largely surgeon opinion and preference
- Determinants of choice
  - **Surgeon preference** → main determinant
  - Dislocation risk
  - Obesity
  - Previous incisions
  - Previously implanted hardware
- Each surgeon → favored approach → defends it

Classical Advantages and Disadvantages of Each Approach Ask AI

Posterior (Posterolateral) Approach Ask AI

Posterior

- Advantages
  - Easy
  - Not bloody
  - Very quick
  - **Abductor mechanism** not violated
  - **Extensile** → good femoral and acetabular exposure
    - → Extendable for revision surgery
- Disadvantage → **high dislocation rate**

Direct Lateral Approach Ask AI

Lateral

- Advantages
  - Easy
  - Excellent **acetabulum** exposure
  - Dislocation rate → lower than posterior
- Main disadvantage → **postoperative limp**, Trendelenburg gait

Direct Anterior Approach (DAA) Ask AI

DAA

- Newer approach → "younger sibling" of lateral approach
  - Described as modification of lateral approach
  - Enters hip from front
- Main advantage → **less muscle damage** → faster recovery
- Main disadvantage → **technically difficult** → more problems
- Associated **neuropraxia** → lateral femoral cutaneous nerve

Classical View at a Glance Ask AI

Approach	Advantages	Disadvantage
Posterior	Easy, not bloody, quick, abductors intact, extensile	High dislocation rate
Direct lateral	Easy, acetabular exposure, dislocation < posterior	Limp, Trendelenburg gait
Direct anterior	Less muscle damage → faster recovery	Technically difficult, LFCN neuropraxia

- Open question → evidence behind each claim
  - Correct, incorrect, or weakly supported?

Patient-Reported Outcomes Ask AI

- **Patient-reported outcomes (PROs)** → patient satisfaction with operation

JBJS 2014 — Total Hip Replacement Ask AI

THA

- Journal of Bone and Joint Surgery (JBJS), 2014
- Slightly better PROs → **posterior approach**
- Attributed to **abductor weakness** with lateral approach

JBJS 2019 — Hemiarthroplasty Ask AI

Hemi

- Fracture patients, elderly > 75 years
- Approach → **no apparent effect** on outcome
  - Elderly satisfied with hemiarthroplasty itself, regardless of approach
- Contrast: THA → posterior slightly better for PROs

Functional Recovery: Direct Anterior vs Posterior Ask AI

CORR 2018 RCT — 101 Patients Ask AI

DAA

- **Randomized controlled trial (RCT)**, Clinical Orthopaedics and Related Research (CORR)
- Smaller than larger series discussed before
  - ~2,000 and 42,000 patients → likely Swedish registry
- Comparison → **mini-posterior approach** vs anterior approach
- Findings
  - Slightly faster early recovery with DAA → up to first 2 months
  - PROs, radiographic parameters, complications → equal

Journal of Arthroplasty 2017 RCT — 70 Patients Ask AI

DAA

- DAA vs posterior approach
- Outcome scores → no difference
- **Functional recovery** better with DAA
  - Putting on socks
  - Tying shoes
- **Hip flexion activities** better with DAA → at 6 weeks

- Flexion strength, by contrast
  - **Hip flexors weaker** with DAA → dissection close to them
  - Flexion activities still better → up to 6 weeks

Meta-analysis 2018 — ~1,000 Patients Ask AI

DAA

- **Systematic review and meta-analysis,** Journal of Arthroplasty
- DAA patients
  - Fewer **narcotics**
  - Less pain
  - Better functional outcome
- Explanation → accompanying RCTs
  - Lower **CRP levels**
  - MRI and (probably cadaveric) studies → less **muscle edema** with DAA
    1. Less muscle affection
    2. Less tissue damage

Meta-analysis 2015 Ask AI

DAA

- Second systematic review and meta-analysis
- Better **functional rehabilitation** → DAA
- Cost → much **longer surgical time** with DAA

Gait and Trendelenburg Gait Ask AI

- Question: does direct lateral approach truly produce **Trendelenburg gait**?

Cochrane Systematic Review 2010 Ask AI

Gait

- **Cochrane Database**; large number of patients (exact number not recalled)
- Trendelenburg gait, lateral vs posterior → **not significantly different**
- Dislocation rate, posterior vs lateral → not different
- Authors → **data insufficient** → no firm conclusion
  - Trendelenburg gait unsettled either way

Gait Analysis RCT 2018 — 30 Patients Ask AI

Gait

- 10 anterior, 10 posterior, 10 lateral
- Lateral approach → higher **pelvic tilt**
  - Pelvis drops on opposite side + **ipsilateral trunk lean** → Trendelenburg gait
- Reported as affecting patients' outcomes (hedged)
- Measured → first 6–12 weeks

Smartphone Accelerometer RCT 2020 Ask AI

Gait

- **Smartphone accelerometer**
  - Carried while walking → counts steps, records gait
- RCT, ~70 patients, DAA vs posterior
  - **Very minimal differences** → in favor of DAA
  - Limited → first month

Exam Pearl

- DAA functional advantage → early only: 2 months, 6 weeks, 1 month
- Hip flexors weaker after DAA, yet flexion activities better to 6 weeks
- Cochrane 2010 → Trendelenburg gait lateral vs posterior not significantly different
- Gait analysis 2018 → lateral approach: pelvic drop, ipsilateral trunk lean

Dislocation Ask AI

- Key question → dislocation higher or lower → which approach better?

Acta Orthopaedica 2014 — Hip Fracture Ask AI

Fracture

- Hip fracture patients > 75 years
- Posterior approach → markedly higher **risk of re-operation due to dislocation**
- Fracture patients → PROs good and similar across approaches
  - Yet posterior → much higher re-operation risk → dislocation
- Supporting papers → 2016; 2008 (Acta Orthopaedica)

Lateral vs Posterior and Augmented Repair Ask AI

Posterior

- JBJS paper → **direct lateral approach**
  - → Dislocation risk lower than posterior
- Second paper, Acta Orthopaedica → **augmented posterior repair**
  - → Decreases dislocation rate/risk
  - Still does not reach anterolateral or lateral approach rates

Swedish Hip Register 2019 — 156,000 Patients Ask AI

Posterior

- Two periods → 1999–2006 and 2007–2014
- Endpoints
  - Risk of re-operation due to dislocation
  - Risk of **all-cause re-operation**
- 1999–2006 → dislocations higher with posterior approach
- 2007–2014 → dislocation risk similar, lateral vs posterior
  - Attributed → **augmented posterior repair**
- Posterior approach → lower risk of all-cause re-operation
  - Dislocation set aside, all causes included

2016 Study — Posterior vs DAA Ask AI

DAA

- Dislocation risk, posterior vs DAA → probably **no difference**
  - Probable trade-off → more femoral complications with DAA

16,000-Patient Comparison Ask AI

Dislocation

- Posterior vs lateral vs DAA
- Posterior dislocation rate → 10 times **lateral approach**
- Posterior dislocation rate → 2 times **DAA**
- Anterior approaches → increased **femoral failure** risk (femoral fractures)
- Conclusion → **lateral approach safest**
  - No femoral fractures, no dislocations *[Q1 · As spoken; lateral approach lowers but does not abolish dislocation or femoral fracture; absolute claim overstated.]*

- → Lowest rate of mechanical complications

Dislocation Evidence at a Glance Ask AI

Study	Population / comparison	Dislocation finding
Acta Orthop 2014	Hip fracture > 75 years	Posterior → higher re-operation for dislocation
JBJS	Lateral vs posterior	Lateral lower
Acta Orthop	Augmented posterior repair	Lower, still > anterolateral/lateral
Swedish Hip Register 2019	156,000; 1999–2006	Posterior higher
Swedish Hip Register 2019	156,000; 2007–2014	Lateral = posterior
2016 study	Posterior vs DAA	Probably no difference
16,000 patients	Posterior vs lateral vs DAA	Posterior 10× lateral, 2× DAA

Must-Know

- Elderly hip fracture → posterior approach → higher re-operation for dislocation
- Augmented posterior repair → lowers posterior dislocation risk
- Swedish 2007–2014 → dislocation risk lateral = posterior
- Posterior approach → lower all-cause re-operation (156,000 patients)
- Anterior approaches → increased femoral failure risk

Other Complications Ask AI

- Beyond dislocation → rates of other complications?

Journal of Arthroplasty 2019 — Three Groups Ask AI

Lateral

- ~2,000 patients
- Lateral group
  - Longer **surgical time**
  - More general anesthesia
  - Longer hospital stay
  - Surprising → expected less
- Major and minor complications → **no difference** among 3 groups
  - Lateral, posterior, DAA

RCT 2017 — 70 Patients Ask AI

DAA

- RCT, 70 patients
- Same findings
- **Minor adverse events** slightly higher → DAA
  - Cause → neuropraxia of **lateral femoral cutaneous nerve** (lateral cutaneous nerve of thigh)
  - Very **transient** → ~12 weeks

Traction Table and Femoral Fracture 2017 Ask AI

DAA

- ~500 patients; **traction table** use vs non-use
- Endpoint → incidence of **femoral fractures** with DAA
- Traction table vs none → no difference
- Fractures in both groups → low rate, ~2.5%
- Fracture vs no-fracture patients → no difference, including **gender**
- Fracture patients → very small proportion

Readmission Ask AI

- **Journal of Orthopaedics**, 2020 → readmission after DAA at 30 and 90 days
- Overall readmission rates → low
- **Obese patients** → higher risk of early wound complications → readmission
  - Superficial wound problem
  - **Hematoma formation**
  - Similar issues

Long-Term Implant Survival and Revision Rates Ask AI

- Question → long-term **implant survival** better with any approach?

Norwegian Registry Thesis 2008 Ask AI

Survival

- **Doctoral (PhD) thesis**, Norway, Norwegian Registry
- Lateral approach + trochanteric osteotomy → **best long-term survival**
  - i.e., **digastric trochanteric osteotomy** (explained by Dr. Hazem)
  - Probably → implant positioned very well
- Posterior approach → **highest revision rate** → dislocation
- Trochanteric osteotomy vs none → no difference in **abductor strength**
- Abductor strength improves up to 12 months
  - Marked improvement → first 6 months
  - Continued improvement → 6–12 months
  - → **Trendelenburg gait** improvement expected up to one year

Swedish Registry 2019 — 38,000 Patients Ask AI

Survival

- Posterior vs lateral → **no difference**
  - Reoperation rate
  - Implant survival

Swedish Registry 2017 — Mini Approaches Ask AI

Mini

- **Mini approaches** → increased risk
  - More mini approaches → higher reoperation rate
  - → Lower implant survival
- Probable cause → implant not in optimal position
  - Less-exposed (mini / direct anterior) approaches → lower survival

Swedish vs American vs Australian Registries Ask AI

Registries

	Sweden	United States / Australia
Commonest approach	Posterior (probable)	Posterior (probable)
DAA use	Essentially none	A proportion of surgeons
Revision rates	Better	Worse
5- and 10-year implant survival	Better	Worse

- **Observational** finding only
- Authors → possibly related to non-use of DAA in Sweden

Learning Curve of the Direct Anterior Approach Ask AI

- Practical question → switch from lateral/posterior to DAA?
  - Motive → early recovery, current trends
- Instructive paper → highly experienced **posterior approach** surgeon switched to DAA
  - First 1,000 cases reviewed
- Published 2018
- Formal **training** + fellowship → proficiency → then 1,000 DAA cases
  - Comparable to Dr. Ahmed Samir
- **Total OR time** (setup + surgery) → 30% initial increase
- **Procedure time** (surgery only) → equal to posterior
  - → After first 400 cases
- Operative time shorter than posterior → after ~150 further cases (hedged)
- Total OR time equal to posterior → 900 cases
- **Early femoral fractures**
  - Femoral fractures and femoral complications → higher in early learning curve
- → **Very steep learning curve** (1,000 cases)

Learning-Curve Milestones Ask AI

DAA

Milestone	Cases
Total OR time, initial phase	30% higher
Procedure time = posterior	First 400
Operative time < posterior	~150 further (hedged)
Total OR time = posterior	900
Early femoral fractures / complications	Higher in early curve

Pitfall

- DAA → very steep learning curve, even after fellowship
- Procedure time matches posterior only after 400 cases
- Total OR time matches posterior only after 900 cases
- Femoral fractures higher early in learning curve

Revision Surgery Ask AI

- **Extended trochanteric osteotomies** → revision surgery (also discussed by Dr. Hazem)
- **Review article** (not original research), 2015
- Revision → may need entirely different approach
  - Careful **preoperative planning** (Dr. Mahmoud Abdel Karim)
- Options beyond posterior or lateral
  - Possibly direct exposures
  - **Digastric trochanteric osteotomy** approach
  - **Extended trochanteric osteotomy** approach
- Author → trochanteric osteotomy (and possibly direct) approaches best
  - → Allow all problems to be solved

Summary: Take-Home Message Ask AI

- **THA** → successful, largely unrelated to approach
  - Most cases → over 95% satisfied
  - Provided everything else done correctly
  - Surgeon uses approach he performs well
- Settled points
  - Elderly **hip fracture** → direct lateral approach → best results
    - Genuinely reduces dislocation rate
  - Posterior → careful repair of **capsule and short external rotators**
    - → Dislocation rate greatly reduced
    - Some authors → rate similar to direct lateral
- **Trendelenburg gait** → still a problem
  - Not definitively settled

  - Certainly more frequent with lateral approach
- DAA → **very long learning curve** → results close to posterior and lateral
- Better DAA functional recovery → **time-limited,** early period only
- **Obese** or revision patient → posterior approach preferred
- Best approach for each surgeon → the one he performs well

High Yield

- THA successful regardless of approach → over 95% satisfied
- Elderly hip fracture → direct lateral approach → lowest dislocation
- Posterior + capsule/short external rotator repair → dislocation near lateral
- Lateral approach → more Trendelenburg gait; DAA → early-only recovery gain
- Obese or revision patient → posterior approach

Complex primary THA; Acetabular side

(Acetabular Anatomy, Defect Classification and Reconstruction Options)

Prof Dr Ahmed Hazem

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Lecture Objectives Ask AI

- Complex primary THA material → scattered across sources
- Objectives
  - What is a **complex primary** case?
  - How to **recognise** it → complex vs routine primary
  - How to evaluate it → management **algorithm**
- Aim → algorithm, or at least handling principles

Acetabular Anatomy Ask AI

Cave Analogy Ask AI

Anatomy

- Acetabulum ≈ **cave**
  - Two pillars → **anterior column** + posterior column
  - Columns join → **dome** (superior region)
  - Columns → anterior wall + **posterior wall**
  - Build sequence: anterior + posterior columns → walls close the front

Terminology Ask AI

Anatomy

- Same structures → basis of classification throughout
- **Rim** → peripheral margin
- **Floor** = medial wall → middle of acetabulum
- Anterior wall + posterior wall + dome → surround acetabular entrance
- Terminology used throughout lecture

Definition of Complex Primary Acetabulum Ask AI

- Native acetabulum → **not a true hemisphere;** irregularities
- Reaming → hemispherical → accepts hemispherical cup
  - Cemented or cementless
- Cannot be made hemispherical

→ **additional step** → reconstruct normal hemispherical shape

- Definition → **not clearly stated** in literature
- Working definition → any additional step to reconstruct a **hemispherical acetabulum**
  - Site: wall, column, dome or floor

→ **complex case**

High Yield

- Complex primary → any **additional step** to restore a hemispherical acetabulum
- Site of that step → wall, column, dome or floor
- No clear literature definition

Scope Ask AI

Scope

- This lecture → **bone side** only
- Non-bone side (likely implant) → Dr. Mohamed Younis
- Complex case → not only a bone problem
- Many conditions → deficiency of walls, columns, rim, dome, any part

Classification of Acetabular Deficiency Ask AI

AAOS Classification Ask AI

AAOS

- **AAOS classification** → used in this lecture
  - Developed 1989; American Academy of Orthopaedic Surgeons
  - Defects → several categories
- **Type I — segmental deficiency** → loss of part of rim
  - Peripheral or central
  - Peripheral → anterior, posterior or superior segmental loss
  - Same peripheral/central subdivision → applied to cavitary type
- **Type II — cavitary deficiency** → loss of volume within bone
  - Rim preserved; inside enlarged with missing area
  - Central → part of **floor** missing; rim present → contained
  - Peripheral → anterior, superior or posterior
- Segmental and cavitary → two separate entities

	Segmental (Type I)	Cavitary (Type II)
Containment	**Uncontained** — missing piece, not surrounded	**Contained** — enlarged, containment retained
Rim	Piece lost	Preserved
Defect	Rim segment	Loss of volume
Subtypes	Peripheral (anterior, posterior, superior) / central	Peripheral (anterior, superior, posterior) / central

- **Type III — combined deficiency** → cavitary + segmental in one acetabulum
  - Example → superior rim loss + substance loss
  - Complex because

→ part of **rim** missing

- **Type IV and V** → atypical in complex primary
  - **Type IV — pelvic discontinuity** → acetabulum/pelvis in two unconnected pieces
    - Example → post-traumatic cases
  - **Type V — arthrodesis** → not deficiency but **excess bone**
    - Included → cannot be treated as routine

Alternative Radiological Classification Ask AI

Radiologic

- Mentioned → so as not to upset its proponents
- Primarily a **revision** classification → osteolysis + deficiency of revision
  - Some surgeons → use in primary

  - Likely the Paprosky classification
- Simple; relies mainly on **radiology**
  - **Type 1** → minimal deficiency; rim intact, no migration; small cavitary defect
  - **Type 2** → defect with some superior involvement, not large
  - **Type 3** → **migrated** defect → cup migrated superiorly
- This lecture → keeps to AAOS

Recognising a Complex Case Ask AI

History Ask AI

History

- First clue → **patient's history**
- **Childhood onset** → "many pelvic operations when very young"

→ most likely **post-dysplastic** → deficiency in 100% of cases

  - Deficiency → superior or anterior
- **Post-traumatic** → usually deficient (in our country)
  - Exception → fracture fixed, and fixed well
  - Planned hip replacement → proper evaluation → recognised as complex, not simple
- **Rheumatological** → rheumatoid arthritis; possibly psoriatic arthritis
  - Usually **no deficiency**
  - Exceptions → steroid use / osteoporosis; long-standing disease

→ head migrates into acetabulum → **protrusio** (central cavitary defect)

- **Osteoarthritis** → usually routine primary
  - Long-standing + **osteoporosis** → usually not routine → check for deficiency

Pitfall

- **Post-dysplastic** hip → deficiency in 100% of cases
- Post-traumatic → deficient unless fracture well fixed
- Long-standing RA or **osteoporosis** → protrusio
- Long-standing OA + osteoporosis → not routine

Examination Ask AI

Exam

- **Examination** → little information on deficiency

Required Imaging Ask AI

Imaging

- **Radiology** → most important; tells deficiency or not
  - Detailed radiology → Dr. Mahmoud Abdel-Karim (afternoon)
- Radiographs → AP + lateral + **Judet views**
- Judet views added
  - Good for **columns** assessment (e.g. anterior column)
  - True AP / true lateral → very difficult
    - Deformed pelvis, **pelvic inclination**, other factors

→ all available views → complete picture

- **CT scan** → very important → defines defects precisely
  - Axial cuts + **coronal cuts**

Clinical Pearl

- Judet views → **columns** assessment
- Deformed pelvis → true AP / true lateral hard → add all views
- CT → axial + **coronal** cuts in every complex case

Plain Radiograph Assessment Ask AI

Reference Lines Ask AI

X-ray

- Familiar lines → related to **centre of the hip**

Line	Structure	Normal	Abnormal
Anterior wall line	Anterior wall; continuous with inferior border of superior pubic ramus, crosses acetabulum	Just in front of (medial to) hip centre	Further → **anterior wall deficient**
Posterior wall line	Posterior wall	≈ centre of head, ≈ lateral to it	Less → **posterior wall deficient**
Dome / sourcil	Roof	Starts + ends in same **sclerotic zone**	Open (e.g. positive acetabular index) → **dysplastic**
Teardrop	Floor	Follows floor	—
Ilio-pubic line	Anterior column	—	—
Ilio-ischial line	Posterior column	—	—

- All lines → must be checked; assessment not always easy

Templating the Hip Centre Ask AI

Template

- Confirm X-ray correctly taken → **horizontal reference line**
- → three basic reference lines
- Measure roof-to-reference distance → lines → where **roof** should be
- Midpoint between the two → mark area → measure distance
- Lines simulated on **contralateral side** → normal acetabular relationships
  - Detail → Dr. Mahmoud (afternoon)
- Point at intersection → ilio-ischial line + ilio-pubic line + **teardrop** (floor)
- Centre point → vertical + horizontal distances, simulated from contralateral side

→ **centre of hip**

= centre of cup

- Cup → placed at desired **inclination**
- Bone missing over templated cup → **deficient bone**

Case Examples Ask AI

Cases

- Case 1 → reconstructed
  - Reason → **superior + posterior** deficiency

- Case 2 → deficient acetabulum or not?
  - Undecidable until lines drawn
  - Lines drawn → **not deficient** → routine primary
  - Components excellent; only slight **medialization** → good cup seating

CT Assessment Ask AI

Axial Cuts Ask AI CT

- **Axial cuts** → show column deficiency
  - Anterior / posterior wall → present or absent
  - Anterior and posterior columns
  - **Floor thickness** → how far to ream → adequate cup medialization

Coronal Cuts Ask AI CT

- **Coronal cuts** → superior coverage
  - Cup uncovered > 50% → **reconstruction** needed
- Basic principles of radiological assessment

3D CT Ask AI CT

- **3D CT** → better idea of defect
- Example
  - Anterior view → good anterior column; protruding **anterior osteophytes**
  - Lateral view → **superior dome** deficiency
  - Inferior / posterior view → good posterior column + good floor

Goals of Acetabular Reconstruction Ask AI

- **Short term** → reduce **joint reaction forces**
  - High hip centre of rotation → normal position
  - Normal **biomechanics** restored

→ very good long-term results; less wear and loosening

- **Long term** requirements
  - Good **fixation** (cemented or cementless) → good host-bone contact
  - Adequate **cup positioning**
  - Rigid fixation of cup → **host bone**

Cavitary Defects — Management Ask AI

Small Cavitary Defect Ask AI Cavitary

- **Small cavitary defect** → ignore
- Case: **posterior wall + posterior column** fracture, very well fixed → small residual defect
  - Cup ~1 mm larger
  - That alone → very good outcome
- Anterior approach case → small cavitary defect
  - First → confirm **no anterior deficiency** on X-ray or CT

  - Then → proceed anteriorly → place cup

Large Cavitary Defect Ask AI

Cavitary

- **Larger cavitary defect** → e.g. protrusio
  - **Morselized graft** (bone chips) → autograft or allograft
  - Pieces ≈ 3–5 mm
  - Impact → **impaction bone grafting** → good support
  - Condition → cup contact with ≥ 50% **host bone** → good fixation

Extensive Cavitary Defect Ask AI

Cavitary

- Additional devices
  - Cup with **external flanges** → prevents cup migrating onto graft
  - More **screws**
  - Devices
    - **Müller ring** → graft-protecting device
      - Prevents excessive graft extrusion
      - Protects graft until **incorporation**
- Most important → 50% of **porous surface** against host bone

Must-Know

- Small cavitary → **ignore**
- Large cavitary → **impaction bone grafting** (3–5 mm chips)
- Extensive → flanges, screws, **Müller ring**
- Always → 50% of porous surface on **host bone**

Segmental Defects — Management Ask AI

- **Segmental defect** = uncontained → assess location + size

Location Ask AI

Segmental

Segment	Role	If deficient
Anterior / inferior	Good rim still surrounds acetabulum	Tolerated → may be ignored
Superior / posterior	Main cup support; intact → **press-fit** / stable fixation	Cannot be relied on → **acetabular augmentation**; must be reconstructed

- Superior / posterior defect → never neglected; region supports cup
  - No fixation without augmentation
  - Different reconstruction methods for this region
- Anterior / inferior ignored → only with very good quality of remaining **supporting bone**

Reconstruction Methods Ask AI

Segmental

- Methods
  - **Structural allograft**
  - Structural graft fixed with **screws**
  - **Augments**

- Key principle → **uncontained** → **contained**
- Approach used in practice

Exam Pearl

- **Superior + posterior** segments → main cup support → must reconstruct
- Anterior + inferior → may ignore, if remaining bone good
- Segmental principle → **uncontained** → **contained**

AAOS Types III–V — Management Ask AI

Combined Defects Ask AI

Combined

- **Combined defects** → uncommon in primary; more in revision
- Each defect separately

→ segmental (uncontained) → **contained**

→ then cavitary defect

Pelvic Discontinuity Ask AI

Discont.

- **Pelvic discontinuity** → un-united / non-united acetabular fractures
  - Very common post-traumatic
- Principle → **unstable** → **stable** construct
- Fix discontinuity first → stable → then cup or **ring/cage**
  - No cup straight away

Hip Fusion Ask AI

Fusion

- **Hip fusion** → not bone deficiency → problem is **excess bone**

Summary — Management Algorithm Ask AI

- Complex primary → not precisely defined → **additional step** to reconstruct acetabulum
- Classification → **AAOS**
- Evaluation → history, examination, **radiology**
- Radiographic evaluation → good X-rays + axial and coronal **CT cuts**

Defect	Management
Cavitary, small	Ignore
Cavitary, large	**Impaction grafting**
Segmental	Structural graft / augment; anterior + inferior → may ignore; superior + posterior → must reconstruct
Combined	Segmental first → convert to cavitary → then cavitary defect
Pelvic discontinuity	Stabilize columns first → then treat as segmental or cavitary

Complex primary THA; Femoral side

(Femoral Deformity, Retained Hardware and Reconstruction Strategies)

Prof Dr Mohamed Gobba

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Complex Primary THA: The Mindset Ask AI

- Opening case: condition for eight months; radiograph at decision to operate
- Clearly **not a normal femur** → specific mindset required
  - Question: how to get this patient through surgery?
- Particular features → think ahead → prepare additional equipment
- Additional steps not used in routine **primary total hip**
- **"Complex"** = not an ordinary total hip
- Three commonest femoral-side problems
  - **Failed intertrochanteric fixation**
  - Hip **fusion (arthrodesis)** → patient wants conversion to total hip
  - **DDH**

Failed Intertrochanteric Fixation: Preoperative Planning Ask AI

Broken Hardware Ask AI

ITF

- Many challenges; first → **broken hardware**
- On reaching implant → remove hardware first
  - Some operations: removing construct alone → two hours
  - Prolonged removal → surgeon did not prepare well
- Plan exactly in advance how hardware will be removed
- Anticipated problems → plan before theatre; be well prepared
- Proper **preoperative planning** mandatory

Imaging Ask AI

ITF

- **AP X-ray** alone not enough
  - Add a traction view
  - Add **CT**
- CT → location of **united** vs ununited fracture parts
  - → changes type of implant used
- Prepare all before surgery; no intraoperative discovery

Stability Ask AI

ITF

- Greater **stability** than usual required in this situation
  - Option giving more stability → **dual mobility**
  - Not simply the standard **28 mm head** used routinely

Bone Quality and Defects Ask AI

ITF

- **Bone quality** on films → imagine bone left after **plate** removal
- **Defects**: present? management? **graft** needed?
  - All answered before going in
- Fracture may be **united** → healed in malunited position

Canal Block Ask AI

ITF

- **Proximal femur**: pieces of **cortical bone** may block the canal
  - Equipment set → team prepares **high-speed burr** to remove obstructions
  - Otherwise → stem **rasps** cannot be introduced
- Rasps designed to remove hard **cancellous bone,** not cortical bone
- Rasp impacted hard against cortical bone → femoral **fracture**

Pitfall

- Cortical pieces blocking proximal canal → rasp impaction → femoral fracture
- Prepare high-speed burr in equipment set

Infection Ask AI

ITF

- **Exclude infection** before taking patient to theatre
- Details and steps: Dr. Mahmoud Abdel Karim, **PGI** lecture

Hardware Removal Set Ask AI

ITF

- Complete **instrument tray** solely for removing screws, including broken ones
  - Many surgeons unaware it exists
- No tray → no entry into **complex primary** case
  - Unavailable → **postpone the patient**
- Otherwise → two hours removing plate alone
  - → full concentration lost for rest of operation
- Hardware removal = **preliminary step** leading into main procedure

Must-Know

- Hardware removal set unavailable → postpone the patient
- Hardware removal = preliminary step, not the operation

Failed Intertrochanteric Fixation: Intraoperative Technique Ask AI

Long Stem and Screw Holes Ask AI

ITF

- **Long-stem prosthesis** → must **bypass the screw holes**
  - Bypass distance → **two cortical diameters**
- Method
  - Most distal screw hole → draw line → measure **canal diameter** at that level
  - Stem passes beyond by `2` cortical diameters, e.g. `4 cm`
- Purpose → avoid **stress riser**
  - Stem tip at level of hole → femur fractures

High Yield

- Long stem bypasses most distal screw hole by two cortical diameters
- Stem tip at screw-hole level → stress riser → fracture

Dislocation Ask AI ITF

- Take time → adequate release before dislocating
- Direct dislocation → **osteoporotic bone** fractures
- Fracture = last thing wanted in this operation
  - Surprises already expected on entry and **canal preparation**
  - Goal → finish case safely

DHS in Situ at Dislocation Ask AI ITF

- Dilemma: remove **lag screw** and plate first, or leave them?
- Lag screw **cut out** and impinging on **acetabular dome** → leave it too?
- Planned decision → written in advance at **planning meeting**

Situation	Hardware before dislocation
Lag screw cut out	Remove plate and screws first, then begin
Femoral head badly destroyed or ununited, hardware in place	Remove plate and screws first, then begin
Lag screw in good position	Keep in → facilitates rotating femoral head

Canal Block and Soft-Tissue Repair Ask AI ITF

- **Canal block** (as above) → be prepared
- Have **Ethibond** or FiberWire available
  - Greater trochanter often involved more than expected
  - → repair of capsule and **gluteus medius**
  - = main effort at this stage

Conversion After Failed Fixation: Evidence Ask AI

- Studies compare **total hips after DHS** vs after PFN
  - Conversion after PFN → **more complications** than after DHS
  - Converting PFN → expect somewhat higher complication rate
- Other studies → **worse outcomes** after intramedullary (IM) devices
- Above = minimum preparation for **primary total hip after failed intertrochanteric fixation**

Arthrodesis Conversion: Indications Ask AI

- **Arthrodesis** conversion: fused hip → total hip
- Very uncommon in our practice; not routine
- **Indications**
  - **Mechanical back pain** → e.g. fused at age 11, now cannot walk from back pain
  - **Generalized loss of function** → wants to work and be active
  - **Failed fusion**
  - **Shortening**
  - **Pain in adjacent joints**
  - **Social problems** → e.g. girls of marriageable age cannot live normally

- All factors weighed; decision neither easy nor simple

Arthrodesis Conversion: Technical Challenges Ask AI

Landmarks and Hardware Ask AI

Fusion

- Identify **anatomical landmarks**
  - Normal hip → evident
  - Here → **distorted anatomy,** especially after previous surgery
- **Remove the hardware**

Fusion Takedown and Neck Cut Ask AI

Fusion

- Takedown **osteotomy** → **intraoperative fluoroscopy** for exact position
- Anatomy very distorted, especially after hardware removal
  - → fluoroscopy when making the **neck cut**

Acetabular Reaming Ask AI

Fusion

- Choose **center of rotation** → begin **reaming** there
  - **Smallest reamer** first → best estimate
  - Usually a marker at presumed center of rotation
  - Identify anterior and posterior columns → go between them
  - Ream until seated in bone → leave reamer → **image intensifier** shot
  - Accept, or too high / too medial → correct → continue enlarging
- Never ream, insert **cup,** then discover wrong position on X-ray

Soft-Tissue Release Ask AI

Fusion

- Consider iliopsoas **tenotomy**
  - Muscles not moving → all **fibrosed** → motion problems

Femoral Component Ask AI

Fusion

- Usually **porous-coated** stem → metaphyseal fixation
- Must **bypass the stress risers**
  - Previous plate
  - Previous osteotomy

Abductors Ask AI

Fusion

- **Abductors** = key problem in arthrodesis → assess from beginning
- Completely replaced by white (fibrous/fatty) tissue → **transfer** options
  - Vastus lateralis transfer
  - **Gluteus maximus** transfer
  - Details outside this lecture
- Deficient even after reconstruction → future **instability**
  - Plan from start → **dual mobility** or constrained implant

High Yield

- Abductors = key problem in arthrodesis conversion
- Deficient after reconstruction → dual mobility or constrained implant

Arthrodesis Conversion: Outcomes and Counselling Ask AI

- Evidence: two or three papers → same conclusion
- Total hip groups compared by **functional scores**

Group	Functional outcome
Primary THA	High scores; patients very satisfied
THA after revision	Somewhat lower
THA after fusion	Not satisfied, even after total hip

- Conclusion → conversion patients
  - **Inferior functional outcomes**
  - Higher complication rates
- Most important factor → **patient expectations**
  - E.g. mild back pain, simple analgesics, years unchanged
  - Symptoms neither progressing nor improving
  - Imagines a magical state → walking normally like ordinary people
- Counsel patient → does he really need the total hip?
- **Systematic review** → complication rate more than 54%
- Lower patient expectations → very important

Clinical Pearl

- Fusion conversion → complication rate above 54% (systematic review)
- Lower patient expectations before offering conversion

Total Hip Arthroplasty in DDH Ask AI

- Detailed coverage → lecture by Dr. Mohamed Younes; brief note here
- **Indications for surgery**
  - **Pain**
  - Interference with **ADL**
- Dislocated since birth → abnormal **hip anatomy**
  - **Excessive femoral anteversion**
  - Femur: very high anteversion + very **small canal**
- → **modular stem** → femoral version set → stem implantable

Subtrochanteric (Rotational/Shortening) Osteotomy Ask AI

Trial Before Osteotomy Ask AI

DDH

- After exposure → **broaching** femur to maximal size
- **Trial prosthesis** inserted → fits well
- Fit confirmed, but 3–4 cm still to shorten

  - → **subtrochanteric shortening osteotomy**
- After this step → remove trial

Osteotomy Ask AI

DDH

- Level: below **lesser trochanter** (subtrochanteric)
- Configuration: **chevron** or transverse; many methods
- Essentials
  - Osteotomy
  - Iliopsoas tenotomy
  - Separate the fragments
- No femoral resection yet → amount determined after this step

Determining Resection Length Ask AI

DDH

- Trial prosthesis in → **assistant** applies traction on femur
  - → tension → fragments in final position
- Mark **overlap** with diathermy = excess segment
- Excess removed → femur brought together
- Result → best tension
  - Avoid **over-tensioning** or under-tensioning
  - → desired **soft-tissue tension**
- Images: before vs after operation

Opening Case Revisited Ask AI

- End of lecture → revisit opening X-ray

Fixation Level Ask AI

Case

- Options
  - **Metaphyseal** fixation
  - Metaphyseal-diaphyseal fixation
  - Diaphyseal-diaphyseal fixation
- Region very badly damaged
- Company contacted → **long stem** prepared = first planned point

Neck Cut and Second Osteotomy Ask AI

Case

- Second point → site of **neck osteotomy**
- Usual neck cut? → answered by **templating**
- Rasp in canal → would exit through the **trochanter**
  - Trochanter became medial → overhanging the canal
  - Not acceptable → cannot start with rasp in this area
- Decided at planning → **second osteotomy** before starting
- Second osteotomy → window opened → canal directly in front
- After prosthesis insertion → **fragment fixation**
  - Fragment shaped at an angle → fits onto prosthesis

Conclusion Ask AI

- **Complex primary THA** = big topic → never enter unprepared
- Self-check before operating
  - Required **skills**
  - Necessary **knowledge**
  - Available equipment and anaesthesia support
- Know your **limits**
- Closing quote
  - "A good surgeon knows how to operate."
  - "A better surgeon knows when to operate."
  - "And the best surgeon knows **when not to operate**."

THA after dysplasia; Soft tissue releases

(Classification of Dysplasia and Soft-Tissue Release Strategies)

Prof Dr Mohamed Youness

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 208) Transcript garbled here; a high hip centre usually shortens the abductor lever arm, which raises joint forces

Q2 (p. 208) Same point: a high hip centre usually shortens the abductor lever arm, it does not lengthen it

THA in the Dysplastic Hip — The Challenge Ask AI

- Key questions → difficulty → can it be overcome → does patient benefit?
  - Comprehensive problem → **femur** and acetabulum
  - Ultimate aim → patient benefit
  - Patient → young, **high-demand**
- **Soft tissue** problems → mostly present; must be addressed
  - Wants active life, long-term use of joint
  - Wants no **early revision**

Pathological Anatomy on the Radiograph Ask AI

- Single X-ray → summarizes all problems of the condition
  - Acetabulum → not **normal anatomy**; not dealt with easily
  - Femur–acetabulum relation → not that of a normal joint
  - Problems expected on both sides
    - **Dysplastic acetabulum** → abnormal appearance
    - Femur → hypoplastic, small
- **Neck** → shortened, not in true orientation
  - Intra-op → don't expect to find the neck in front of you
- **Medullary canal** → small, narrow
  - Cortex → probably very thin
  - Marked **anteversion**
  - Trochanter → not in normal position

Classification Ask AI

- First step → **stratify** type of dysplasia
  - First classification → Hartofilakidis

Hartofilakidis Classification Ask AI

- **Hartofilakidis** → A, B, C by degree of dislocation

Type	Description
A	Subluxation (dysplasia)
B	Low dislocation
C	High dislocation

Crowe Classification Ask AI

- **Crowe classification** → most widely used
- Four groups → by amount of head dislocated outside acetabular socket

Group	Head outside acetabulum
1	< 50%
2	More (approx. 50–75%)
3	Up to approx. 100%
4	Femur high up; no relation to acetabular socket

- **Group 4** → worst

High Yield
- Crowe → most widely used classification
- Basis → % of head outside acetabular socket
- Group 4 → femur high, no relation to socket

Radiographic Examples Ask AI

- X-ray series → simpler → more complex degrees
- **Crowe IV** → femur rides high in pelvis, away from acetabular socket
- Further X-ray examples

Indications for Surgery Ask AI

- **When to operate** → very important point
  - Typical → patient in forties or fifties with dysplasia
  - Questions → recent history; operate on every **acetabular dysplasia**?
- Tell patient today → operation + joint replacement?
  - Ordinary complaint vs incidental **DDH** / acetabular dysplasia
  - Surgery → not the only option
  - Management → very **conservative**
- Reason → considerable technical problems and complications
  - Required → **pain** + disability
    - Ambulation difficulty
    - Major functional problems
  - Many problems → still may not **justify** surgery
  - **Indication** → must be very clear at assessment
- Goal → patient benefits 100%
  - Hip replacement → good results
- **Total hip** for acetabular dysplasia → only when benefit is 100%
  - Must confirm significant level of **disability**
  - Gait problem

Must-Know
- Dysplasia alone → not an indication
- Indication → pain + significant disability
- Technical problems, complications → considerable

Femoral and Acetabular Deformity Ask AI

- Problems in brief → both sides

Femoral side	Acetabular side
Head → deformed, displaced	Deficiency
Neck → shortened	Oblong, vertically oriented
Trochanter → posterior	False acetabulum
Medullary canal → narrow	Deficient bone stock for cup

- Cup in **false acetabulum** vs true → makes a difference
  - Restore **biomechanics** → patient outcome depends on it

Pitfall
- Open, start preparing → cup placed in false acetabulum
- Correct target → true acetabulum

Soft Tissue Problems Ask AI

- **Soft tissue** → most important issue
  - Forces **extra procedures** during THA in dysplastic hip
- All muscle groups around hip → involved
  - **Abductors** → weak, contracted
  - Rectus femoris
  - Iliopsoas
  - Hamstrings
- Cause → long-standing shortened state in dislocated position
- **Capsule** → elongated, thickened
  - Part interposed where head should sit
- Knowing these points → be prepared
- Unknown → unprepared → problems (Dr. Ahmed, Dr. Gouda)
- Otherwise → unnecessary steps or poor results
  - You cannot see what you do not know

Goals of Reconstruction Ask AI

- **Assessment** → very important
  - Expected → shortening
  - Restore normal anatomy → regain length → plan compensation
- Regain length → without stretching muscles
  - **Neurovascular structures** → at risk
- Principles known before surgery → goals for good results

Acetabular Side Ask AI

Acetabulum

- First goal → restore **center of rotation**
- Large difference → early results + **long-lasting** prostheses
- Normal center reached → **cup** stable initially and long term
- Reconstruct acetabular bed
- → **stability** in correct position

Femoral Side Ask AI

Femur

- After acetabular side → femoral side
- Femoral side restores
  - **Offset**
  - Abductor mechanism
  - Limb length

Must-Know

- Restore center of rotation
- Stable cup → initially and long term
- Femoral side → offset + abductor mechanism
- Femoral side → limb length

Preoperative Planning and Special Sets Ask AI

- **Preparation and planning** → avoid unnecessary steps
  - Standard **primary** set → cannot be relied on
  - Special set → fits anatomy and deformity
  - Anticipate problems → **special implants** + special sets
- Goal → don't make the difficult case harder
  - **Surgical technique** → crucial for good results
- **Special sets** → used
- Sound **principles** → know why each technique, why each step
  - Practical steps → release small muscles → bring acetabulum to its place
- Depends on → acetabulum location + plan for **acetabular reconstruction**
  - **Bicanic** (2014) → current concepts in acetabular and femoral reconstruction
- Different concepts → reach **four pillars** → good results

Acetabular Reconstruction Ask AI

High Hip Center Ask AI

Acetabulum

- Early studies → cup placed high, before tackling true acetabulum
- Reason → **contracture** + soft tissue problems around hip
  - Cup → slightly high center
  - Femur reconstructed onto it → good results reported
- Later study → **stresses** and forces on acetabular component
- Long-term results → not as that school claimed

  - Higher center → ↑ **lever arm** of surrounding muscles (abductors) *[Q1 · Transcript garbled here; a high hip centre usually shortens the abductor lever arm, which raises joint forces]*
  - → ↑ stresses on construct → failure

Exam Pearl

- High hip center → ↑ abductor lever arm *[Q2 · Same point: a high hip centre usually shortens the abductor lever arm, it does not lengthen it]*
- → ↑ stresses on construct → failure

Placing the Cup in the True Acetabulum Ask AI

- Options → cup in **true position**

Technique	Key point
Bulk graft	Cup in true position
Partly uncovered cup	Up to 30–40% uncovered, within true acetabulum → osseointegration
Augments	—
Different shells	Reconstruction
Cotyloplasty	Cup placed medially

High Yield

- Up to 30–40% of cup → may be left uncovered
- Condition → cup within true acetabulum

Soft Tissue Release and Balancing Ask AI

- Study (2012) → **soft tissue release** + balancing with acetabulum in true position
- **Stepwise technique**
  - Not everything around hip at once
  - Assess release needed → reduced, stable hip
- Release sequence → stop once result good
  - Step 1 → adductors (probable) + part of **superior attachment of gluteus maximus** at greater trochanter
  - Step 2 → **iliopsoas**, part of rectus, sartorius from ASIS
  - Step 3 → **piriformis** + hamstrings

Clinical Pearl

- Release stepwise, not all at once
- Stop once hip reduced, stable

Femoral Reconstruction Ask AI

- Then → **femoral reconstruction techniques**
  - **Distractor**
  - **Trochanteric osteotomy** → different modalities
  - **Subtrochanteric osteotomy**
- Shortening → e.g. Dr. Gouda's **subtrochanteric osteotomy**
  - **Trial** in position → determine segment to resect → compensate length
  - Then modular components placed
- Segment → **modular fixation**
- **Distal femoral osteotomies** + shortening

Key Message and Conclusion Ask AI

- Second key point → know **pathological anatomy** + expected problems
  - → prepare **special instruments** + special sets
- → avoid many technical problems
  - Correct planning → good results → patient benefits

Dysplastic hip

(A Case-Based Approach to the Dysplastic Hip)

Prof Dr Fouad Zamel

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 220) “white line” → “radiolucent (white) line”

Mildly Dysplastic Hip: Case 1 Ask AI

Context Ask AI

Case 1

- Pathology, **pathological anatomy**, acetabular reconstruction → covered earlier by Dr. Mohamed Younis
  - This session → case-based discussion
  - First case → simple, from private practice

Presentation Ask AI

Case 1

- Woman, presenting at 51–52 years
  - Complaint → **limp**
- Early presentation; very active, horse rider; hip pain
  - Dysplastic or not? → little **dysplasia**, visible on full radiographic views
- Diagnosis → hip arthritis secondary to **mild dysplasia**
  - Mild dysplasia → mechanics not much disturbed, despite long-standing course

Planning Ask AI

Case 1

- Plan → **total hip replacement**
  - Little **subluxation** → some difficulty possible
- Question raised: **CT** needed?
  - First step → list risks expected in a dysplastic case
  - Mildly dysplastic acetabulum → little difficulty expected
- Opposite hip → replaced 1 year earlier
- Second hip replaced → no problems
  - Acetabulum → essentially **normal**
  - Uncovering / lateralization → arguable, minimal

Principles of Reconstruction Ask AI

Center of Rotation and Bone Stock Ask AI

- Main principles → repeated throughout (also covered by Dr. Younis)
- Main mechanical principle → cup at **anatomical (true) center of rotation**
  - Cup at center of rotation → work with bone present there
  - Best bone stock → at true **center of rotation**, contrary to expectation
  - Holds despite region being unloaded
- Bone and acetabulum never fully loaded → still best **bone stock**
- Higher position → slightly more bone, not much more
  - Best bone around center of rotation → use it for any cup type

Sequence of Goals Ask AI

- Reconstruction order:
  - Restore **center of rotation** → place cup there first
  - Secure cup **fixation**
  - Finally → restore **global offset**
- Global offset → affects **leg length** and offset → affects **stability**

  - These → goals of hip replacement in dysplasia
  - First goal → **contained cup**
  - Good, contained cup → good hip, good patient outcome
- Achieved in Case 1 → not a problem
- Not always this straightforward across the spectrum

High Yield
- Cup at anatomical (true) center of rotation
- Best bone stock → at true center of rotation, though unloaded
- Order: center of rotation → cup fixation → global offset
- Global offset → leg length + offset → stability
- First goal → contained cup

Spectrum of Dysplasia Ask AI

- Wide range of radiographic appearances
- Range of dysplastic hips → different grades
  - Classification → not discussed here
  - Dysplasia starts → changes in **acetabulum** and **proximal femur**, hip still at center of rotation
- Increasing severity → head rides higher → additional problems
  - End stage → head completely dislocated, never reduced
- What not to do → cup in high **false acetabulum** (example case, discussed at end)
  - Correct site for cup → **true acetabulum**

Pitfall
- Cup placed in high false acetabulum → wrong
- Cup belongs in true acetabulum

Subluxated Dysplastic Hip: Case 2 Ask AI

Radiographic Findings Ask AI

Case 2

- Change in **proximal femur**
  - **Subluxation** → **Shenton's line** broken
- Bone deficient inferiorly
  - Subluxation → present, not major

Proximal Femur in Dysplasia Ask AI

- Proximal femur findings:
  - Increased **anteversion**
  - **Narrow canal**
- Proximal femoral anatomy → always abnormal in dysplasia
  - Common misconception → problem acetabular only
  - Problem → proximal femur as well
- Clinically → marked **internal rotation** (anteversion)

- Intraoperatively → **excessive anteversion**, consistently found
  - Stem positioned by native femoral version → stem in excessive anteversion
  - **Small canal** → confirmed

Pitfall

- Dysplastic femur → excessive anteversion, always
- Stem positioned by native femoral version → stem in excessive anteversion
- Problem not only acetabular → proximal femur abnormal too

Iliopsoas and Capsule Ask AI

- Capsule in dysplastic hip → not short
  - Two parts; one part adherent above acetabulum → needs **release**
- Plan → **total hip**; type to be chosen

Cup Sizing Ask AI

Case 2

- Need for special stems, special cups, small cup?
  - Cup size determined by → **anteroposterior dimension**
  - Relevant part → lower / anterior wall, not upper part

Acetabular Reconstruction Ask AI

Case 2

- Young patient → **cemented reconstruction**, acetabular side
  - Reason → very small cementless components unavailable
  - Not because cemented cups work better
  - Preferred bearings → **highly cross-linked polyethylene** or **ceramic**
- Other techniques acceptable; shown later
- **Bone graft** from small femoral head → ample bone
- Good-quality graft piece
  - Cemented reconstruction → speaker's former technique; different technique now
  - Performed 12–14 years ago
  - No graft **resorption**; graft lies posteriorly
- Femoral canal → very narrow

High Dislocated Hip: Case 3 Ask AI

Acetabular Side Ask AI

Case 3

- High dislocated hip with **false acetabulum**
  - Plan → **total hip**
- Planning → list expected problems, divide them
  - Main problems → acetabular side
- Identifying **true acetabulum**:
  - False acetabulum vs lower part
  - Lower part → **anterior wall**, posterior wall behind
  - Anterior dimension → not narrow
  - Small cup → not needed
- **Mushroom-shaped** femoral head → large anterior dimension

  - Large anterior dimension → larger cup required
  - Questions: reduction and component problems?
- This bone stock → accepts larger cup
  - Cup here → good size, any design

Femoral Side Ask AI

Case 3

- **Shortening osteotomy** → acceptable option
  - Technique must be defined
- Different procedures; amount of shortening → per surgeon
  - Special stems?
- Options:
  - Direct shortening
  - Extensive shortening
  - **Cemented stem** or **cementless stem**
- Choice → technique best known, best results in surgeon's hands
- This case → **femoral head** harvested; large amount of bone

Reconstruction Ask AI

Case 3

- Graft fixed with **screws** superiorly; **cementless cup**
  - Operated 14–15 years ago; doing well

Completely Dislocated Hip: Case 4 Ask AI

Problems Ask AI

Case 4

- Completely dislocated hip
  - **Leg-length discrepancy**
  - Very large displacement → **sciatic nerve** problems
- Capsule origin → guide to true acetabulum
- Capsule position → site of **false articulation**
  - False socket → very small, small diameter

Locating the True Acetabulum Ask AI

- True acetabulum → difficult to find
  - Follow **ligamentum teres**
  - Intraoperative **fluoroscopy** → easy alternative
- Walls → flat, **saddle-shaped**
  - Anterior and posterior columns → suitable position
  - Columns obscured → covered by bone / soft tissue
- Area closed off by bone overlying columns
  - True acetabulum → very, very small
  - Do not ream as in a normal hip
- Previous case → large acetabulum, because loaded
  - This case → very small acetabulum
- Walls → not clear
  - Landmarks → follow **sciatic notch** and **iliopsoas**
  - True acetabulum → between them

  - Overlying soft tissue → resembles exostosis / protrusion
- Not a true exostosis, not large → marks route to true acetabulum

	Loaded acetabulum (Case [illegible])	Unloaded true acetabulum (Case [illegible])
Size	Large	Very, very small
Walls	Anterior and posterior walls identifiable	Not clear; flat, saddle-shaped
Reaming	Standard	Not as in a normal hip
Landmarks	Lower part, anterior wall	Ligamentum teres; sciatic notch and iliopsoas; fluoroscopy

Floor Thickness Ask AI

Case 4

- Floor **thickness** → speaker's technical trick:
  - Small **drill** through the area only
  - Drilling → estimate of thickness
- Thickness cannot be palpated → drilling easy, estimates thickness in centimeters
- Reconstruction / **grafting** → required
- Patient from **Yemen;** operated many years ago

Clinical Pearl

- True acetabulum → follow ligamentum teres
- Alternative → intraoperative fluoroscopy
- Between sciatic notch and iliopsoas → true acetabulum
- Floor thickness → small drill hole

Sciatic Nerve and Lengthening Ask AI

- Cup at true acetabulum + reduction of **femoral head** → risk
  - Risk → **sciatic nerve** injury, at reduction
  - Lengthening > 3.5–4 cm → risk of sciatic nerve injury
- Intraoperative check:
  - **Palpate sciatic nerve** → feel tension with leg pulled down
  - Nerve much tighter at reduction → problem

Must-Know

- Lengthening > 3.5–4 cm → sciatic nerve injury risk
- Intraoperative test → palpate sciatic nerve, leg pulled down
- Nerve much tighter on reduction → problem

Cemented Reconstruction with Bulk Autograft Ask AI

- Technique → **meticulous cemented technique**
- Combined with **shortening osteotomy**

  - Osteotomy + reconstruction → **bulk femoral-head autograft** fixation
  - Learned from **Dr. Maher** many years ago; still favoured, useful
  - No autograft reconstruction revised by speaker yet
  - Some cases possibly revised by others
- No graft case revised since return home 20 years ago
  - First revision → now
  - Graft reconstruction → works well; this patient > 8–9 years out
- Follow-up → **nine-year follow-up**
  - **Cemented reconstruction** → works well
  - **Standard poly, 28 mm heads** → work
  - Prerequisite → **meticulous technique**

Subtrochanteric Shortening: Case 5 Ask AI

- Woman, 36 years; current hip problems
  - Hip reconstructed; fixation with 6 screws (6-hole DCP)
  - Technique → **rasp (broach)** in proximal femur first
- Rasp inserted before femoral **osteotomy** → then **subtrochanteric shortening**
- Femur back in place → reduce → measure fragment **overlap** → resect that amount
  - Some techniques → trial stage before cutting
  - Speaker → no trial stage

Exam Pearl

- Rasp in proximal femur before osteotomy
- Subtrochanteric shortening → reduce, measure overlap, resect overlap

Leg-Length Discrepancy and Pelvic Tilt Ask AI

- Additional problem beside **leg-length discrepancy** → **pelvic tilt**
  - **Long-standing pelvic tilt** → usually not easily corrected
  - Patients struggle to correct tilt
  - Even after shortening → often end up with more length

Bilateral Dysplasia: Case 6 Ask AI

History Ask AI

Case 6

- Woman from **Port Said**; both hips affected
- Left hip replaced 4 years earlier
  - Postoperative dislocation → reduced → walking
  - Told bed rest 3 months → walked 3 months → dislocated again → reduced again
- Then hip **spica** for a while → mobilized → no further dislocation
  - Painful, but walking on it for almost 4 years
  - Question: cup stable or unstable?
  - **Osseointegration** → superior part

- At least on x-ray
  - Other hip → **Crowe** group 3–4, at most 3

Side Selection Ask AI

Case 6

- Operate? Which side first?
- Question anticipated by **Dr. Abu Saud**
  - Most choose left first
- Reason → otherwise second side has a problem
  - Right more painful than left → right done first
  - Speaker's rule → listen to patient's complaint

Anticipated Problems Ask AI

Case 6

- Femur → **small canal**
- Options for small canal → shortening? osteotomy?
  - Two-stage procedure?
- Main bone stock; acetabulum viewed from other side
- **Standard cup** vs **augmentation** and preparation?
- False acetabulum → look inside for true walls

CT Assessment Ask AI

Case 6

- True acetabular walls
- CT **axial cuts**
- Axial cuts → good idea of bone stock type
  - Good area; walls small and very low

Implants Ask AI

Case 6

- Usual technique → **cemented stem** + standard **cementless cup**
  - Cup → 46 mm, standard
  - Cemented stem → easier with shortening **osteotomy**
  - Insert, good cementation → worked
  - No problem with cement
- Why not **modular stem**?
  - Some reservations; does well
  - Not better than cemented stem
  - Modular stem → acceptable alternative here
- Choice → surgeon's preference

	Cemented stem	Modular stem
Use here	Chosen	Acceptable alternative
Rationale	Easier with shortening osteotomy; good cementation	Does well
Speaker's view	No problem with cement	Some reservations; not better than cemented

Second Side Ask AI

Case 6

- After first side → walked month after month → then dislocation (after outside advice)
- Other side → now required
- Time limit noted by Dr. Abu Saud
  - Cup → covered, not uncovered
  - Covering bone stock → lowest, most medial part of previous acetabulum
- Bone at dislocation site → hip had been dislocated, not reduced
  - Patient walking on dislocated hip → abnormal gait
- **Close-up view** → **bone stock** covering cup
  - **Standard cup**; no **bone grafting**
  - Same for other side → size adequate
  - Radiolucent (white) line → surprises some *[C1 · transcript: "white line"]*
- Line closed; not the line below → **dome** fine
- Preoperative image → comparison of both hips intended
  - Outcome → did well

Conclusions Ask AI

- **Dysplastic cases** → challenging; require special preparation
- Proper **assessment** + proper **planning** → decide needs
  - Most cases → **standard implants;** no special or very large implants
  - Most cases → no **grafting**
  - Both techniques work → if reconstruction done well
- Cemented vs cementless → per preferred technique
  - **Cemented reconstruction** works → case with > 15–16 years follow-up
  - **Cementless techniques** → also work
  - Key → perform chosen technique well

High Yield

- Dysplasia → challenging; special preparation, proper assessment, planning
- Most cases → standard implants
- Most cases → no grafting
- Cemented and cementless → both work if done well

Case Summary Ask AI

Case	Pattern	Key problems	Reconstruction	Follow-up
1	Mild dysplasia; woman, 51–52 years	Little subluxation	THR; acetabulum essentially normal	Contralateral THR 1 year earlier; no problems
2	Subluxated; Shenton's line broken	Bone deficient below; excessive anteversion; narrow canal	Cemented cup + femoral-head graft	12–14 years; no resorption
3	High dislocated; false acetabulum	Large anterior dimension (mushroom head)	Femoral-head graft + screws; cementless cup	14–15 years; doing well
4	Completely dislocated (Yemen)	Very small true acetabulum; sciatic nerve	Meticulous cemented + shortening osteotomy + bulk autograft	Nine years
5	Woman, 36 years	Hip problems	Rasp first → subtrochanteric shortening	—
6	Bilateral (Port Said); Crowe 3–4	Recurrent contralateral dislocation; small canal	Cemented stem + 46 mm standard cementless cup; no graft	Did well

Periprosthetic Fractures

(Classification and Management of Periprosthetic Fractures)

Prof Dr El Sayed Kassem

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 230) “less than 2 diameters” → “at least 2 cortical diameters”

C2 (p. 232) “6 to 8 months” → “6–8 weeks”

Overview and Definition Ask AI

Lecture Scope Ask AI

- Summarized review of **periprosthetic fractures** → essentials only
- Prepared with **Dr. Mahmoud Abdel-Karim**
- Objectives
  - Risk factors and classification after THA
  - Intervention → reducing occurrence
  - Management and take-home messages

Definition and Epidemiology Ask AI

- Scope includes avoiding **intraoperative periprosthetic fractures**
- Fracture around implant after **total hip arthroplasty (THA)**
  - Sites → acetabulum or femur
- Incidence rising
  - Increasing number of arthroplasties performed
  - Indications extended → younger, more active patients
- Timing → **intraoperative** or postoperative

General Principles Ask AI

Risk Factors Ask AI

- Patient-related
  - Female sex (probable)
  - **Advanced age**
  - Reduced bone quality
    - **Osteoporosis**, osteopenia
    - Rheumatoid arthritis, **Paget's disease**
    - DDH, similar conditions
- Implant-related
  - Main → **osteolysis** and loosening
  - Implant type → cemented vs cementless

Patient Evaluation Ask AI

- History and examination first
  - Pre-fracture symptoms (e.g., pain before injury)
    - Suggest pre-existing **loosening** or osteolysis
- Medical → elderly with comorbidities
  - Identify and optimize before surgery
- Radiological → **AP and lateral X-rays**
  - Fracture classification
  - Implant stability in that region
- **Infection workup** → exclude periprosthetic joint infection
  - Detailed in next lecture (Dr. Mahmoud Abdel-Karim)

- Soft tissue / skin condition (probable)
- **Neurovascular examination**
- Existing implant details → type, company, size
  - Purpose → full preparation

Aims of Management Ask AI

- **Fracture union**
- Restoration of alignment (probable)
- Preservation of function (probable)
- **Implant stability**

Intraoperative Periprosthetic Femoral Fractures Ask AI

- Femoral fractures → intraoperative vs postoperative

Incidence Ask AI

Intraop

- **Cementless** > cemented systems
- Higher in **revisions**

Risk Factors Ask AI

Intraop

- **Inadequate exposure** → poor visualization
  - See well before **femoral neck cut,** rasping, acetabular reaming
- **Excessive force** → rasping or impaction
  - Especially **osteoporotic** patients
  - Hurried hammering → sudden femoral fracture
- **Press-fit stem larger than medullary canal**
  - Rasped to size `6` → size `6` unavailable
  - Size `7` forced → larger than canal → fracture
- **Long stem mismatching femoral bow**
  - Femur → **anterior bow**
  - Long stem must match bow → else fracture
- Revisions → higher incidence
  - **Compromised bone stock**
  - Cement removal
  - Impaction bone grafting

Modified Vancouver Classification (Intraoperative) Ask AI

- Most famous → **Modified Vancouver Classification**
- Three main types × three subtypes

Type	Site
A	Proximal metaphyseal
B	Diaphyseal, around stem
C	Distal to stem

Subtype	Lesion
1	Cortical perforation
2	Non-displaced linear crack
3	Displaced / unstable fracture

Type A — Metaphyseal Ask AI

Intraop

- **A1** → bone graft
  - Source → femoral head or acetabular reamer reamings
  - Stem stable in metaphysis
- **A2** → cerclage around metaphysis

Type B — Diaphyseal Ask AI

Intraop

- **B1** → bypass perforation with longer stem
  - Bypass length → **2 cortical diameters**
- **B2**, stable stem → cerclage alone acceptable
  - Fracture fixation preferred
- B2, **unstable stem** → longer stem revision + fixation
- **B3** → revision by longer stem + fixation
  - ± Graft

Type C — Distal to Stem Ask AI

Intraop

- **C1** → cortical strut graft
  - Bypasses defect (probable)
- **C2** → fixation, cerclage (probable)
- **C3** → fixation (ORIF)

Prevention of Intraoperative Femoral Fractures Ask AI

- Lecturer → most important point of the lecture
- **Adequate exposure** → no step without good visualization
- Adequate soft tissue release (probable)
  - Difficult exposure, e.g., **protrusio**
- Osteoporotic → no **excessive force** in preparation and impaction
- Proper **templating**
- Check trial / broach size (probable) before final implant
  - Confirms canal fit
- Revision, stem not coming out → **extended trochanteric osteotomy**
  - Controlled osteotomy > uncontrolled fracture
  - "By my hand, not by Amr's hand"
- **Preoperative planning and templating**
  - Patient size known → avoids size 6 / size 7 scenario

Pitfall

- Size 6 unavailable → **size 7** forced → femoral fracture
- Long stem ignoring **anterior bow** → fracture
- Hurried impaction in **osteoporotic** femur → sudden fracture

Clinical Pearl

- Stuck stem in revision → **extended trochanteric osteotomy**
- Controlled osteotomy > uncontrolled fracture

Postoperative Periprosthetic Femoral Fractures Ask AI

- Incidence (no intraoperative fracture)
  - Primaries → 1%
  - Revisions → 4%
- Main risk factors → **osteolysis,** loosening
- Mechanism → **low-energy trauma**
- **Swedish National Hip Arthroplasty Register** → mean time to fracture
  - Mean 7.3 years (recording unclear: "7 3.9")
- **Mayo Clinic** study → mean 8.1 years after index operation

Vancouver Classification (Postoperative) Ask AI

- Most famous for postoperative periprosthetic fractures
- Based on three criteria
  - **Stem stability** → loose vs well fixed
  - Fracture site
  - Bone quality

Types Ask AI

Postop

Type	Location	Subtype	Defining feature
A	Trochanteric area	AG	Greater trochanter
		AL	Lesser trochanter
B	Around or just distal to stem	B1	Well-fixed stem
		B2	Loose stem, adequate bone quality
		B3	Loose stem, poor bone quality
C	Well below stem	—	—

Management by Type Ask AI

Postop

- **Vancouver A** → trochanteric region
  - Non-displaced / minimally displaced → **conservative**
  - Displaced (unstable) → fixation, wires and plate (probable)
- **Vancouver B1** → fixation around stable stem
  - Plate or cerclage (probable)
  - i.e., **ORIF**
- **Vancouver B2** → around or just distal to loose stem
  - Stem removal → **longer stem** → fracture fixation
- **Vancouver B3** → loose stem, poor bone quality
  - Longer stem + proximal femur reconstruction, grafts (probable)
  - Reconstruction impossible → **proximal femoral replacement**
- **Vancouver C** → stem untouched → ORIF

Management Algorithm Ask AI

Postop

Type	Management
A, non-displaced	Conservative
A, displaced	ORIF
B1	ORIF
B2	Longer stem + fixation
B3	Longer stem + proximal femur reconstruction, or proximal femoral replacement
C	ORIF

High Yield

- Vancouver criteria → stem stability, fracture site, bone quality
- Well-fixed stem (B1, C) → **ORIF**
- Loose stem, adequate bone (B2) → **longer stem** + fixation
- Loose stem, poor bone (B3) → **proximal femoral replacement** if no reconstruction

Fixation of Periprosthetic Femoral Fractures Ask AI

Indications Ask AI

- **Well-fixed stem** → B1 and C

Problems Ask AI

- Stem and cement → prevent **rigid fixation**
- **Poor bone healing** around stem
- **Bone defects**

Fixation Principles Ask AI

- **Relative stability** → especially osteoporotic bone
- **Long spanning plates**
- **Bicortical screws** when possible
  - Otherwise → unicortical screws + cables
  - Or purpose-designed plates
- **Minimally invasive techniques** → less soft tissue stripping
- Plate–stem overlap → at least 2 cortical diameters (probable) *[C1 · transcript: "less than 2 diameters"]*

Devices Ask AI

- Very new devices on the market
  - Multi-plane devices
  - **Periprosthetic locking screws** → short
    - Pass beside stem (probable)
  - **Attachment devices**

  - **Positioning pin** for cable system
  - **Proximal femoral hook plates**
  - **Distal femoral locking plates**
  - **Retrograde locked intramedullary nail**

Exam Pearl

- Fixation alone → only with **well-fixed stem** (B1, C)
- Osteoporotic bone → **relative stability**
- Long spanning plates, minimally invasive technique

Periprosthetic Acetabular Fractures Ask AI

Overview Ask AI

Acetabular

- Uncommon → less than femoral
- When present → **catastrophic**
- Cemented and cementless cups → more with **cementless**
- Intraoperative or postoperative

Risk Factors Ask AI

Acetabular

- **High impaction force** → cup insertion or reaming
  - Avoid → prevents periprosthetic fracture
- **Revision cases** → higher risk
- **Cup design** → elliptical cups, threaded cup systems
- Acetabular under-reaming (probable)
- **Osteoporosis**, radiation, dysplasia
- Each → ↑ intraoperative or postoperative fracture risk

Classification Ask AI

Acetabular

- Several classifications; this one → **5 types**

Type	Description
1	Intraoperative, during cup insertion
2	Intraoperative, during cup removal (revision)
3	Traumatic → patient fall
4	Spontaneous (probable)
5	Pelvic discontinuity (probable)

- Focus → intraoperative and postoperative fractures

Diagnosis and Management of Acetabular Fractures Ask AI

Diagnosis Ask AI

Acetabular

- Identify and assess fracture
  - **Fracture stability**
  - **Cup stability**
- Intraoperative suspicion (primary THA)
  - **Stress testing** of cup and pelvis (probable) → cup motion?
  - **Intraoperative imaging**
  - Still suspicious → **remove cup** → inspect acetabulum directly
- Postoperative
  - **AP and lateral hip views**
  - **CT** → fracture assessment
  - **CT angiography** → intrapelvic components
    - Assess vessels before component removal

Management Principles Ask AI

Acetabular

- Depends on **implant stability**
- Depends on **fracture pattern and displacement**
- Aims
  - Rigid fracture fixation → **bone union**
  - Stable cup integration
  - Restore **acetabular bone stock**
  - Re-establish **center of rotation,** offset, limb length
- Revision cases → full preparation
  - Bone graft, bulk / chip (probable)
  - **Meshes,** rings, augments
  - **Protrusio cups,** jumbo cups

Intraoperative Fracture Ask AI

Acetabular

- Assess fracture first
- Non-displaced + **stable cup** → nothing major
  - **Additional screws**
  - Close follow-up
  - Protected weight bearing
- Displaced + **unstable cup**
  - Remove cup → **plate** fixation → reinsert cup

Postoperative Fracture Ask AI

Acetabular

- Stable cup → supplementary fixation (probable) → prevents cup movement
- Non-displaced → **conservative**
  - **Protected weight bearing** → 6–8 weeks *[C2 · transcript: "6 to 8 months"]*
  - Serial follow-up
  - Watch for cup loosening (probable) despite union
- Displaced + **unstable cup** → seen on X-ray and CT

- **Cup revision** after acetabular fixation
- Acetabular **ORIF** first (e.g., posterior column (probable))
- Then cup insertion
- Stable cup unattainable → cup-cage construct (probable)

Conclusion Ask AI

- Periprosthetic fractures after THA → no longer uncommon
- Correct **classification** + implant stability assessment → essential
- **Prevention** → best, especially intraoperative fractures
- Planning → important
- Full revision preparation → no unexpected surprises
- Main goals → **fracture union** and implant stability

Must-Know

- **Adequate exposure** → before every intraoperative step
- Preoperative planning and templating → avoids size mismatch
- Management → driven by **implant stability** and fracture pattern
- Main goals → fracture union + implant stability

Diagnosis of PJI

(Criteria, Investigations and Diagnostic Algorithm for Periprosthetic Joint Infection)

Prof Dr Mahmoud Abdel Karim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 239) As spoken; ICM 2018 preoperative score: 2–5 inconclusive, 0–1 not infected, not 3–5 and <3.

Q2 (p. 242) As spoken; leukocyte esterase sensitivity is about 80%; the ~100% figure is its specificity.

Q3 (p. 244) As spoken; 28 of 65 is 43%, not 61% (61% of 65 is about 40).

Diagnostic Criteria: Background Ask AI

Origin of the Criteria Ask AI

- Central question: why does the diagnosis not work? (Dr. Mulham)
  - Criteria first introduced as the American **MSIS** (Musculoskeletal Infection Society) criteria
- Then → **International Consensus Meeting (ICM),** Philadelphia 2013
  - Dr. Wadid Obeid took part
  - Output → the blue book; excellent reference
  - Consensus session design → demanding

Why Diagnosis Is Difficult Ask AI

- Diagnosis not straightforward; can be wrong
  - Many tests used → almost none **specific** on its own

2013 Definition: Major and Minor Criteria Ask AI

Need for an Algorithm Ask AI 2013

- All available tests described in literature; many commercial kits
  - All tests for every patient → not possible
  - → Prospective **algorithm** needed → infected or not infected

Criteria Ask AI 2013

- **Major criteria** (either one)
  - Sinus tract **communicating with the joint**
  - **Two positive cultures** → same organism, two separate samples
- **Minor criteria** → three of five
  - Elevated CRP and ESR
  - Elevated synovial fluid WBC count
  - Positive **leukocyte esterase**
  - Elevated synovial **PMN percentage**
  - Positive histological analysis
  - Single positive culture

2018 ICM Scoring System Ask AI

What Changed Ask AI 2018

- 2018 → International Consensus definition changed
- No need to memorize the slide → keep it on a phone
  - Major criteria → essentially unchanged
  - Minor criteria → **weighted scoring system**

Scored Items Ask AI 2018

- Serum CRP → acute **> 100**; chronic **> 10**
- **D-dimer** → also included

- ESR → not relevant in acute
  - Chronic → **> 30**
- Points per item → summed at the end
- Preoperative + postoperative scores → combined

Item	Threshold	Points
Serum CRP	Acute > 100; chronic > 10	2
ESR	Chronic > 30; none in acute	1
Synovial WBC count	—	3
Leukocyte esterase	—	3

Score Interpretation Ask AI

2018

- Total score → three outcomes

Total score	Interpretation	Next step
≥ 6	Infected	—
3–5	Inconclusive	Molecular techniques (not yet in Egypt) *[Q1 · As spoken; ICM 2018 preoperative score: 2–5 inconclusive, 0–1 not infected, not 3–5 and <3.]*
< 3	Not infected	—

Consensus Vote Ask AI

2018

- Debated at Philadelphia → more than 2 hours
- Vote → 68% agreed, 28% disagreed, 4% abstained
- **Proceed with caution** in
  - **Adverse local tissue reaction (ALTR)**
  - **Slowly growing organisms**
  - Inconclusive results → **molecular diagnosis**

High Yield

- Major criteria → sinus tract to joint, or two positive cultures of same organism
- 2018 minor criteria → weighted score: CRP 2, ESR 1, synovial WBC 3, LE 3
- ≥ 6 infected; 3–5 inconclusive → molecular; < 3 not infected
- Caution → ALTR, slowly growing organisms

Preoperative Screening Ask AI

Clinical Assessment Ask AI

- Main message → **every patient infected until proven otherwise**
- Full history + physical examination

  - Discharge; prior history of infection
  - → **High index of suspicion**

ESR and CRP Ask AI

Screening

- Universal preoperative screen → **ESR and CRP**; no disagreement
- Both normal ≠ no infection
  - **Negative predictive value** → 95%
  - ~5% → still infected
- Still very useful
- Marker thresholds
  - ESR → chronic 30; no specific value in acute; 1 point
  - CRP → chronic 10; acute 100; 2 points
- Guidelines → both normal, patient may still be infected
- Common belief: normal ESR + CRP → enough; incorrect
- ESR → falling out of use; countless causes of elevation
- CRP → more specific
- Both elevated/abnormal → **aspiration**
  - Early aspiration favoured by some → additional results

Pitfall

- Normal ESR and CRP do not exclude infection
- NPV 95% → ~5% still infected
- ESR → many non-specific causes; CRP more specific

Joint Aspiration Ask AI

Indication and Culture Yield Ask AI

Aspiration

- Clinical judgment; abnormal ESR or CRP → **aspiration**
- Raising **culture yield**
  - **Off antibiotics 2 weeks** (14 days) before aspiration
  - Request **extended (prolonged) culture**
  - Incubation → 9–14 days
- Ordinary culture container → incorrect
  - Inoculate into **blood culture bottles** → higher yield
  - European Bone and Joint Infection Society (EBJIS) microbiologist → yield increase, possibly up to 9×
- Both points → very important

Dry Tap Ask AI

Aspiration

- Dry tap → injecting saline **not acceptable**
  - Risk → introducing infection

Requested Analyses Ask AI

Aspiration

- Request **total and differential leukocyte count incl. PMN**
- Plus culture and sensitivity

- Common error → culture and sensitivity only

Synovial Thresholds Ask AI

Aspiration

- Synovial WBC → chronic 3,000; acute > 10,000
  - > 10,000 → possible **acute infection**
- **PMN** → subset reflecting infection; reason for differential count

Synovial marker	Acute	Chronic
WBC count (cells)	> 10,000	3,000
PMN percentage	> 90%	70% or 80%

Clinical Pearl

- Off antibiotics 14 days before aspiration
- Extended culture → 9–14 days incubation
- Aspirate → blood culture bottles, not ordinary container
- Request total + differential count with PMN, not culture alone
- Dry tap → never inject saline

Culture-Negative Infection Ask AI

- Culture-negative rate → 5% to 42%
- Negative cultures → possible repeat sample
  - Hold antibiotics 14 days
  - Or **molecular technique** → e.g. **PCR**
- Negative culture ≠ no infection
  - Many returning patients (up to five) → bacteria despite negative cultures
- Molecular techniques → address this gap

Other Tests Ask AI

Not Supported Ask AI

- **Gram stain** → no
- Systemic WBC → no
- Intraoperative Gram stain → no supporting evidence
- CT and MRI → no evidence; possibly evolving
  - Requested for other purposes, not to confirm infection

Nuclear Imaging Ask AI

- **Bone scan** → commonly requested
  - **False positive** 1–2 years after prosthesis placement
  - Positive scan ≠ infection
- Role → when diagnosis not established by other means
  - **Indium scan, sulfur colloid** scan

D-dimer and Plasma Fibrinogen Ask AI

- D-dimer → threshold **850 ng/mL**
  - Single group only
- Later study (last year) → **plasma fibrinogen** performs better
- D-dimer → not better than conventional serum markers; not superior
- Further tests → leukocyte esterase, alpha-defensin

Leukocyte Esterase Test Ask AI

Principle Ask AI LE

- Urine strip (sugar/acetone) → **leukocyte** pad → turns **pink**
- Quite good test
- Enzyme secreted by **activated neutrophils** migrated to infection site
- Likened to a **pregnancy test** → aspirate applied to strip as with urine

Reading Ask AI LE

- Colour → one-plus (+) or two-plus (++) pink
  - Sensitivity → 80% or 100% *[Q2 · As spoken; leukocyte esterase sensitivity is about 80%; the ~100% figure is its specificity.]*
  - **Two-plus** → very important finding
- Positive and negative predictive values → very high

Practical Use Ask AI LE

- Blood-contaminated sample (> 3,000 cells) → caution
- Keep a box of strips in the **operating room**
- Sample → **no blood**

Alpha-Defensin Ask AI

Properties Ask AI AD

- Covered in detail by Dr. Fouad
- **Not affected by antibiotics**
- Guidelines → good **complement** to existing diagnostic tests
- Not 100% perfect
  - **ALTR** → false positive

Synovial CRP Combination Ask AI AD

- 2014 key article → add **synovial CRP** to avoid false positives
  - Synovial, not systemic CRP
  - Combined → resolves equivocal results
- Test cost → 500, 5,000 or 4,000 (company-dependent)
  - Speaker → requests synovial CRP

Exam Pearl

- LE → enzyme from activated neutrophils; ++ pink most important
- LE sample → must contain no blood
- Alpha-defensin → not affected by antibiotics
- Alpha-defensin false positive → ALTR; add synovial CRP

Intraoperative Sampling Ask AI

Number and Sites Ask AI

Sampling

- Workup from first presentation with pain → surgical stage
- **Three to five samples;** preferably **five; odd numbers**
- Different sites → **capsule, acetabulum, femur;** both sides
- → Most accurate yield for infection diagnosis

Every Revision Ask AI

Sampling

- Sample in every revision → yes
- Prior aspirate cultures → repeat still matters
  - Aspirate organism may differ from intraoperative tissue organism (two different sites)
  - Valid in **polymicrobial infections**
  - Especially **low-virulence organisms**

Why Odd Numbers Ask AI

Sampling

- Single sample → may be **contaminated**
- Two → one against one
- Three → two against one
- Five → three or more against two or one → higher yield rate

Bone Culture Ask AI

Sampling

- Two-stage revision → bone culture to raise yield?
  - Consensus → **inconclusive;** can't recommend for or against

Must-Know

- 3–5 intraoperative samples, preferably 5, odd numbers
- Sites → capsule, acetabulum, femur, both sides
- Sample in every revision; aspirate organism may differ
- Bone culture in two-stage → no recommendation for or against

Frozen Section and Sonication Ask AI

Frozen Section Ask AI

- Useful test → if eminent, well-versed pathologist available
- Threshold → **5–10 PMN per high-power field**
- UK fellowship experience → reliable pathology service
  - Result by phone → within 15–20 minutes

  - Meanwhile → clean and debride
  - Then → proceed, or **staged (two-stage) revision**

Sonication Ask AI

- Good test → positive = high likelihood of infection
- Cultures commonly negative → sonication adds yield
- Removed **implant** → device → **disrupts biofilm** → fluid extracted
  - Fluid → culture
  - Yield → better and more targeted than ordinary culture
- Use when possible; especially **culture-negative** cases

Molecular Diagnostics Ask AI

- Last resort → still uncertain after all other tests
  - New technique; the future; already available abroad
- Ordinary cultures → 28 of 65 infected cases positive (61%) *[Q3 · As spoken; 28 of 65 is 43%, not 61% (61% of 65 is about 40).]*
- **Next-generation sequencing (NGS)** → ~90%
  - 61% → 90%
- Detects more infected cases
  - Rarely positive in non-infected cases → not 100% perfect

ICM Philly Application Ask AI

- **ICM Philly** (Philadelphia) → Android application
- Diagnosis section → enter available data → calculates **score**
- Continues data entry → suggests **next step**
- Strongly advised for everyone

Take-Home Message Ask AI

- Diagnosis → many elements: aspiration, culture, ESR, CRP
- **Multifactorial process**; no single test enough
  - Not **Synovasure** alone
  - Not ESR alone; not CRP alone
- **Every patient infected until proven otherwise**
  - Painful total hip → assume infection
- Read the guidelines → incorporate and integrate into practice

Must-Know

- Every patient infected until proven otherwise
- Painful total hip → assume infection
- No single test sufficient → multifactorial diagnosis

Management of PJI

(Surgical Strategies for Periprosthetic Joint Infection)

Prof Dr Mohamed Abdel Moneim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 251) “40 g pack: more than 4 g” → “40 g pack: no more than 4 g”

C2 (p. 252) “patient has a sinus” → “no sinus”

Aim of the Lecture Ask AI

- Infected joint → close follow-up → early intervention, before infection establishes
- First presentation: clinic, deteriorating joint → surgical management
- Single goal: **management plan** of periprosthetic joint infection (PJI)
  - Focus: treatment, not diagnosis of infection
- Infected patient presents → surgeon needs a definite plan, not discharge without one
- Aims
  - Concept of PJI management
  - Management techniques
  - Advantages of each technique
  - Disadvantages of each technique
  - When each technique should not be used

Classification of Infection Ask AI

- Management starts with classification → **acute** vs chronic
- Acute — definition 1: presentation within 4 weeks of **index operation**
- Acute — definition 2: hip replacement present for years
  - Sudden acute symptoms, duration < 3 weeks
  - Example: hip replaced 5 years ago → problematic within one week → **acute hematogenous** infection

Irrigation and Debridement Ask AI

- First of two options in acute infection: **irrigation and debridement**
- Wound opened within 4 weeks → debridement
  - **Superficial debridement:** skin + subcutaneous tissue
    - Limited superficial infection → may stop, no deeper exploration
  - **Deep debridement:** fascia opened → joint itself
- Prosthesis not removed → debridement around **prosthetic joint**

Irrigation Methods Ask AI

I&D

	Pulsatile lavage device	Bulb syringe
Mechanism	Irrigation + suction simultaneously	Irrigation by squeezing syringe
Pressure	High; adjustable level	Low
Preference	Not preferred	Preferred

- High-pressure device problems
  - Drives microorganisms into deeper areas otherwise unreachable
  - Picks up organisms from one area → seeds another
  - Possible damage to surrounding tissue and bone

Pitfall

- High-pressure **pulsatile lavage** → organisms driven deeper
- Organisms carried from one area → seeded into another
- Low-pressure syringe preferred

Determinants of Eradication Ask AI

I&D

- Eradication by I&D depends on
  - Good **immunity** + good general condition
  - Early intervention

Outcome Ask AI

I&D

- Success rate much lower in this patient than in others: ≈ 65–70%
  - **High-risk** patient → 16–28%

DAIR Ask AI

- Second option: **DAIR** = Debridement, Antibiotics, and Implant Retention
  - Re-intervention with debridement
  - Antibiotic administration
  - Exchange of **modular components**
  - **Implant retention**
- Joint opened → **femoral head and cup liner** always accessible → easy removal → exchange
  - → reduced **biofilm** load → higher success rate
  - Implant preserved → stable, well fixed

Success Factors Ask AI

DAIR

- Most important factor determining DAIR success?
- → **interval between index operation and DAIR**
- Modular component exchange
  - No exchange → reportedly ~4× failure risk
  - Exchange → **survival rate** 52% → 79%

High Yield

- DAIR = Debridement, Antibiotics, and **Implant Retention**
- Most important success factor → interval from index operation to DAIR
- Femoral head + cup liner exchanged → reduced biofilm
- Modular exchange → survival 52% → 79%

Chronic Infection Ask AI

- **Chronic infection**
  - Presentation > 4 weeks after surgery
  - Acute hematogenous infection present for some time (e.g., symptoms 2–3 months) → managed as chronic
- Management → **revision** surgery

- One-stage or two-stage

	Acute	Chronic
Postoperative	Within 4 weeks of index operation	> 4 weeks after surgery
Hematogenous	Long-standing hip; sudden symptoms < 3 weeks	Symptoms for some time (e.g., 2–3 months)
Management	Irrigation and debridement; DAIR	Revision: one-stage or two-stage

Two-Stage Revision Ask AI

- Classic technique → **gold standard** for chronic PJI

Indications Ask AI

Two-stage

- Chronic PJI
- Acute PJI or acute hematogenous infection → sometimes used from the start
- Severely **immunocompromised** patient
- Previous failed attempts
  - Repeated operations → ↑ morbidity, ↑ cost, ↑ **arthrofibrosis**, worse final outcome
- Acute PJI → DAIR + other measures all failed → ultimately two-stage revision

Technique Ask AI

Two-stage

- **First stage**
  - Removal of all implants + all cement
  - Good debridement
  - Spacer of **antibiotic-loaded cement**
- **Intermediate (interim) period** → wait until patient completely free of infection
  - Nutritional optimisation helpful
  - **VAC** (vacuum-assisted closure) — like an implant — stated as contraindicated in PJI
- **Second stage:** spacer removal → definitive reimplantation

Must-Know

- Two-stage revision → **gold standard** for chronic PJI
- Stage 1: all implants + cement out → debridement → antibiotic-loaded spacer
- Interim: wait until infection-free
- Stage 2: spacer out → definitive reimplantation

Three-Stage Exchange Arthroplasty Ask AI

- When is **3-stage exchange arthroplasty** considered?
- Indication: highly resistant/virulent organisms + **severe infection**
- Technique
  - Spacer insertion
  - Repeat debridement after 2–4 weeks, regardless of wound state → eradication
  - Wait for improvement → definitive reimplantation

- Spacer → repeat debridement → reimplantation: three operations

Antibiotic-Loaded Cement for Spacers Ask AI

- Cement: highly **porous**
- **Hand mixing** preferred over vacuum mixing → more pores → ↑ antibiotic elution
- Antibiotic requirements
  - **Heat-stable** → withstands heat of cement polymerisation
  - **Water-soluble** → elutes into surrounding tissues
- **Combination of antibiotics** better than single antibiotic
  - Second antibiotic → ↑ elution of first (**synergism**)
  - Organism gentamicin-sensitive → not **gentamicin**-loaded cement alone; add second antibiotic
- Dose: antibiotic < 10% of total **cement weight**
  - Example: 40 g cement pack → ≤ 4 g antibiotic *[C1 · transcript: "40 g pack: more than 4 g"]*
  - Absolute maximum 8 g → more weakens cement **mechanically**
  - Recommended maximum 4 g; range 0.5–4 g by antibiotic type

High Yield
- **Hand mixing** > vacuum mixing → porosity → elution
- Antibiotic: heat-stable + water-soluble
- Combination > single antibiotic → synergism
- Antibiotic < 10% of cement weight: 40 g → 4 g
- Never > 8 g → cement weakened

Systemic Antibiotics Ask AI

- **IV antibiotics** → 2 weeks
- Organism identified + general condition improved
  - → switch to **oral antibiotics** → 4 weeks

No Drug Holiday Ask AI

Antibiotic

- Classic approach: antibiotics for a period → stopped for a period → confirm no persistent infection → reimplantation
- Found to be a poor idea
  - ↑ **drug resistance**
  - ↑ wound complications / reinfection
- → antibiotics not stopped before reimplantation (**no drug holiday**)

Pitfall
- Stopping antibiotics before reimplantation → **drug holiday**
- → ↑ drug resistance, ↑ wound complications / reinfection
- Antibiotics continued up to reimplantation

Spacers Ask AI

- Many antibiotics addable to cement → must be heat-stable + water-soluble
- Antibiotic form → **powder**, not liquid
- Spacer type → static or articulating
- **Static spacer:** cement block inside acetabulum, nothing more
- **Articulating spacer:** several types
  - **PROSTALAC** (Prosthesis of Antibiotic-Loaded Acrylic Cement): very simple total hip → patient moves
  - **Preformed** (ready-made) spacer
  - **Intraoperatively moulded** spacer → shaped by hand during surgery
    - Cement over **Steinmann pin** or wires → Thompson shape

Advantages Ask AI

Spacer

- Articulating spacer
  - ↑ **joint mobility**
  - Prevents soft-tissue contracture
  - Facilitates reimplantation
  - High local bactericidal antibiotic levels

One-Stage Revision Ask AI

- **One-stage revision:** single operation, everything revised

Selection Criteria Ask AI

One-stage

- **Organism** criteria
  - Identified, not unknown
  - Not highly virulent/resistant
  - Sensitive to antibiotic addable to cement
- **Patient** criteria
  - Generally stable condition
  - No sinus *[C2 · transcript: "patient has a sinus"]*
  - Good soft-tissue/bone quality and viability

Goals and Advantages Ask AI

One-stage

- Goal: operation → **mobilisation** next day → finished
  - ↓ time
  - ↓ cost
  - ↓ hospital stay
  - Rapid return to function
  - Better outcome → no multiple operations, no interim period
- Results comparable, very good → **selected cases** only
- Not every case; "saw someone do it and it worked" → no justification
  - e.g., **Staphylococcus** case → one-stage, went well → not a rule

Exam Pearl

- Organism: identified, not highly virulent/resistant, sensitive to cement antibiotic
- Patient: stable, **no sinus**, good soft-tissue/bone quality
- Results comparable → selected cases only

Resection Arthroplasty Ask AI

- Other management types → very selected cases
- **Resection arthroplasty** (Girdlestone)
- Everything removed → hip left free, no prosthesis

Indications Ask AI

Girdlestone

- **Non-ambulatory** from the beginning
- Very poor bone stock
- Poor soft-tissue coverage
- **Highly resistant** organism
- Poor general condition → cannot undergo multiple operations, repeated attempts
- Multiple previous failed attempts (e.g., two-stage revision)
- Image: patient after resection arthroplasty

Chronic Suppressive Antibiotics Ask AI

- **Chronic suppressive antibiotics** → antibiotics only
  - Single indication: patient unfit for surgery → would die if operated on
  - Reason: antibiotics stopped → 80% failure; no eradication of infection

Clinical Pearl

- Suppressive antibiotics → **unfit** patient only
- Antibiotics stopped → 80% failure

Summary Ask AI

- Know management **decision factors**
- Key questions: operation timing, symptom onset → acute vs chronic
- Management **tailored** to each case → good surgical experience
- Antibiotics: major role → careful choice of agents

Option	Setting / indication
Irrigation and debridement	Acute; within 4 weeks; prosthesis retained
DAIR	Acute; modular components exchanged; implant retained
Two-stage revision	Gold standard for chronic PJI; immunocompromised; failed attempts

Option	Setting / indication
Three-stage exchange	Highly resistant/virulent organism; severe infection
One-stage revision	Selected cases: known, non-virulent, cement-sensitive organism; stable patient; no sinus
Resection arthroplasty	Non-ambulatory; poor bone stock/soft tissue; resistant organism; failed attempts
Chronic suppressive antibiotics	Unfit for surgery

Loosening

(Osteolysis, Aseptic Loosening and Clinical Work-up)

Prof Dr Sherif Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Overview: From Wear to Loosening Ask AI

- **Aseptic loosening** → mechanism step by step
- Wear → **particulate debris** → inflammatory response
- Radiological expression → **osteolysis** → loosening
- Sequence: wear → debris → osteolysis → loosening on **X-rays**

High Yield

- Wear → particulate debris → inflammatory response → osteolysis → loosening

Learning Outcomes Ask AI

- **Osteolysis** → brief description
- **Clinical picture** of loosening
- Identifying loosening:
  - **Cemented** total hip
  - Cementless total hip

Osteolysis Ask AI

Definition Ask AI

- **Time-dependent process**
- Inflammatory reactions → mainly **cell-mediated**
- Activated by **wear debris**

Debris Sources Ask AI

- Commonest → **polyethylene**
- Others:
  - **Cobalt**
  - Titanium
  - **Methyl methacrylate** cement

Biologic Response Ask AI

- Mediated by:
  - **Macrophages**
  - Chemical mediators:
    1. Interleukin (likely IL-1, as heard)
    2. **Interleukin-6**
    3. Prostaglandin (likely E2, as heard)
- Mediators = **bone-resorbing cytokines** → periprosthetic resorption → loosening

Debris Pathway Ask AI

- **Femoral osteolysis** → debris tracks down canal:
  - Bone–cement interface → cemented
  - Bone–prosthesis interface → **press-fit** prosthesis
- Same mechanism → **acetabular** side

  - Acetabular loosening → often not diagnosed on X-rays

Particle Generation Ask AI

- Wear modes (as heard): adhesive, abrasive, fatigue
- Effects (as heard): local or systemic
- Two particle sources → **wear** and corrosion
- Lecture focus → identifying loosening, not wear or osteolysis

Clinical Picture of Loosening Ask AI

- Presentation → hip replacement + **pain**
- Loosening → **septic** or aseptic

Feature	Septic loosening	Aseptic loosening
Pain pattern	**All the time**	Activity-related
Timing	Worse **at night**	**Start-up pain**; worse on movement
Fever	Possible; most **afebrile**	—
Diagnosis	**Laboratory** investigations + X-rays	—

- Pain site (as heard) → groin or thigh
- Sometimes anterior pain (as heard) → examination localises true source
- Pain ↑ with **weight bearing**
- Investigations → ↑ **ESR** and CRP, + further work-up

Investigations and Radiographic Assessment Ask AI

- **CRP** and ESR → important
- X-rays → **AP** + lateral view → look for loosening
- Treatment → **revision**
  - Outside scope → basic total hip course
- Diagnosis easier → **femur** than acetabulum
- Acetabular cases → next session case presentation
- Many cases → loosening easy to diagnose
- **Diagnostic accuracy:**
  - Femoral → 90%
  - Acetabular → 65%
- Greater problem → **cemented** components
  - Cemented → easier to see around femoral component

Radiolucent Lines Ask AI

Lucency

- Radiolucent lines ≠ necessarily **loosening**
- Common around **cementless** hips → may reflect design
- Key point → lines **not progressive**

> **Pitfall**
> - Radiolucent lines around a cementless stem may be design-related
> - Loosening concern → only if lines are progressive

Follow-up Ask AI

- **Follow-up** mandatory → patients in Egypt disappear once well
- Schedule → yearly → every 3 years → every 3–5 years
- Practice → rarely seen after 3 years unless **symptomatic**

Radiographic Zones Ask AI

- Zones needed → **measure** loosening

Femoral — Gruen Zones Ask AI

- **Gruen zones** → zone 1 to zone 7
  - Zone 1 → lateral
  - Zone 4 → tip
  - Zone 7 → medial

Acetabular Zones Ask AI

- Zone 1 → lateral (**superior third**)
- Zone 2 → middle third
- Zone 3 → inferior third
- Always → superior, middle, inferior thirds

Cementless Stem Loosening — Engh Classification Ask AI

- Cementless stem fixation → **Engh classification**

Engh type	Stem status
Stable bony ingrowth	Fixed; not loose
Stable fibrous ingrowth	Stable
Unstable fibrous ingrowth	Loose

Stable Bony Ingrowth Ask AI

Engh

- Used to judge **cementless prosthesis** → loose or not
- Signs of accepted (fixed) stem:
  - **Spot welds** → bone hypertrophying at site
  - **Reactive bone** lines around stem
  - **Remodeling** → reactive bone between stem and cortex
- Example shown → **stable bony ingrowth**
- Minor signs:
  - **Stress shielding**
    1. Calcar → compare bone quality / **cortical density** site to site

    - **Calcar atrophy** labeled
    - **Diaphyseal fitting** → fixed distally → no proximal load
    - Unloaded area → **resorption;** load through diaphysis
    - Normal finding → diaphyseal fitting
  - **Absence of pedestal**
- **Pedestal** → defined below

Stable Fibrous Ingrowth Ask AI

Engh

- No **spot welds**
- No **migration**
- **Parallel sclerotic lines**
- X-ray → no significant sign of loosening, nor of **fixation**

Unstable Fibrous Ingrowth Ask AI

Engh

- **Progressive radiolucent lines**
- **Subsidence**
- **Pedestal formation**
- **Subsidence** → unfixed prosthesis moves downward → position change
  - Stem descends → measurable distance
  - Drifts **laterally** → varus
- **Pedestal** → bone hypertrophy at stem tip → attempt to stop subsidence

High Yield

- Engh: stable bony / stable fibrous / unstable fibrous ingrowth
- Fixed stem → spot welds, reactive bone, remodeling
- Minor signs → stress shielding, absence of pedestal
- Loose stem → progressive lucencies, subsidence, pedestal

Cementless Cup Loosening Ask AI

- **Migration** > 2–5 mm
- Radiolucent lines → **all three zones**: superior, middle, inferior
- Lines > 2 mm
- **Progressive** over 2 years
- Pain + very obvious lines → **loose cup**
- Key idea → lines must be **progressive**
- **Change in inclination** (± version, as heard)
- Case → inclination (± version) change after **four years**
  - No other sign present
  - Change in inclination alone → **loose cup**

Exam Pearl

- Change in cup inclination alone defines a loose cup

Cemented Stem Loosening Ask AI

Classification Ask AI

Cemented

- Most textbooks → **possible / probable / definite** loosening
- **Possible loosening** → radiolucent lines in 50–100% of interface
  - Case → cement–femur lucency, 50–100% of total interface
  - Progression check → followed **5 years**
  - Re-presented → **fracture neck of femur**, other side
  - Index side → no complaints, appearance unchanged at 5 years
  - Cause → poor initial **cementing technique** → radiolucency
- **Probable loosening** → radiolucent line around whole stem (as heard)
  - ± Surrounding changes
- Debris tracks down → **osteolysis** in this area
- **Definite loosening** → four X-ray signs:
  - Stem fracture
  - Cement fracture
  - Radiolucency → **cement–implant interface**
  - Change in stem position
- Each sign → own X-rays

Grade	Radiological finding
Possible	Radiolucent lines in 50–100% of interface
Probable	Radiolucent line around whole stem (as heard) ± surrounding changes
Definite	Stem fracture / cement fracture / cement–implant lucency / stem position change

Definite Signs Ask AI

Cemented

- Sign 1 → **stem fracture**
  - Broken stem → sure diagnosis of loose cemented stem
  - Mechanism → **bending cantilever**
    - Distal stem well fixed in bone → proximal part breaks

Exam Pearl

- Mechanism of cemented stem fracture → bending cantilever
- Distal stem well fixed → proximal part breaks

- Sign 2 → **cement fracture** → prosthesis loose
  - Key in **periprosthetic fractures** → loose or not changes classification
  - Stable stem → **fracture fixation** without stem revision
  - Cement fracture + radiolucent lines → loose → **revise stem**
- Sign 3 → **radiolucent lines** between cement and implant

Clinical Pearl

- Periprosthetic fracture: loose or fixed stem changes the classification
- Fixed stem → fracture fixation without stem revision
- Cement fracture or lucent lines → revise stem

Change in Stem Position Ask AI

Cemented

- Sign 4 → **change in stem position;** three types
- **Subsidence** → prosthesis subsides downward
  - Expected: **air-filled centralizer** → stem subsides 1–2 mm (per Dr. Ahmed Hazem)
  - Purpose → fit tightens
  - 1–2 mm → normal in **first year**
  - 2–5 mm → concern
  - > 2 mm → abnormal; centralizer allows no more space
  - Measured: **greater trochanter tip** or head–neck junction → stem tip
  - Two types:
    1. Cement + stem subside together within femur
    2. Stem subsides within cement mantle → **pistoning**
- **Medial midstem pivot**
  - Fixed pivot point → medial, mid-stem
  - Prosthesis rocks around it → moves **laterally**
  - Rocking → radiolucent lines appear
  - Lucency → proximal medial (**superomedial**)
  - Lucency → distal lateral (**inferolateral**)
- **Calcar pivot** (distal toggle)
  - Only calcar fixed → whole distal part loose
  - Distal stem moves right and left → **windshield-wiper effect**
  - **Ballooning** → distal stem
  - Distal area widens
  - Supported proximally, loose distally → keeps wiping back and forth

Mode	Fixed point	Motion	X-ray sign
Subsidence	None	Downward; stem alone in cement = pistoning	GT tip / head–neck → stem tip distance
Medial midstem pivot	Medial mid-stem	Rocking → lateral	Superomedial + inferolateral lucency
Calcar pivot	Calcar	Distal windshield-wiper	Distal ballooning

Must-Know

- Definite cemented stem loosening → four signs
- Stem fracture, cement fracture
- Cement–implant radiolucency
- Change in stem position: subsidence, medial midstem pivot, calcar pivot
- Subsidence 1–2 mm in first year → normal

Cemented Cup Loosening Ask AI

- Example → cemented cup **postoperatively** vs 7 years
- Sign 1 → **radiolucency**
  - Normal **cement mantle** → lucency / bone resorption all around cup
  - Lines > 2 mm, extensive, **progressive**
  - Over years → **all three zones:** superior, middle, inferior
- Sign 2 → **migration** or subsidence
  - Direction → superior or medial
  - Medial → entering **protrusio** → loose cup
  - ± **Medial wall fracture**
- Sign 3 → **change in inclination** → loose cup
- Other signs:
  - **Cup wear**
  - Cup or cement fracture → rare

Sign	Cementless cup	Cemented cup
Migration	> 2–5 mm	Superior or medial → protrusio ± medial wall fracture
Radiolucent lines	All three zones; > 2 mm; progressive over 2 years	All around cup; > 2 mm; extensive; progressive; all three zones
Inclination	Change (± version, as heard) → loose	Change → loose
Other	—	Cup wear; cup or cement fracture (rare)

Take-Home Message Ask AI

- **Osteolysis** → definition
- Clinical picture:
  - **Aseptic vs septic** loosening
  - Start-up pain
  - Related laboratory investigations
- **Radiological diagnosis** → loosening of cemented
  - And cementless components; management

Instability

(Causes, Diagnosis and Management of Dislocation After THA)

Prof Dr Ahmad Azzam

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 268) as spoken; 70% of the first-year 75% is ~52%, not 90% — two conflicting figures

Instability After THA — Overview Ask AI

Learning Objectives Ask AI

- Instability → many contributing factors
- Objectives
  - Magnitude of the problem
  - **Risk factors** for instability
  - **Multifactorial** nature of the problem
  - **Management algorithm** for instability after total hip arthroplasty (THA)
- **Javad Parvizi:** despite decades of progress in the profession
  - No answer yet to instability after hip arthroplasty
  - Advances in implants and surgical technique → problem not yet behind us

Magnitude of the Problem Ask AI

- **Incidence:** ~3% of primary THA in most series
  - Higher after revision THA
- **Timing:** early vs late dislocation
  - Late → after ~5 years from surgery
  - Early → before that
- 75% of instability → **first year** (12 months)
  - Of these, ~70% → first 3 months
  - Also quoted: ~90% of dislocations within first 3 months *[Q1 · as spoken; 70% of the first-year 75% is ~52%, not 90% — two conflicting figures]*
- **Cumulative risk** persists after first year
  - ~1% per year → first 2–3 years
  - Then ~1% every 5 years
- **Late dislocation** (> 5 years)
  - More than half recurrent
  - Most → revision surgery

High Yield

- Incidence ~3% after primary THA
- 75% within first year; ~70% of these within first 3 months
- Cumulative risk ~1%/year for 2–3 years, then ~1% per 5 years
- Late (> 5 years) → mostly recurrent, mostly revision

Diagnosis Ask AI

- Same as native hip dislocation
  - Sudden **pain**
  - **Loss of function**
  - **Rotational deformity** → apparent immediately
  - **Limb shortening**

Risk Factors Ask AI

Categories Ask AI

- Two axes → patient-related vs surgical; modifiable vs non-modifiable

Category	Modifiable?	Note
Patient-related	Essentially non-modifiable	Can still be optimised
Surgical	Modifiable	Totally avoidable

Patient-Related Factors Ask AI

- **Spinopelvic pathology** / lumbar spinal fusion → ↑ instability
  - Alters hip mechanics and stability
  - Intraoperative effect → **component positioning**
  - Postoperative effect → hip **dynamics**
    - Sitting and standing → ↑ chance of instability
- **DDH** → higher incidence of instability (see Hip Dysplasia)
- Anatomic factors
  - **Abductor insufficiency**
  - **Trochanteric malunion / non-union** — esp. with migration
    - **Fibrous non-union** → better prognosis than non-union with migration
  - Higher preoperative **range of motion**
  - **Spinopelvic dysfunction**

Surgical Factors Ask AI

- Modifiable, totally avoidable
  - **Component malposition** → most important, highest incidence, commonest cause of dislocation
  - **Primary stability** of implant (primary fixation)
  - **Surgical approach** — posterior vs anterior
  - **Soft tissue tensioning** and capsular repair
  - **Larger femoral head** use
  - **Surgical expertise**
    - Low annual THA volume → ~50% higher dislocation rate
    - **Learning curve** → very important

Clinical Assessment and Work-up Ask AI

History Ask AI

- Beyond diagnosing the dislocation, establish
  - **Time** from surgery to dislocation
  - **Surgical approach** used
  - **Incident position** — position in which hip dislocated
  - Clinical signs of **infection**

Laboratory Ask AI

- **Infection markers** → if infection suspected
  - **Aspiration** → positive markers or clinical judgement

Imaging Ask AI

- Standard AP and **cross-table lateral** radiographs
- **Standing and sitting views** → spinopelvic mechanism, possible missed problem
- **CT** → accurate component position
  - Esp. **version** of acetabular and femoral components
- **MRI with MARS** (metal artifact reduction sequence) → late instability
  - Suspected **osteolysis** and wear

Classification of Instability Ask AI

Dorr Classification Ask AI

- Type 1 → **Positional**
- Type 2 → **Soft tissue imbalance**
- Type 3 → **Component malpositioning**

Wera Classification Ask AI

- Based on 75 revisions of total hips for dislocation → six types

Type	Cause
1	Acetabular malposition
2	Femoral malposition
3	Abductor insufficiency
4	Impingement
5	Late wear
6	Unresolved (unknown) etiology — no diagnosis or main factor after full work-up

- Highest recurrent dislocation after revision → **abductor insufficiency** group
  - Lecturer's view: revision protocol only partially addresses abductor insufficiency

Exam Pearl

- Wera: six types, from 75 revisions for dislocation
- Worst recurrence after revision → abductor insufficiency
- Dorr type 3 → component malpositioning

Spinopelvic Dynamics Ask AI

Normal Mechanism Ask AI

- Standing → sitting → accommodative change in pelvic position
  - Pelvis tilts posteriorly ~20–35°
- **Radiological indices** of spinopelvic dynamics
  - **Pelvic incidence**
  - **Sacral slope**
  - **Pelvic tilt**

Dynamic Component — Fusion or Stiff Spine Ask AI

- **Spinopelvic fusion**
- Limited spinal **range of motion**
- Either → **accommodative mechanism** fails
- Sitting normally → pelvis tilts backwards
  - ↑ **acetabular anteversion**
  - Anterior acetabular rim moves away from femoral neck → no **impingement** in sitting
- Mechanism lost → ↑ instability in sitting

Static Component — Fixed Sagittal Deformity Ask AI

- **Fixed sagittal spinal deformity** with loss of lumbar lordosis (flat back) → altered mechanics
- Compensation for upright posture → pelvis tilts backwards
  - → ↑ instability in standing

	Dynamic	Static
Cause	Spinopelvic fusion / limited spinal ROM	Fixed sagittal deformity (flat back)
Pelvic behaviour	No posterior tilt on sitting	Compensatory posterior tilt
Unstable position	Sitting	Standing

Outcome and Planning Ask AI

- Instability after **spinal fusion** ~8%
  - Despite correct technique and component position
- → address more than one factor
- Preoperative plan
  - Assess **deformity**
  - Assess spinal **mobility** vs stiffness
  - Plan **implant design**

Hip Dysplasia Ask AI

- **DDH** → higher instability rate; more than one factor
- **Component positioning** → highly relevant
  - **Shallow acetabulum** → more medialized cup

  - **Deficient walls** → exaggerated anteversion
    - Both acetabular and femoral components

Component Positioning Ask AI

- **Component malposition** → most important, most modifiable surgical factor

Lewinnek Safe Zone Ask AI

- Target zone for cup position
  - **Anteversion** 15 ± 10°
  - **Inclination** 40 ± 10°
- Within zone ≠ guaranteed stability
- Later studies questioned safe zone → other targets proposed

Combined Anteversion Ask AI

- **Combined anteversion** = acetabular + femoral anteversion
  - One component compensates for the other
  - Optimal → 30–50°

Must-Know

- Lewinnek: anteversion 15 ± 10°, inclination 40 ± 10°
- Safe zone ≠ guaranteed stability
- Combined anteversion target → 30–50°

Pitfalls in Achieving the Safe Zone Ask AI

- Narrow safe zone → risk in more than one direction → precise positioning needed
- Patient factors (**obesity**, female pelvis) → ↑ error
- **Lateral decubitus** → misjudged cup position
- **Gender** differences in pelvis
- Pelvic position change → lateral decubitus vs supine
- **Residual osteophytes**, esp. anterior wall → intraoperative misjudgement of cup version
- **Wall deficiencies**
- **Distorted femoral version** (table)

Femoral version	Causes
Increased anteversion (native)	DDH, juvenile rheumatoid arthritis
Less anteverted / retroverted (native)	Previous trauma, Legg–Calvé–Perthes disease
Neutral / less anteverted neck	Lower neck cut

Ensuring Correct Position Ask AI

- Intraoperative reference aids → cup oriented to fixed anatomical landmark
  - e.g., **transverse acetabular ligament**

Surgical Approach Ask AI

- Dislocation rate → anterior approach lower than **posterior approach**
- More epidemiological and **anatomical risk factors** → lower-risk approach preferred

Soft Tissue Tension Ask AI

- **Soft tissue tension** → maintains coaptation of femur and acetabulum
  - Influenced by many factors
- Restore offset and leg length
  - Offset in vertical and anteversion planes
  - Unqualified "offset" → usually **medial offset**
  - Medial offset + leg length restoration → key to correct tension
- Not a discrete surgical step → depends on **preoperative planning**
  - Component size
  - Component position
  - Offset and leg length restoration
- **Posterior soft tissue repair** → better tensioning
- **Trochanteric non-union** with migration → more instability than fibrous non-union
  - **Trochanteric advancement** → may improve stability in some cases

Patient Education and Support Ask AI

- Patient repeats **instructions** before discharge
  - Restates them at follow-up visits
- Maintain patient **independence**
  - **Assistive tools** for getting up from / sitting on bed
  - → no violation of instruction safe zone when getting out of bed
- **Caregivers** → major help
  - Dementia, alcoholism, cognitive impairment

Femoral Head Size and Capsular Repair Ask AI

Femoral Head Size Ask AI

- **Larger femoral heads** → among best measures against instability
- Biomechanics of larger head
  - Greater **jump distance**
  - Greater range of motion before **impingement**
  - → lower dislocation risk
- Wear concern
  - Conventional polyethylene → higher wear
  - **Highly cross-linked polyethylene** → improved
  - Concern persists → wear / trunnion problems with metal heads

Capsular Repair Ask AI

- **Capsulectomy** without capsular repair → major cause of instability
  - Unrepaired capsule → early postoperative dislocation, ↑ instability

Management Ask AI

First Step Ask AI

- Rule out **infection** first

Treatment by Type Ask AI

- Dorr types 1–2 (positional, soft tissue imbalance) → best treated without component revision
- Options
  - **Liner exchange**
  - **Elevated-rim liner**
  - Larger head
  - **Dual mobility** or constrained liner
  - **Trochanteric advancement**

Closed Reduction Ask AI

- **Closed reduction** → most successful in early, first-time dislocations
  - **Sedation and analgesia** usually enough; general anesthesia rarely needed
  - Longitudinal traction under **image intensifier** → abduction → usually reduces
  - Traditional hip reduction methods also usable
- Polyethylene liner may **dissociate** during dislocation, before patient arrives

Pitfall

- Liner may dissociate during the dislocation episode
- Capsulectomy without repair → early postoperative dislocation
- Always rule out infection first

Post-Reduction Mobilization Ask AI

- Recurrent dislocation → **hip abduction brace**
  - ~20° abduction
  - Flexion limited to 60°
  - Duration → 6 weeks to 3 months

Isolated Head/Liner Exchange Ask AI

- Literature: **isolated head/liner exchange** with larger head → well-positioned components
- Lecturer: selective use
  - Even 10° malposition → component revision required
  - Shell not changed → larger head preferred

Increased Constraint Ask AI

- Methods to increase constraint → **dual mobility**, constrained liners
  - Useful in **multifactorial** instability
  - Many patients → multifactorial or **idiopathic** (cause unknown)

Soft Tissue Reconstruction Ask AI

- Preferred in proven **abductor deficiency**
  - **Achilles tendon allograft** or synthetic graft

Resection Arthroplasty Ask AI

- **Last resort**
  - Multiple dislocations
  - Severely debilitated
  - Failed appropriate management
  - Non-ambulatory
  - Severe cognitive impairment

Clinical Pearl

- Closed reduction → best in early, first-time dislocation
- Traction under image intensifier, then abduction
- Recurrent → brace: ~20° abduction, flexion ≤ 60°, 6 weeks–3 months

Take-Home Message Ask AI

- Instability → important, common, highly **multifactorial**
- **Prevention** better than treatment → do the primary THA correctly from the start
- Search for the **true cause** → apply appropriate management
- **More than one treatment modality** → liner and soft tissue improvement best options
- **Component malposition** and loosening
  - → best treated by **revision**

Cause	Treatment
Positional / soft tissue imbalance	Liner exchange, elevated rim, larger head, dual mobility / constrained liner, trochanteric advancement
Multifactorial / idiopathic	Dual mobility / constrained liner
Abductor deficiency	Soft tissue reconstruction (Achilles allograft, synthetic graft)
Component malposition / loosening	Revision
Failed management, debilitated, non-ambulatory	Resection arthroplasty

LLD Nerve Palsy

(Neurovascular Injury, Leg-Length Discrepancy and Nerve Palsy)

Prof Dr Khaled Fawzy

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

- **C1** (p. 283) “shortening less than 1 cm, 8%” → “shortening more than 1 cm, 8%”
- **C2** (p. 286) “shortening less than 1 cm, 8%” → “shortening more than 1 cm, 8%”

Vascular Injury — Anatomy, Incidence and Mechanisms Ask AI

- Session: two talks → **neurovascular injury**, then leg length discrepancy (LLD)
- Hip muscles stripped → complex regional neurovascular anatomy exposed
  - Not limited to **sciatic nerve**, femoral and external iliac vessels
  - Numerous other vessels present
  - **Superior gluteal vessels** → not to be forgotten
- **Vascular injury** incidence: 0.2%–0.3%
- Commonly injured vessels
  - **Iliac**
  - **Femoral**
  - **Superior gluteal**
  - **Obturator**
- Mechanisms of injury
  - **Sharp instruments** → knife, osteotomes
  - **Blunt instruments** → retractors → compress, stretch or tear vessel
  - **Screw placement** → through acetabular shell of cementless cup

Acetabular Quadrant System Ask AI

- Acetabulum → four quadrants → **safe zone** for screws: posterior-superior
- Safe zone — structures still at risk
  - **Sciatic nerve**
  - **Superior gluteal nerve and vessels**
  - → Screws within the lines, not at their periphery
- Construction
  - Line 1: **ASIS** → center of acetabulum
  - Line 2: perpendicular to line 1
- Four zones

Zone	Quadrant	Structures at risk
Safe	Posterior-superior	Safest → screws placed here
Death	Anterior-superior	Iliac vessels → perforation → death on table or later
Cautious	Posterior-inferior	Screws up to 20 mm; over-penetration → sciatic nerve, inferior gluteal vessels, pudendal nerve and vessels
Danger	Anterior-inferior	Obturator vessels

- **Posterior-superior quadrant** → safest → screw placement site
- Pelvis viewed from inside → screw perforations and structures affected
  - **Superior gluteal** vessels
  - **Iliac** vessels

  - **Femoral** vessels
  - **Obturator** nerve
  - **Pudendal** nerve
  - → Screws placed with great care

High Yield

- Quadrant lines → ASIS to acetabular center + perpendicular
- Safe zone → posterior-superior; sciatic and superior gluteal still at risk
- Death zone → anterior-superior → iliac vessels
- Danger zone → anterior-inferior → obturator vessels
- Cautious zone → posterior-inferior → screws up to 20 mm

Vascular Injury — Risk Factors, Presentation and Management Ask AI

Predisposing Factors Ask AI

Vascular

- **Revision surgery**
- **Atherosclerosis** → vessel tears easily with malpositioned retractor
- **Previous vascular surgery** or vascular injury
- **Abnormal anatomy** → e.g. dysplasia

Presentation Ask AI

Vascular

- Intraoperative → **bleeding**; never underestimate
- Postoperative
  - **Occult bleeding** → nothing visible externally unless in drains
    - Blood pressure falls, heart rate rises
  - **Ischemia**
  - → Postoperative vigilance; regular examination

Cement-Related Injury Ask AI

Vascular

- Over-reaming → **acetabular floor** breached → cement extrudes into pelvis
  - **Heat of polymerization** → vessel injury → thrombosis
  - Postoperative erosion into iliac vessels → **pseudoaneurysm**
- Case: marked cement extrusion → vessels spared → **obturator nerve** irritated

Management Ask AI

Vascular

- Intraoperative bleeding
  - Control, transfuse, pack field
  - Call **vascular surgeon** or interventional radiologist without hesitation
  - Never underestimate
- Postoperative
  - **Angiography**
  - Repeated clinical examination
  - Vascular surgeon as soon as possible

Nerve Injury in THA Ask AI

- **Nerve injury** incidence: 1% — somewhat higher than vascular
- Nerves at risk
  - **Sciatic** → most commonly injured, especially posterior approaches
  - Then femoral, superior gluteal, obturator
  - Anterior approach → **lateral cutaneous nerve of thigh**
- Mechanisms
  - **Sharp laceration** of sciatic nerve by knife
    - Rare
  - Commoner: **retractor pressure** → traction, stretching, contusion
  - Postoperative **hematoma** → sciatic nerve compression
  - Postoperative **dislocation** → nerve injury
- Risk factors
  - **Female sex**
  - **Revision**
  - **Dysplasia**
- Approach-specific risk

Approach	Nerve at risk
Posterior	Sciatic — slightly more at risk
Lateral	Superior gluteal
Anterior	Lateral cutaneous nerve of thigh

Sciatic Nerve Injury Ask AI

Pattern Ask AI

Sciatic

- **Lateral popliteal (common peroneal) division** → commonly injured
  - Less fat between fasciculi → less protection
  - Lateral, not medial, position
  - Relatively **tethered at sciatic notch**
- → **Foot drop**: common presentation

Exam Pearl

- Peroneal division injured → less interfascicular fat, lateral, tethered at sciatic notch
- Result → foot drop

Outcome Ask AI

Sciatic

- Recovery categories

Outcome	Proportion
Good recovery	40%
Mild residual motor or sensory symptoms	45%
Severe motor or sensory deficit	15%

- Recovery possible up to 2–3 years → long regeneration distance
- **Axonotmesis** → regeneration 1 mm per day

Diagnosis Ask AI

Sciatic

- Clinical examination
- **EMG** and nerve conduction studies → level of lesion, signs of recovery

Management Ask AI

Sciatic

- Any foot drop → **foot drop splint**
- Acute postoperative nerve injury
  - Compressive hematoma → **evacuate**
  - Dislocation → **reduce**
  - Persistent palsy → **explore nerve**
  - Excessive lengthening → reoperate → **shorten leg** (see LLD)

Prevention Ask AI

Sciatic

- Identify nerve intraoperatively where necessary
- **Short external rotators** reflected laterally → sciatic nerve protected
- **Knee flexion** → relaxes sciatic nerve
- Lateral approach → no gluteal split > 5 cm above **tip of greater trochanter**
- Careful **retractor** placement

Neurovascular Injury — Summary Ask AI

- Neurovascular injuries → uncommon
- Know anatomy; care with **knife and osteotomes**
- Retractors → appropriate position
- **Sciatic nerve** → protect in posterior approach
- No screws in acetabular **death and danger zones**
- **Bleeding** → never underestimate

Leg Length Discrepancy — Normal Variation and Incidence Ask AI

- Second talk: **leg length discrepancy (LLD)**
  - Audience poll → majority have seen LLD after hip replacement
  - Test: **femur protrudes forwards** → leg possibly lengthened
- Normal variation
  - Many normal individuals → 5–15 mm difference between legs

  - Not everyone perfectly symmetric
  - Majority unaware if < 1 cm
- LLD after **total hip replacement (THR)**

Finding	Incidence
Shortening or lengthening within 10 mm → unimportant, patient unaware	~72%
Over-lengthening > 1 cm	22%
Shortening > 1 cm	~8% *[C1 · transcript: "shortening less than 1 cm, 8%"]*
Noticeable or consequential difference overall	~30%

Consequences Ask AI

LLD

- Surgeon forced to **revise** hip
- Patient may **sue** surgeon
  - USA: winning case → compensation + **punitive damages**
    - e.g. $10,000 compensation + $50,000–$100,000 penalty for lack of care
  - → Care: risk of **litigation**

Preoperative LLD and Effects of LLD Ask AI

Preoperative LLD Ask AI

- Common; causes
  - **Acetabular erosion**
  - **Femoral head collapse**
  - **Subluxation**
  - **Fixed deformities**
- Example: acetabular articular cartilage erosion + damaged femoral head

Effects of LLD Ask AI

- Especially postoperative
  - **Limp** → dips on shorter side
  - Large **shoe raise** forced → very distressing
- Normal spine + LLD
  - **Pelvic obliquity**
  - **Compensatory scoliosis** or list
  - **Tiptoeing** on shorter side
  - **Knee flexion** on longer side, especially standing
- Other effects
  - **Abductor strain** → pain or ache
  - Decreased **abductor lever arm**
    - → Abductors generate more force → increased **joint reaction force**
      - → Long-term loosening and wear

Causes and Types of LLD Ask AI

Shortening Ask AI

LLD

- Shortening → may lead to **dislocation**
- Causes
  - **Short head**
  - **Acetabular cup** placed too high
  - Deeper **stem insertion**
  - Most important: **failure to restore offset** — horizontal and vertical
    - → Decreased myofascial tension

Lengthening Ask AI

LLD

- Lengthening > 2.5 cm → **nerve palsy**
  - Sciatic, femoral or both → sciatic commonest

Types of LLD after THR Ask AI

- **Type 1 — component-related**
  - Femoral component **too proud** → lengthening
  - Cup too high or too low
  - **Neck cut** too long or too short
- **Type 2** → components malpositioned in retroversion or anteversion
  - → Hip unstable → **long neck** deliberately used for stability

High Yield

- Shortening → dislocation
- Lengthening > 2.5 cm → nerve palsy; sciatic commonest
- Key cause of shortening → failure to restore horizontal and vertical offset
- Type 2 LLD → long neck to stabilize malpositioned components

Acceptable LLD and Preoperative Planning Ask AI

- Acceptable LLD
  - Mean in most studies: ~3–9 mm (figure unclear in recording)
  - **Within 1 cm** → acceptable; reinforced by Charnley
  - More → **unhappy patient**
  - Reported range: 22 mm shortening → 3.5 cm lengthening
- Minimizing LLD
  - **Preoperative examination**
  - **Preoperative planning**
  - **Intraoperative measurement and adjustment**
    - Postoperative → too late; only shoe raise left

Preoperative Examination Ask AI

LLD

- Check **fixed deformities**
- Measure **apparent and true leg length**

Preoperative Planning Ask AI

LLD

- Radiographs → **preoperative planning**
  - **Standard- vs high-offset** prosthesis
  - Stem insertion depth
  - Cup position
  - **Digital planning** → software on market
    - → Restore center of rotation
- Acetabulum → size, coverage, **level of insertion**
  - No medialization or lateralization of cup
- Stem insertion and **neck cut** → planned properly
- Femoral component
  - Restore **center of rotation of head**
  - Check component size and neck cut

Intraoperative Assessment of Leg Length Ask AI

- **Supine position** (some surgeons) → easy
  - Square pelvis → palpate medial malleoli → almost same level
- **Lateral position** → difficult, tricky
  - **Longitudinal traction** → joint surfaces separate a few mm; depends on
    - Force applied
    - Muscle relaxation by anesthetist
    - Extent of capsular release
    - e.g. lateral approach → one third glutei detached → traction against two thirds → more separation
  - **Soft-tissue tension** → longer leg → tight hip
  - **Knee-flexion test:** lateral, hip extended, knee flexed → leg kicks back on release → too long
- Measurement devices
  - Dedicated or simple: **Steinmann pin** in ilium → distance to tip of greater trochanter
    - Especially after trial reduction
  - More complex devices
  - **Computer navigation** → used in the West
    - Not every operation; not available in every hospital
  - Simple alternative: **intraoperative X-ray** on trial reduction

Restoring Offset Ask AI

LLD

  - Very important: restore **center of rotation** + horizontal and vertical offset
  - Not done → lax soft tissues
    - e.g. short neck cut + lax tissues → tendency to lengthen leg → stabilize joint, minimize dislocation
  - Longer leg → **unhappy patient**

Pitfall

- Short neck cut + lax soft tissues → leg lengthened to stabilize hip
- Longer leg → unhappy patient

Postoperative LLD and Management Ask AI

- Majority of postoperative inequality → **functional**
  - Resolving flexion soft-tissue contractures
  - Flexible pelvic obliquity
- **True LLD within 1 cm** → asymptomatic
- Symptomatic LLD > 1.5 cm → most probably **shoe lift** or shoe insert
- **Bilateral hip disease** → further chance to equalize at second operation

Revision for LLD Ask AI

- Avoid if possible → shortening risks **destabilizing joint**
- Indications
  - Significant shortening → dislocation → revise to **longer head**
  - **Severe nerve palsy** from significant lengthening → reoperate, shorten leg
    - → Nerve recovery hoped for
  - Patient intolerant of large **shoe raise**
- Shortening leg → great care not to **destabilize hip**

Leg Length Discrepancy — Summary Ask AI

- **Lengthening** commoner than shortening after THR
  - Majority within 1 cm; many patients unaware
- **Shortening** tolerated better than lengthening
  - > 1 cm not tolerated
- LLD after THR → cannot be eliminated, **can be minimized**
- **Revision** → avoid where possible

Must-Know

- Normal individuals → 5–15 mm difference; < 1 cm unnoticed
- After THR → over-lengthening > 1 cm 22%, shortening > 1 cm ~8% *[C2 · transcript: "shortening less than 1 cm, 8%"]*
- Acceptable LLD → within 1 cm (Charnley)
- Symptomatic > 1.5 cm → shoe lift or insert

PJI

(Case-Based Discussion of Periprosthetic Joint Infection)

Prof Dr Mohamed Abdel Moneim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Queried - kept as spoken, but looks wrong

Q1 (p. 296) As spoken; colistin is the nephrotoxic drug, tigecycline needs no renal adjustment - reasoning looks reversed.

Case 1 — Chronic PJI After Implant Removal Ask AI

Presentation Ask AI

Case 1

- Not yet on dialysis; potassium reportedly acceptable
- General condition reasonable for 1 year
- Previous surgeon → **hip joint removed** ~6 months ago
- Current status: **wound closed**, patient bedridden
- Difficult case: wound closed + **no prior work-up** available

Pre-operative Imaging Ask AI

Case 1

- Pre-operative images at clinic presentation
- Unusual finding: prosthesis removed but **cement left in femur**
  - Retained cement → no blood supply → avascular foreign material

Pre-operative Planning Ask AI

Case 1

- Direct re-implantation → no
- Planned **second stage** → what to request before theatre?
- **Aspiration** first → establish and confirm diagnosis before surgery
- Investigations
  - **ESR** 80 (figure 125 also quoted — unclear)
  - **CRP** 48 (normal 6)
- Hip **aspiration** — reasons
  - Wound closed → no sinus
  - No documentation of previous **cultures**
  - Without it → **organism** and antibiotic unknown
- Aspiration done → culture results obtained
- Management: **two-stage** approach → starting with first stage

Clinical Pearl

- Closed wound, no sinus, no prior cultures → **aspirate** before surgery
- Unknown organism → no targeted antibiotic

First Stage — Cement Removal and Spacer Ask AI

Cement Removal Options Ask AI

- Instruments: **burr** or others
- Two options
  - **Intramedullary removal** → through femoral canal, femur not osteotomized
  - **Femoral osteotomy** → cement removal

Case 1 Procedure Ask AI

Case 1

- Retained cement removed → thorough **debridement**
- Antibiotic-loaded cement spacer → **PROSTALAC**
- Cup: standard cemented cup, **snap-fit** type

  - Snap-fit → head held firmly → dislocates less than standard cup
  - Slightly different from a **constrained** cup

Spacer Stem Selection Ask AI

PROSTALAC

- Always choose **extra-small (thin) stem** → maximum cement around it
  - Goal: whole prosthesis **covered with cement**
  - ↑ antibiotic **elution,** ↑ antibiotic surface area
  - Allows **immediate mobilization**
- Image after first stage

Interval Antibiotic Protocol Ask AI

- Question: how are antibiotics given after first stage?
- Case 1 operated 2 years ago → older protocol (2013–2016 practice)
  - Antibiotics per **culture** results
  - Targeted by **sensitivity**
  - 6 weeks: 2 weeks IV → 4 weeks oral
  - End of 6 weeks → antibiotics **stopped**
  - ESR/CRP repeated → confirm return to normal
  - After stopping → ESR/CRP repeated again
- Antibiotic-free interval → **drug holiday**
- Current practice: **no drug holiday** → proceed to revision
  - 2 weeks IV + 4 weeks oral antibiotics
  - Then check **ESR/CRP** → usually proceed to revision
  - Option: **extend antibiotic course**, even beyond 4 weeks
  - Never an antibiotic-free rest period
- Case 1 → **revision** chosen; pre-op image re-shown

	2013–2016 protocol	Current protocol
Antibiotic choice	Culture + sensitivity	Culture + sensitivity
Course	2 weeks IV → 4 weeks oral	2 weeks IV → 4 weeks oral, may extend
After course	Stopped → drug holiday	No drug holiday
ESR/CRP	Repeated at end and after stopping	Checked → usually revision

High Yield

- Interval antibiotics: 2 weeks IV → 4 weeks oral
- **Drug holiday** → no longer recommended
- Course may be extended beyond 4 weeks
- ESR/CRP → then usually proceed to revision

PROSTALAC Spacer — Principles and Technique Ask AI

Why the Spacer Is Placed Loose Ask AI

- Q: why does a spacer become **well fixed** → hard removal → more bone removed?
- Answer: inserted at first stage, when ESR/CRP not yet good
- **Lucency** all around the cup on first-stage image
  - Cup not pressurized → simply placed → comes out easily
  - Nothing fixing it initially
  - First stage → cement only used to place it
- Construct differs from **G-stem** shape
- Cement spacer shapes familiar; hemi (head-only) form also referred to (unclear)

Rationale Ask AI

PROSTALAC

- Functions in the **interim** period; simple principle
- Stem, cup, all components completely **loose** → easy removal
- Aim: allow **weight bearing**
- **Dislocation** incidence lower than G-stem
- **Acetabular erosion** lower than hemi spacer
  - Hemi spacer moves in acetabulum → erodes it if left longer

	PROSTALAC	Comparator
Dislocation	Lower	G-stem
Acetabular erosion	Lower	Hemi spacer → moves in acetabulum, erodes if left longer

Spacer as Foreign Body Ask AI

- Antibiotic-loaded **G-stem** → elution lasts 4–6 weeks
- After 4–6 weeks → cement = **foreign body**
  - Both spacer types → foreign body
- ↓ elution → uncleared bacteria adhere → **biofilm**
- Accept G-stem → no major objection to **PROSTALAC**
- Second point (any implanted joint)
  - Patient **cured** → single revision possible → remove all, place joint, mobilize
  - Patient **still infected**
    1. → must remove a **fixed** prosthesis
    2. Bone stock loss, **osteotomies**, revision implants
    3. Unneeded procedures at both first and second stages

Loose Insertion Technique Ask AI

PROSTALAC

- Construct deliberately **loose**
  - Cup: hold in hand → apply cement by hand
  - Place in acetabular position without **pressurization**
  - Not cement first then pressurize
  - Stem: hold in hand → coat with cement → insert into femur when cement near setting

  - Result: no **interdigitation** with bone → easy removal
- Allows movement for a limited period
  - e.g. 2–3 months
  - Cannot be left long → loosening, **periprosthetic fracture**

Pitfall

- Pressurized cement → interdigitation → **fixed** spacer
- Fixed spacer + persistent infection → bone stock loss, osteotomies, revision implants
- Spacer left too long → loosening, periprosthetic fracture

Case 1 — Second Stage Ask AI

- Patient **renally impaired** → creatinine 7, about to start dialysis
- → **whole cemented revision**
- End of Case 1

Case 2 — Early Post-operative Infection Ask AI

Presentation Ask AI

Case 2

- Age 30 years; no comorbidities
- **AVN** → total hip arthroplasty 3 weeks earlier
- Pre-operative image
- Post-operative image
- Post-op joint appears good
- Presented 3 weeks after surgery
  - **Tenderness**
  - **Pain on weight bearing** (± at rest — unclear)
  - **Hotness**
  - **Serous discharge** from sutures
- At 3 weeks → **DAIR**
- Q: straight to theatre, or culture first?
- Participant answer → judged correct

Management — DAIR Ask AI

Case 2

- **Do not wait**
  - Theatre → **debridement**
  - Change **modular components** → retain implant
  - Intra-op **cultures** → targeted antibiotics post-op
- Must not delay; review whether index operation performed properly
- Signs evident: high temperature, discharge
- Delay (a week, 5 days, 9 days) → infection established → **chronic** infection
- Catch early; otherwise situation lost
- Outcome: DAIR → patient did well

Exam Pearl

- Early infection at 3 weeks → **DAIR**
- Debridement + modular exchange + implant retention + cultures
- Delay → **chronic** infection

Case 3 — Chronic PJI With a Sinus Ask AI

Presentation Ask AI

Case 3

- Male, 36 years; **multiple operations** for total hip replacement
- Last procedure 5 years ago (image shown)
- 1 year later → **sinus**
- Presents with image + sinus → management?
- Q: what first before theatre?
- Participant proposal: first stage

Pre-operative Work-up Ask AI

Case 3

- **No aspiration** needed → sinus present
  - **Culture** from sinus
  - Point of disagreement with participant ("aspiration")
- **ESR/CRP** drawn
  - Not to prove infection → sinus itself confirms it
  - Purpose: post-operative **follow-up**, as in other case
- ESR/CRP = **baseline**, even if very high with obvious clinical picture
  - Post-op repeat improved → reassuring
  - Not improved → another risk not eliminated
- Done: ESR/CRP + **culture & sensitivity**

	Closed wound, no sinus (Case)	Sinus present (Case)
Aspiration	Required	Not needed
Culture source	Hip aspirate	Sinus
Infection proof	Aspiration confirms diagnosis	Sinus confirms infection
ESR/CRP	Requested pre-op; repeated in interval	Baseline for post-op follow-up

Clinical Pearl

- Sinus present → **no aspiration**; culture from sinus
- ESR/CRP → baseline, not proof of infection

First Stage Ask AI

Case 3

- Culture & sensitivity obtained
- **2-stage revision** → spacer
  - **Articulating** spacer preferable to static spacer

  - Options: G-stem, custom-made, PROSTALAC → surgeon preference
- **PROSTALAC** used
- Large **acetabular defect** → all filled with cement
  - Unlimited cement possible
  - Much cement around stem → avoid **fixed** prosthesis
- Image after first stage → proceed to second stage
- Q: imaging needed before second stage?

Case 3 — Second-stage Planning and Reconstruction Ask AI

Planning Ask AI

Case 3

- **CT** → plan second stage
- Prepare **revision implants**
- Patient image
- CT findings
  - Coronal cut → **cavitary defect** + superior defect
  - Axial → anterior and posterior walls; mostly **cavitary**, small part possibly segmental

Reconstruction Ask AI

Case 3

- **Allograft**
- **Reinforcement ring** + cemented cup
  - Ring → cup **must be cemented**
  - No metal on metal
- **Cemented stem** preferred → possibly for extensive antibiotic loading
- Cement separates ring from cup
- **Press-fit** cup cannot sit directly on ring

Exam Pearl

- Reinforcement ring → cup **must be cemented**
- No press-fit cup on ring → no metal on metal

Case 4 — Infected Hemiarthroplasty Ask AI

Presentation Ask AI

Case 4

- **Hemiarthroplasty**, not THA → concept similar
- Female, 65 years; **renal** and cardiac disease (abdominal disease also mentioned — unclear)
- **Thompson** hemiarthroplasty 10 months ago
- 3 months of deterioration, possibly discharge
- Significant **renal** disease
- Image: Thompson prosthesis at presentation
- Previously mobile for simple activities (e.g. visiting her children)

Work-up and Antibiotics Ask AI

Case 4

- **Culture** first
- ESR/CRP high
- Culture → highly resistant **Enterobacter**
  - Sensitive only to colistin and tigecycline → the two traditionally available
  - **Tigecycline** excluded → renal concerns → colistin given *[Q1 · As spoken; colistin is the nephrotoxic drug, tigecycline needs no renal adjustment - reasoning looks reversed.]*
- Options: second stage, or **Girdlestone** (participant) → two-stage attempted

First Stage — Implant Removal Ask AI

Case 4

- Removal **without osteotomy** (same principle)
  - Ream cement gradually → **tibial locking-nail reamers**
  - Remove prosthesis gently → **preserve bone stock**
- Prosthesis removed without problems
- **PROSTALAC** → first stage
- **IV antibiotics** 2 months in hospital → wound healed, well
- Then **partial wound dehiscence** → readmitted 2 weeks, then another week → large wound
- Q: decision if she were 25 years old?
- Three weeks on → decision differs from a 25-year-old?
- Participant suggestion → **another stage**

Decision — Third Stage vs Excision Arthroplasty Ask AI

Case 4

- Options: **three-stage** procedure or Girdlestone
- **Counsel** patient and family
  - High-risk → each surgery carries specific risk
  - Third stage (change spacer, restart) vs **excision arthroplasty**
- Family decision → excision arthroplasty
  - Reason: afraid of multiple further operations
- Joint removed → **excision arthroplasty**

> **Clinical Pearl**
> - High-risk patient, repeated failure → **counsel** family
> - Options: third stage vs excision arthroplasty (Girdlestone)

Periprosthetic Fractures

(Case-Based Discussion of Periprosthetic Fractures)

Prof Dr Mahmoud Abdel Karim

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Case Presentation and History Ask AI

The Case Ask AI

- **Hip arthroplasty** in place → 7 years
- Group exercise: classify the fracture — B, C, D, or other?
- Age: 75 years
- Full history gathered before the answer

Pre-Fracture History Ask AI

- **Ambulatory status** before the injury
- **Pre-fracture pain**
  - Prior pain → loosening or **osteolytic process** already started
- **Anticoagulants** → relevant to both surgery and anaesthesia
- Trauma case → surgery not postponed a week
  - Exception: unfit for both general and **spinal anaesthesia** → wait
- **Neurological disorder** — very important
  - → drives implant choice: large head / tripolar / **dual mobility**
- Diabetes
- Hypertension
- **Diabetes** in this setting
  - Uncontrolled (glucose ~400) → postpone surgery
  - Internist or endocrinologist + anaesthesia team
  - Up to 300 sometimes accepted by anaesthetists → off guidelines
  - **Acute adjustment** of glucose sometimes performed
- Key point, above the case itself
  - Pain + **limited range of movement** since end of surgery
  - Pain since index operation → suspect **infection**
  - → ask about implant type and whether it is the relevant type

Must-Know

- Pain since the index operation → suspect infection
- Prior pain → loosening or osteolysis already started
- Neurological disorder → large head, tripolar, or dual mobility

Imaging Work-up Ask AI

Plain Radiographs Ask AI

- Imaging mandatory → X-rays before CT
- No planning on CT alone
- Required views
  - **Long lateral**
  - **Long AP**
  - Most important besides the long films

    - **AP pelvis** (both hips) → basic requirement
        - Both sides visible → contralateral comparison in **templating**
        - → gives the **offset**
- AP pelvis alone insufficient → **distal end** of prosthesis not visible
- **Long film** of entire prosthesis → AP + lateral

Other Investigations Ask AI

- **Bone scan** → loose or not; infected or not
    - False positive here (fracture) → not needed
- Limited range of movement → reason for further studies
- **CT scan** — important
    - **Bone stock**
    - Fracture **extension** not seen on X-ray
    - Displacement
    - **Implant seating**
    - **Wear process**
    - Finer bone detail → fragments, comminuted fragments
    - Extent of **comminution**

Before Calling B1 vs B2 Ask AI

Item	Purpose
Pre-fracture status	Walking? Pain?
Radiography of entire femur	AP + lateral
CT	Prosthesis stability; wear of prosthetic components
Intraoperative test	Implant stability

Pitfall

- Bone scan after periprosthetic fracture → false positive
- AP pelvis alone → distal end of prosthesis missed
- CT alone → never the basis for planning

Vancouver Classification Ask AI

Classification Systems Ask AI

- **Vancouver classification** → presented by Dr. Sayed; the more important
- **UCS** (Unified Classification System) → relevant for exams
- Two systems differ considerably → both must be known
- **Vancouver** → A, B, C only; no D
- This case → type B; B1 vs B2 to be justified

Applying the Subtypes Ask AI

- **B** → fracture around the stem
- Audience proposals, challenged by the lecturer
  - Comminution → B2
  - Displaced → B3
- Treatment based on classification → brainstorming matters
- This case → **B2:** around the stem, loose implant, good bone
- B3 proposed because "unstable" → revisited below
- Unstable prosthesis → B2, not B3
- B1 excluded → is the stem loose?
- Stem not loose → **B1**

Three Elements Ask AI

- **Location** → A / B / C
  - Proximal → greater or lesser trochanter
  - Around the stem
  - Distal
- **Prosthesis fixation** — the most important
  - Well fixed + trochanteric → A
  - Well fixed + around the stem → B1
  - Unstable / not well fixed / loose → **B2**
- **Bone quality** → loose stem may be **B3**
- Algorithm: **location** → **fixation** → **host bone**

Type	Location	Stem	Bone
A — AG	Greater trochanter	Well fixed	—
A — AL	Lesser trochanter	Well fixed	—
B1	Around or just distal to stem	Stable	—
B2	Around or just distal to stem	Unstable (loose)	Good
B3	Around or just distal to stem	Loose	Poor
C	Well distal to stem	—	—

High Yield

- Vancouver → A, B, C; no type D
- Always: location → fixation → host bone
- Fixation is the most important element
- Loose stem + good bone → B2; poor bone → B3

Ruling Out Infection Ask AI

- Poor bone possibly present from the start
- **Infection** → must be ruled out, per the infection lecture
- Preoperatively → look for active inflammation; check **inflammatory markers**
- **Examination** + history

Test	Note
Alpha-defensin	Confirmation if still needed; expensive → not routine; acceptable if affordable and in dilemma
Leukocyte esterase	Confirmation if still needed
Frozen section	Needs in-hospital expertise → very difficult in Egypt
Cultures	Always taken, even if findings normal

- **Any revision** → rule out infection
- Suspicion → **spacer** on standby, or know how to make one

Must-Know
- Any revision → rule out infection first
- Cultures sent even when the field looks normal
- Suspicion → spacer on standby

Preoperative Preparation Ask AI

Multidisciplinary Team Ask AI

- **Multidisciplinary team** → mandatory; patient can die easily
  - **Neurology** → neurological disorder; age 75–80
  - **Cardiology**
  - **Anaesthesia** → assessment; possible **ICU** need
- All must be involved
- Any history of illness or **comorbidity**

Informed Consent Ask AI

- Family told about everything
- Lecturer's consent form → sent via WhatsApp to all hospitals
- One form per operation type (4–5 types) → all known **complications** listed
- Serves as a reminder → stated to patient and family

Planning Checklist Ask AI

- **Planning** → most important step before theatre
- Never enter the operating room without knowing the plan
- Personal **checklist** → what to bring, do, have available
- Start a plan checklist → add to it with every patient

Prevention of Periprosthetic Fracture Ask AI

- Best treatment → **prevention** (augmenting Dr. Sayed's lecture)
- Better than the most successful treatment → exam answer
- During arthroplasty
  - Avoid **cortical defects**
  - Avoid **malalignment**
- Regular **radiological follow-up**
- "Catch the patient before he catches you" → detect **osteolysis** and loosening before fracture
- **Loosening** seen (Dr. Sherif Khaled) → fracture prevented → best treatment

Exam Pearl

- Best treatment of periprosthetic fracture → prevention
- Avoid cortical defects and malalignment
- Regular radiological follow-up → catch osteolysis and loosening early

Treatment of the B2 Fracture Ask AI

Revision Plan Ask AI

- Agreed type → **B2**
  - Loose implant + **bone stock** present
- Algorithm (Dr. Sayed): fix or revise → **revision ± fixation ± augmentation**
- Steps
  - Remove the **implant**
  - Remove the **cement** → may or may not be easy
  - Fractured femur → **extended trochanteric osteotomy** possible
  - **Long stem** bypassing the fracture
    - By twice the femoral diameter, or 4–6 cm (yesterday's figure)
    - Lecturer uses 6 cm → **more cortical contact**
    - Large gap → less stable construct; consensus → more cortical contact

Fixation Ask AI

- Long stem alone → fracture union?
- → **fixation** needed in addition
- Options → **plate, cable,** or cable-plate
  - All three → added expense
  - Evidence of superiority → questioned
- Cables alone may suffice in such fractures vs added plate
- Intraoperative principle
  - Place **cables** or a couple of wires
  - Assess construct **stability**
  - Add fixation only if needed

High Yield

- B2 → revision ± fixation ± augmentation
- Long stem bypasses fracture → lecturer uses 6 cm
- More cortical contact → more stable construct
- Cables first → add plate only if unstable

Instruments and Implants Ask AI

Prepared Kit Ask AI

- **Instruments** → very important; any revision fully prepared
- Plan included
  - **Dual mobility cup** → size unknown in advance
  - **Cerclage**, wire passers, Biznoska bands
- **Cables** → unavailable at that hospital
  - Alternative resembling a cable → **Biznoska bands**
- **Wagner stem** → shape shown
- Simple **plate** included → not intended for use
- Standby only; plan already known
  - **Long-stem cementless** revision (e.g., Zimmer)
  - **Long-stem cemented** stem
  - **Cemented snap-fit cups** → if dual mobility unavailable
  - **NCB periprosthetic femoral plate**
- No reinforcement **ring** → cup had to be removed in this case
- Everything may turn out loose
- Company representative → always puts a **ring** or similar items in the box
- Work in your own setting with everything available
- Cleaning and **cement removal** → may be easy in this case

Additional Equipment Ask AI

- **Flexible drill** → very important; fancy foreign instruments not available to all in Egypt
  - Drill → makes holes
- Power saw and bone graft (probable) → stored even if not intended
  - Unknown intraoperative findings, e.g.
  - Very poor **bone quality**
- **Screws** → 3.5 and 4.5 mm; all sizes and lengths
- Company representative (Hussein, Zimmer) → always present; must hold your profile or reliably bring items
- Surgical drains (probable)

Clinical Pearl

- Any revision → full instrument preparation
- No cables → Biznoska bands as alternative
- Flexible drill → very important
- Screws 3.5 and 4.5 mm, all sizes and lengths

Operative Principles and Steps Ask AI

Surgical Goals Ask AI

- Fracture + implant → addressed equally
  - **Stable fixation**
  - Preserve **fragments** and their soft-tissue attachments
  - Restore **length, offset, orientation** → as templated

Plate Options Ask AI

- Plate shown → used in another case, not this one
  - **Screws from above** + anterior and posterior screws (as Dr. Sayed showed)
  - Alternative → **Locking Attachment Plates** (Synthes)

Operative Steps Ask AI

- First step, before **reaming** → **distal cerclage**
  - Most important step; 75-year-old woman → caution
  - **Shattering** risk → prophylactic wiring very important
- Bypass the fracture by 6 cm
- End of case, rotating the leg → **greater trochanter** fracture
  - Managed with screws and cerclage available

Follow-up Ask AI

- Patient returned last week by coincidence → follow-up
- Now about 80 years (was 77; now 78 or 79)

Clinical Pearl

- Distal cerclage before reaming → prevents shattering
- Greater trochanter fracture possible while rotating the leg

Take-Home Message Ask AI

- Careful handling of **bone fragments**
- Preservation of **soft-tissue attachments** (probable)
- **Stable fixation** of fragments
- Restoration of **length, offset, rotation**
- Follow the **treatment algorithm**; prepare all instruments and assistance
- **Attention to detail**

Loosening

(Case-Based Discussion of Aseptic Loosening)

Prof Dr Sherif Khaled

Watch Original Lecture

Lecture transcript processed with AI assistance.

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Cemented Femoral Stem Loosening Ask AI

Reading the Radiograph Ask AI

- Case series → prosthetic **loosening**
- Per case → loose or not?
- **Full X-ray** → examine each zone around implant
  - Note extent of involved zones → **zones 1 and 7**

Case 1 — Cement Fracture and Radiolucent Lines Ask AI

Cemented

- Key finding → **cement fracture**
  - Site → **lateral cement mantle**
- Plus **radiolucent lines** almost all around implant
- Verdict → **definitely loose**
- Diagnosed because **symptomatic**
  - Asymptomatic → would have been **followed up**
- No symptoms → prosthesis still **definitely loose**

Case 2 — Improper Cementing, Then a Fall Ask AI

Cemented

- Presentation → **periprosthetic fracture**
  - Same topic as Mahmoud's preceding lecture
- Timing → **one year after the surgery**
- Problem → **cement mantle**
  - Cementation not as it should be → **improper cementing**
- Old X-rays unavailable
- Only definite event → **fall**
- Fall → prosthesis failed

Case 3 — Subsidence, Then Pistoning Ask AI

Cemented

- Two similar mechanisms
- **Subsidence** → cement and stem migrate downward together
- **Pistoning** → stem moves within the cement
- Distal opacity → **cement plug**, not distal stem
- Sequence:
  - Whole construct subsided
  - → stem pistoned within cement
  - → **cement fracture** → further sinking
- Summary of findings:
  - **Cement fracture**
  - **Subsidence of cement and stem**
  - **Radiolucent lines all around**

Pitfall

- Stem here → cemented
- **Pedestal** sign → cementless stems only
- Distal opacity → cement plug, not distal stem

High Yield
- Cement fracture, lateral cement mantle → loosening
- Radiolucent lines all around implant
- Subsidence of cement and stem together
- Stem pistoning within cement

Cementless Acetabular Loosening Ask AI

Case 4 — Broken Hardware Ask AI

Cup

- **Cementless** acetabular component
- Finding → **broken hardware**
- Broken hardware → **subsidence**
- Loosening in some zones around cup

Signs of Cementless Cup Loosening Ask AI

- **Breakage of screws**
- **Medialization**
- **Protrusio**
- **Proximal migration**
- **Radiolucent zones** all around
- This cup:
  - **Broken screws**
  - Radiolucent lines > 2 to 5 mm
  - Involvement of **all three zones** of acetabulum
  - Cup in **protrusio**
  - Cup **migrated upward**

High Yield
- Broken screws
- Medialization / protrusio
- Proximal migration
- Radiolucent lines > 2 to 5 mm, all three zones

Femoral Loosening Patterns Ask AI

- Femoral failure modes include **cantilever** (bending)
- Confusing patterns → e.g. **windshield wiper effect**
- Distinguish **medial midstem pivot** from windshield wiper effect

	Windshield wiper effect	Medial midstem pivot
Fixed part	Proximal — held firmly above	Midstem pivot point
Loose part	Distal	Above and below the pivot

	Windshield wiper effect	Medial midstem pivot
Motion	**Distal toggle**	Swings back and forth
Bone effect	—	Progressive erosion of bone on one side

- Medial midstem pivot → stem swings → **erodes** bone on one side

Femoral Revision Options Ask AI

Options Discussed Ask AI

80 years

- Patient aged 80 years
- Option → **long stem Wagner**
- Cup → **dual mobility**, cemented or cementless?

Option	Note
Long stem Wagner	Suggested by audience
Long stem locked	Modular
Long stem cemented	Surgeon's choice
Short stem	Depends on its actual length

- **Implant sets differ** → stem length varies by system
- Some systems → length changes from size to size
  - **Size 1** → length 13.3
  - **Size 6** → length 17.9
- **Revision stem** vs standard → depends on actual stem length
- Adequate length → standard stem can be done

Procedure Performed Ask AI

- Surgeon's choice → **long stem**
- **Long stem cemented**
- **Tripolar** bearing
- **Cup cemented**
- Age 80 years → less demanding construct acceptable

Composite-Beam Stem with Lost Cement Ask AI

Findings Ask AI

75 years

- Friend's father, aged 75 years
- Finding → **subsidence**
- Known **cemented stem** → cement visible only in **zone 1**
- At removal → cement found inside
- Original surgery ~16 years earlier → **cement disappeared**

- Stem type → **composite beam**, not load taper
  - Recognized from **design of the stem**
- Plan → revision

Revision Choice Ask AI

- **Wagner** → proposed
- Must be **long stem**
  - Reason → **original stem already long**
- Performed → **long stem cemented**
- Cup → **dual mobility cup**

Ceramic-on-Ceramic Failure in a Young Patient Ask AI

Presentation Ask AI

39 years

- **Police officer**, 39 years old
- Injury (likely fall) on 12 December 2016
- Associated **head trauma**
  - → cognitive impairment, like Alzheimer's
  - Retired by police; not doing well now
- Cognitive functions → no longer as high as before

Primary Arthroplasty Ask AI

- **CT** → evaluation
- Workup → **rule out infection**
- Plan → **cementless**
- Performed → **total hip, cementless**
- Assess **inclination** and anteversion
- Cup slightly high → compare with **lesser trochanter**
- Relative to lesser trochanter → **cup set well**
- Surgeon's inclination → 40, not 30
- Most cups → 40 degrees

Failure at 7 Months Ask AI

- 7 months later → failure
- Bearing → probably **ceramic**
- Principle → always blame the **surgeon**
  - Never the patient, circumstances, life or nature
  - Error suspected; unclear if in the operation itself
- Fine for seven months → not an **early failure** (within three months)
- Plan → **revise**
- Revision problem → the **ceramic**
- Bearing → **ceramic on ceramic**, not surgeon's choice

Bearing Choice in a 39-Year-Old Ask AI

Preferred: hard on soft	Avoid
Metal on poly	Ceramic on ceramic
Ceramic on poly	

- **Not ceramic on ceramic** → see what happened here

Revision to Dual Mobility Ask AI

- Intra-operative finding → **ceramic head** failure
- **Cognitively impaired** → strange behaviour → **tripolar** construct
- Performed → **dual mobility** construct
- **Leg lengthened slightly** → compare level around lesser trochanter

Clinical Pearl
- Young (39-year-old) patient → hard on soft bearing
- Metal on poly or ceramic on poly
- Not ceramic on ceramic
- Cognitively impaired → tripolar / dual mobility for stability

Asymptomatic Acetabular Loosening on Follow-up Ask AI

- Longer construct → for **stability**
- **Stem not changed** → stem was good
- Follow-up → August 2018
- July 2019 → **lucency** around acetabular component
- **Zone 1** → superior lucency
- **Zone 3** → obscured on imaging, probably loose
- Revision offered → patient had no pain
- **Acetabular loosening** → often asymptomatic
  - → **follow up with X-rays**
- 7/2019 → no pain → no revision needed

Clinical Pearl
- Acetabular loosening → often asymptomatic
- → mandatory follow-up with X-rays

Failure of the Dual Mobility Construct Ask AI

Presentation in 2020 Ask AI

- 2020 → construct failure
- Approach → "wait until he gets fed up", not "wait until it unites"
- Previous: **dual mobility** cup placed well
- First operation → **cementless cup**

- First operation → **cementless cup**
- Current appearance → failed

Dual Mobility vs Bipolar Ask AI

- **Dual mobility ≠ bipolar**
  - **Large cup** in acetabulum → very large head inside
  - A **total hip** construct
  - Articulation → large metal cup ↔ large polyethylene head
    - Inside → **small metal head**

	Dual mobility	Bipolar
Outer head	**Plastic** (large polyethylene head)	Metal
Articulates with	Large metal cup in acetabulum	—
Inner head	Small metal head	—
Construct	Total hip	—

- **Tripolar** = dual mobility
- Course → **dislocation,** progressive displacement → everything disrupted
- Classroom exchange → student presents the case

Exam Pearl

- Tripolar = dual mobility
- Dual mobility ≠ bipolar
- Outer head plastic, inside large metal cup
- Small metal head inside polyethylene head

Infection Workup and Antibiotic Policy Ask AI

Inflammatory Markers Ask AI

Infection

- Student who knows the case → brief exchange
- **ESR** and **CRP** performed
  - CRP 30
  - ESR 100

Surgical Management Ask AI

- Outcome → not yet known
- **Removed everything**
- **Cultures** taken and sent
- Treatment → starts after culture results
  - No antibiotics yet

Antibiotic Policy Ask AI

- **Antibiotics are withheld until the culture results return** → identify organism
  - Exception → dose pre-operatively + dose post-operatively

  - This → quite sufficient

Tissue Cultures Ask AI

- **Tissue cultures** → organism search:
  - **Gram stain** → Gram-positive / Gram-negative
  - **Aerobic cultures**
  - **Anaerobic cultures**
- **TB** → sometimes investigated
  - Especially **chronic cases** referred from outside
- All investigations requested
- High ESR and CRP → **serial readings**

Must-Know

- Antibiotics withheld until culture results return
- Exception → one dose pre-operatively, one post-operatively
- Gram stain, aerobic and anaerobic cultures
- TB → chronic cases referred from outside
- High ESR / CRP → serial readings

Loose Cementless Cup with Cavitary Defect Ask AI

- Clearly loose
- Signs:
  - **Migration** → change in position, movement of acetabular cup
  - **Radiolucent zones**
- **Cavitary defect**
- Another **cementless cup** → all signs visible
- Presented in the first lecture
- **CT** → localize the defect:
  - **Medial wall** broken or not?
  - Cavitary defect?

Septic Loosening Ask AI

Characteristic Picture Ask AI

Septic

- Diagnosis → **septic loosening**
- This appearance → **no differential diagnosis**
- All signs together → **very rare**
- Signs:
  - **Periostitis**
  - **Reactive bone formation**
  - **Endosteal scalloping**
  - **Large radiolucent zones**
    1. **Expand very rapidly** → week by week
    2. Not every 6 months or a year

- → **characteristic septic loosening picture**
- Included because **rare**
- Dr **Mohamed Abou Saoud** → sees such cases often
  - Lecturer has never seen it himself
- Diagnosis → **septic loosening**

Must-Know

- Periostitis
- Reactive bone formation
- Endosteal scalloping
- Large radiolucent zones expanding week by week
- This picture → no differential diagnosis

Take-Home Messages Ask AI

- Loosening → **septic** or **aseptic**
- Clinical → **pain which increases with weight bearing**
- Covered → **radiological diagnosis of loosening**
- **Beware of septic loosening**
  - Must **exclude it** even if case looks aseptic

Instability

(Case-Based Discussion of Instability After THA)

Prof Dr Ahmad Azzam

Watch Original Lecture

Lecture transcript processed with AI assistance.

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AI detected corrections (Mostly bad transcripts)

Corrected - the text was changed; the transcript said

C1 (p. 319) “5-year-old girl” → “55-year-old female”

Case 1 — Early Dislocation After THA Ask AI

Presentation Ask AI

Case 1

- 55-year-old female, **AVN** → total hip arthroplasty *[C1 · transcript: "5-year-old girl"]*
- 2 days post-op → **posterior dislocation**
  - Limb in internal rotation
  - Dislocated three times
- Managed early
- Radiograph → comment on implant position systematically

Radiographic Assessment of Implant Position Ask AI

Cup Ask AI

- **Inclination**
- **Version**
- **Cup position**
  - Mediolateral
  - Superior/inferior
- **Head eccentric within cup** → not clearly seen in Case 1

Stem Ask AI

- **Medial (femoral) offset** → compare with contralateral side
  - Case 1: no major problem
- **Vertical offset** → likewise assessed
- Case 1: **lateral view** also obtained

CT Ask AI

Case 1

- Radiographs → then **CT**
- CT → component malposition, not apparent on plain films
- **Component malposition** > 10 degrees → management must include revision of malpositioned component

High Yield

- Component malposition > 10 degrees → revise the malpositioned component
- CT may show malposition not apparent on plain films

Case 2 — Late Dislocation with Loose Stem Ask AI

Presentation Ask AI

Case 2

- Female, hip arthroplasty → stable hip initially
- **Recurrent dislocation** 3 years post-op
- Radiograph → sign of **loosening**

Work-up and Revision Ask AI

Case 2

- **Infection markers** → negative
- **Aspiration** → negative
- Stem → **loose**
- Revised with **long stem**
  - Fixation cemented vs uncemented — not specified

Case 3 — Post-traumatic Dislocation Ask AI

History Ask AI

Case 3

- **Trauma** (accident) → hip dislocated
- Several months post-injury → hip unstable
- Patient in fact **arthritic**
- Primary fixation → then **total hip**

Cup Assessment Ask AI

Case 3

- Assess point by point
- **Inclination** ≈ 60 degrees → above 40 degrees
- **Version**
  - Some degree of **anteversion** → does not by itself require revision
  - Anteversion ≈ 15–20 degrees → acceptable
- Difficult situation → possibly no adequate acetabular bone stock
- **Hip center** slightly higher vs contralateral side

Femoral Component Ask AI

Case 3

- Joint in this view → appears reduced
- **Long neck,** high offset
  - → possible minor **impingement** → levering of joint

Medial Offset and Rotation Ask AI

- **Medial offset** → first compare with **contralateral side**
- Comparison requires images **rotationally identical** on both sides → axes comparable
- Slightly rotated → cannot judge
- Later view obtained
- Pelvic rotation difference → check:
  - Shape of **obturator foramen** on each side
  - Configuration of **pelvis**

Pitfall

- Offset and version comparison invalid unless both views rotationally identical
- Rotation clue → obturator foramen shape and pelvic configuration

Comparison Between Views Ask AI

Case 3

- **Rotation** differs between images
  - Cup inclination → approximately the same
  - Cup version → anteversion appears greater on one view

    - Rotational difference → cannot be certain
- **Lucency** between **cement and bone**?
- Change in **metalwork** shape/position at **screws**?
- Change in lucency?
- These findings may be present in any cause of failure

Constrained Liner Ask AI

Indications Ask AI

- Preferable → address instability by treating its cause
- Obvious problem (e.g., **component malposition**) → exclude → must be **revised**
- Cause **multifactorial** or **unclear** → **constrained liner** may be used
  - Not with > 10 degrees of component malposition
- Case 3 → revised with constrained liner

Design Ask AI

- Liner extends beyond **circumference of the head**
  - → constrains head within cup

Failure at One Year Ask AI

Case 3

- Post-op view → **screws** fix the liner
- Did well ≈ 1 year → then re-presented
- **Hardware failure** beginning
- **Cup position** changed
- Compare with previous view → new change in this part
- **Hardware broke**
- Defect in **superior wall**

Lessons Ask AI

Constrained

- Constrained liner → not to be relied on heavily
  - Did well 1 year → still not sufficient
- Not for liberal use in multifactorial or unsuitable situation
- Reserved for **salvage**

Pitfall

- Constrained liner not a substitute for revising malposition > 10 degrees
- Case 3: good for 1 year → hardware broke, superior wall defect
- Reserve for salvage, not liberal use in multifactorial instability

Revision After Constrained Liner Failure Ask AI

Options Ask AI

- **Dual mobility construct** planned → useful to convert to it
- Any problem with **component position** → revise the components

Revision Ask AI

Case 3

- Case 3 → revised
- All components removed
- New reconstruction performed
- Dedicated acetabular construct used
- Follow-up time not yet long
- Follow-up view

Prevention and Management Algorithm Ask AI

Prevention Ask AI

- Reference: **2011**
- Very important → in **initial primary total hip**, surgeon must:
  - Choose patient and implant carefully
  - Technically **restore the anatomy**
    - **Soft-tissue tensioning** across hip joint
- Treatment always far more costly than **prevention**

Management by Risk Factor Ask AI

- Identify **risk factors** → first exclude:
  - **Component malpositioning** → revision
  - **Loosening** → revision
- Absent → other adjuncts:
  1) Increase **head size**
  2) Change the **liner**
  3) **Soft-tissue procedure** (e.g., trochanteric advancement)
  4) **Constrained liner**
- Choice approximately according to the problem

Finding	Management
Component malposition > 10 degrees	Revise malpositioned component
Loosening	Revision
Multifactorial / unclear cause	Adjuncts: larger head, liner change, soft-tissue procedure, constrained liner
Constrained liner failure	Revise components; consider dual mobility

Must-Know

- Prevention in primary THA → patient and implant selection, anatomy restoration, soft-tissue tensioning
- Exclude malposition and loosening first → both need revision
- Multifactorial cause → more than one method

Case 4 — Clear Mechanical Cause Ask AI

Case 4

Presentation Ask AI

- 65-year-old, **fracture neck of femur**
- Operated 3 weeks after trauma
- **Dislocation** ≈ 3 months post-op
- Simple → very clear **mechanical** problem

Principle Ask AI

- Other cases → no **clear cause** known
  - As **Dr. Sherif** discussed in the **loosening** lecture
- Cause not apparent or **multifactorial** → use **more than one method** in treatment
- Case 4 → extensive revision
- Excellent result after revision